V.S.B. ENGINEERING COLLEGE, KARUR

2 MARK AND 16 MARK QUESTION BANK

COMMON TO ALL BRANCHES

CLASS I YEAR/ I SEMESTER

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Professional English I

STUDY MATERIAL

UNIT - I

1. READING COMPREHENSION

Passage 1 - Opera

Opera refers to a dramatic art form, originating in Europe, in which the emotional content is conveyed to the audience as much through music, both vocal and instrumental, as it is through the lyrics. By contrast, in musical theater an actor's dramatic performance is primary, and the music plays a lesser role. The drama in opera is presented using the primary elements of theater such as scenery, costumes, and acting. However, the words of the opera, or libretto, are sung rather than spoken. The singers are accompanied by a musical ensemble ranging from a small instrumental ensemble to a full symphonic orchestra.

1. It is pointed out in the reading that opera ----.

- A) has developed under the influence of musical theater
- B) is a drama sung with the accompaniment of an orchestra
- C) is not a high-budget production
- D) is often performed in Europe
- E) is the most complex of all the performing arts

2. We can understand from the reading that ----.

- A) people are captivated more by opera than musical theater
- B) drama in opera is more important than the music
- C) orchestras in operas can vary considerably in size
- D) musical theater relies above all on music
- E) there is argument over whether the music is important or the words in opera

3. It is stated in the reading that ----.

- A) acting and costumes are secondary to music in musical theater
- B) many people find musical theater more captivating than opera
- C) music in musical theater is not as important as it is in opera
- D) an opera requires a huge orchestra as well as a large choir
- E) opera doesn't have any properties in common with musical theater

Passage 2 - Dolphins

Dolphins are regarded as the friendliest creatures in the sea and stories of them helping drowning sailors have been common since Roman times. The more we learn about dolphins, the more we realize that their society is more complex than people previously imagined. They look after other dolphins when they are ill, care for pregnant mothers and protect the weakest in the community, as we do. Some scientists have suggested that dolphins have a language but it is much more probable that they communicate with each other without needing words. Could any of these mammals be more intelligent than man? Certainly the most common argument in favor of man's superiority over them that we can kill them more easily than they can kill us is the least satisfactory. On the contrary, the more we discover about these remarkable creatures, the less we appear superior when we destroy them.

4. It is clear from the passage those dolphins ----.

- A) don't want to be with us as much as we want to be with them
- B) are proven to be less intelligent than once thought
- C) have a reputation for being friendly to humans
- D) are the most powerful creatures that live in the oceans
- E) are capable of learning a language and communicating with humans

5. The fact that the writer of the passage thinks that we can kill dolphins more easily than they can kill us ----.

- A) means that they are better adapted to their environment than we are
- B) shows that dolphins have a very sophisticated form of communication
- C) proves that dolphins are not the most intelligent species at sea
- D) does not mean that we are superior to them
- E) proves that Dolphins have linguistic skills far beyond what we previously thought

6. One can infer from the reading that ----.

- A) dolphins are quite abundant in some areas of the world
- B) communication is the most fascinating aspect of the dolphins
- C) dolphins have skills that no other living creatures have such as the ability to think
- D) it is not usual for dolphins to communicate with each other
- E) dolphins have some social traits that are similar to those of human

2. 'WH' QUESTIONS (QUESTION WORDS)

Types of questions
There are two types of questions:

- Yes or no questions
- 'Wh' questions

Question words
Question words are also called wh questions because they include the letters 'W' and 'H'.

Question words	Meaning	Examples
who	person	Who's that? That's Nancy.
where	place	Where do you live? In Boston
why	reason	Why do you sleep early? Because I've got to get up early
when	time	When do you go to work? At 7:00
how	manner	How do you go? By car
what	object, idea or action	What do you do? I am an engineer
which	choice	Which one do you prefer? The red one.
whose	possession	Whose is this book? It's Alan's.
whom	object of the verb	Whom did you meet? I met the manager.
what kind	description	What kind of music do you like? I like quiet songs
what time	time	What time did you come home?
how many	quantity (countable)	How many students are there? There are twenty.
how much	amount, price (uncountable)	How much time have we got? Ten minutes
how long	duration, length	How long did you stay in that hotel? For two weeks.
how often	frequency	How often do you go to the gym? Twice a week.
how far	distance	How far is your school? It's one mile far.
how old	age	How old are you? I'm 16.
how come	reason	How come I didn't see you at the party?

3. PARTS OF SPEECH

The **parts of speech** explain how a word is used in a sentence.
There are eight main parts of speech (also known as word classes): **nouns, pronouns, adjectives, verbs, adverbs, prepositions, conjunctions** and **interjections**.

Most **parts of speech** can be divided into sub-classes. Prepositions can be divided into prepositions of time, prepositions of place etc. Nouns can be divided into proper nouns, common nouns, concrete nouns etc.
It is important to know that a word can sometimes be in more than one part of speech. For example with the word **INCREASE**.
INCREASE can be a verb e.g. Prices **INCREASED**
and **INCREASE** can also be a noun e.g. There was an **INCREASE** in the number of followers.
The eight main parts of speech in English are:

NOUN - (Naming word)
A noun is the name of a person, place, thing or idea.
Examples of nouns: Daniel, London, table, dog, teacher, pen, city, happiness, hope
Example sentences: **Steve** lives in **Sydney**. **Mary** uses **pen** and **paper** to write **letters**.
Learn more about the different types of nouns.
PRONOUN - (Replaces a Noun)
A pronoun is used in place of a noun or noun phrase to avoid repetition.
Examples of pronouns: I, you, we, they, he, she, it, me, us, them, him, her, this, those
Example sentences: Mary is tired. **She** wants to sleep. **I** want **her** to dance with **me**.
ADJECTIVE - (Describing word)
An **adjective** describes, modifies or gives more information about a noun or pronoun.
Examples: big, happy, green, young, fun, crazy, three
Example sentences: The **little** girl had a **pink** hat.
VERB - (Action Word)
A **verb** shows an action or state of being. A verb shows what someone or something is doing.
Examples: go, speak, run, eat, play, live, walk, have, like, are, is
Example sentences: I **like** Woodward English. I **study** their charts and **play** their games.
ADVERB - (Describes a verb)
An **adverb** describes/modifies a verb, an adjective or another adverb. It tells how, where, when, how often or to what extent. Many adverbs end in -LY
Examples: slowly, quietly, very, always, never, too, well, tomorrow, here
Example sentences: I am **usually** busy. **Yesterday**, I ate my lunch **quickly**.
PREPOSITION - (Shows relationship)
A **preposition** shows the relationship of a noun or pronoun to another word. They can indicate time, place, or relationship.
Examples: at, on, in, from, with, near, between, about, under
Example sentences: I left my keys **on** the table **for** you.
CONJUNCTION - (Joining word)
A **conjunction** joins two words, ideas, phrases or clauses together in a sentence and shows how they are connected.
Examples: and, or, but, because, so, yet, unless, since, if.
Example sentences: I was hot **and** exhausted **but** I still finished the marathon.
INTERJECTION - (Expressive word)
An **interjection** is a word or phrase that expresses a strong feeling or emotion. It is a short exclamation.
Examples: Ouch! Wow! Great! Help! Oh! Hey! Hi!
Example sentences: Wow! I passed my English test. Great! – Ouch! That hurt.

4. PREFIXES & SUFFIXES

PREFIXES

Prefixes are added to the beginning of a word to make a new word, e.g. mis+take. Some common prefixes are 'mis', 'dis', 're', 'for', 'anti', 'ante', 'sub', 'un' and 'in'. New words are made by placing a prefix in front of a word. It is helpful to know the meanings of prefixes. Prefix ‘**re’** means **again,** therefore **reappear** means to appear **again.**

Prefix

A **prefix** is a group of letters placed before the root of a word. For example, the word "unhappy" consists of the prefix "un-" [which means "not"] combined with the root (stem) word "happy"; the word "unhappy" means "not happy."

A Short List of Prefixes:

PREFIX	MEANING	EXAMPLES
de-	from, down, away reverse, opposite	decode, decrease
dis-	not, opposite, reverse, away	disagree, disappear
ex-	out of, away from, lacking, former	exhale, explosion
il-	not	illegal, illogical
im-	not, without	impossible, improper
in-	not, without	inaction, invisible
mis-	bad, wrong	mislead, misplace
non-	not	nonfiction, nonsense
pre-	before	prefix, prehistory
pro-	for, forward, before	proactive, profess, program
re-	again, back	react, reappear
un-	against, not, opposite	undo, unequal, unusual

Suffix

A **suffix** is a group of letters placed after the root of a word. For example, the word flavorless consists of the root word "flavor" combined with the suffix "-less" [which means "without"]; the word "flavorless" means "having no flavor."

A Short List of Suffixes:

SUFFIX	MEANING	EXAMPLES
-able	able to, having the quality of	comfortable, portable
-al	relating to	annual, comical
-er	comparative	bigger, stronger
-est	superlative	strongest, tiniest
-ful	full of	beautiful, grateful
-ible	forming an adjective	reversible, terrible
-ily	forming an adverb	eerily, happily, lazily
-ing	denoting an action, a material, or a gerund	acting, showing
-less	without, not affected by	friendless, tireless
-ly	forming an adjective	clearly, hourly
-ness	denoting a state or condition	kindness, wilderness
-y	full of, denoting a condition, or a diminutive	glory, messy, victory,

5. TECHNICAL VOCABULARY (SYNONYMS/ANTONYMS)

Match the words with their meanings

1. a) accelerate	1) feasible
b) simultaneously	2) increase speed
c) viable	3) extremely high
d) prohibitive	4) occurring at the same time

Answer: 2,4,1,3

2. a) distort	1) bulky
b) cumbersome	2) twist out of shape
c) comprehend	3) state of being out of date
d) obsolescence	4) understand

Answer: 3,1,4,2

3. a) alien	1) remove
b) discard	2) innumerable
c) myriad	3) foreign
d) advent	4) arrival

Answers : 3,1,2,4

4. a) priming	1) thing or part added to improve
b) indispensable	2) preparing
c) supplement	3) consider before hand
d) anticipate	4) necessary

Answer: 2,4,1,3

5. a) explore	1) including all
b) comprehensive	2) arrangement
c) array	3) unfit to eat
d) inedible	4) examine

Answer: 4,1,2,3

6. (a) Slight	a) Temperate
(b) Penny-wise	b) Generous
(c) Avaricious	c) Heavy
(d) Greedy	d) Extravagant

Answer: c,d,a,b

7. a. Diversity	a) Negligent
b. Inadvertent	b) Hindrance
c. Obstacle	c) Shooting forward
d. Projection	d) Difference

Answer: d, a,b,c

8. a) Dynamism	1) producer
b) Enhance	2) sympathy
c) Breeder	3) heighten
d) Empathy	4) strength

Answer: 5,3,1,2

9. a. Accurate	1) Swift
b. Gigantic	2) Precise
c. Elide	3) Huge
d. Rapid	4) Omit

Answer: 2,3,4,1

10	a) Conventional	1) use up
	b) Appropriate	2)traditional
	c) consume	3)exhibit
	d) display	4)suitable

Answer: 2,4, 1,3

I. **Antonyms**

1) What's the opposite of more? - Few, less, a lot, many
2) What's the opposite of alive? - Asleep, dead, death, sick
3) What's the opposite of loud? - Soft, quite, quiet, noisy
4) What's the opposite of happy? - Glad, ugly, difficult, sad
5) What's the opposite of last? - First, beginning, start, end
6) What's the opposite of now? - Always, never, then, soon
7) What's the opposite of to buy? - to take, to sell, to bring, to lend
8) What's the opposite of better? - Bad, worse, good, worst
9) What's the opposite of dangerous? - Easy, beautiful, safe, pleasant
10) What's the opposite of cool? - Warm, hot, cold, cooler

II.

1) What's the opposite of many? - Less, few, more, most
2) What's the opposite of deep? - Shallow, narrow, thin, low
3) What's the opposite of long? - Broad, small, thing, short
4) What's the opposite of clean? - tidy up, clear, dirty beautiful
5) What's the opposite of empty? - Shut, open, Ugly, full
6) What's the opposite of thin? - Light, slow, fat, more
7) What's the opposite of beginning? - End, start, finish, depart
8) What's the opposite of big? - Huge, small, large, less
9) What's the opposite of modern? - Older, ancient, new, contemporary
10) What's the opposite of poor? - Wealth, rich, poverty, wise

III.

1) What's the opposite of to break? - to join, to fasten, to mend, to build
2) What's the opposite of early? - Soon, late, last, first
3) What's the opposite of to lose? - to forget, to find, to search, to look for
4) What's the opposite of over? - Downstairs, below, under, above
5) What's the opposite of rough? - Sharp, heavy, hard, smooth
6) What's the opposite of heavy? - Light, soft, easy, smooth
7) What's the opposite of to learn? - To know, to teach, to forget, to understand
8) What's the opposite of high? - Under, low, beneath, down
9) What's the opposite of to start? - to stop, to begin, to go on, to continue
10) What's the opposite of to push? - to tow, lift ,to pull, to send

Answers

I.1) less 2) dead 3) quiet 4) sad 5) first 6) then 7) to sell 8) worse 9) safe 10) Warm
II.1) few 2) shallow 3) short 4) dirty 5) full 6) fat 7) end 8) small 9) ancient 10) rich
III.1) to mend 2) late 3) to find 4) under 5) smooth 6) light 7) to teach 8) low 9) to stop 10) to pull

UNIT- II

1. ARTICLES

In English grammar, an **ARTICLE** is a type of determiner that precedes a noun
Traditionally, two articles are recognized in English: the definite article (THE) and the indefinite (A/AN). Generally speaking, the definite article specifies a particular individual in a particular context. The indefinite article indicates that the following noun is a member of a class. (The indefinite article A is used before a consonant sound, AN before a vowel sound). The definite article appears in front of both singular and plural nouns. The indefinite article usually appears in front of singular nouns only.

THE DEFINITE ARTICLE (FOR EXAMPLE: THE BOX)

"Definite" means "clear, obvious."
The **definite article** tells us that the noun is specific. The speaker talks about a particular (or known) thing.
The definite article in English is "**the**."
Examples:
"**The** car I bought broke down."
"He has a son and a daughter. **The** daughter is a doctor."
Examples:
"The flower he gave her was very beautiful."
"The flowers on the table are nice."

THE INDEFINITE ARTICLE

(FOR EXAMPLE: A BOX)

"In-" means "not," and **"definite"** means "clear, obvious."
"Indefinite" means "not clear, not exact."
The **indefinite article** tells us that the noun is *not* specific. The speaker talks about *any one* of that type of thing.
The indefinite articles in English are **"a"** and **"an."**
Examples:
"I need **a** pen."
(The speaker doesn't tell us which pen he needs. He simply needs a pen, any pen.)
"Lisa wants to see **a** movie."
(The speaker doesn't tell us which movie Lisa wants to see. She wants to see some movie.)
"Joe has **a** car."
(The speaker doesn't tell us which car Joe has. He has some car.)
"She is **a** dancer."
(She dances for a living.)
What is the difference between "a" and "an"?
"A" and "an" have the same meaning.
We use "a" before a consonant sound.
Examples:
a dog, a building, a country, a professor, a university
We use "an" before a vowel sound.
Examples:

a - ‘ah’ an apple, an aero plane, an amplifier, an air, an ant etc.
e – ‘eh’ an egg, an eminent, an elephant, an apron, an alien, an ancient, an ape, an escape, an education, an etiquette, an M.A. an M. Phil. an M.L.A, M.P. etc.
i – ‘ih’ an insect, an interest, an igloo, an inflation, an important, an ink bottle, an instrument, an innovation etc.
o – ‘awe/ove’ an hour/honour/honest, an orange, an odour, an oven, an owl, an optimist, an awesome performance etc.
u – ‘uh’ an umbrella, an apology, an amazing, an octopus, an ostrich, an ugly, an accumulation etc.

2. PREPOSITIONS

Preposition	**Meaning**	**Examples**
above	higher than, or over	The sun is **above** the clouds.
across	from one side to the other	It's dangerous to run **across** the road.
against	- in opposition to - in contact with	- Stealing is **against** the law. - The sofa is **against** the wall.
along	from one end to the other	They are walking **along** the street.
among	surrounded by	Peter was a**mong** the spectators.
around	- in a circle - near, approximately	- He walked **around** the table. - It costs **around** 50 euros.
before	- earlier than - in front of	- The day **before** yesterday. - He bowed **before** the king.
behind	at the back of	Passengers sit **behind** the driver.
below	lower than	His shorts are **below** his knees.
beside	next to	The bank is **beside** the cinema.
between	in the space separating two things	Mary sat **betwee**n Tom and Jane.
by	near, at the side of	The restaurant is **by** the river.
down	from higher to lower	She pulled **down** the blind.
for	what is intended	I bought this book **for** you.
from	where something starts or originates	The wind is blowing **from** the north.
in	at a point within an area	The pen is **in** the drawer.
inside	on the inner part of	The bird is **inside** the cage.
into	enter a closed space	He went **into** the shop.
of	belonging to	Master **of** Arts and Science
off	down or away from	He fell **off** the horse.

on	in a position touching a surface	The plate is **on** the table.
onto	move to a position on a surface	The cat jumped **onto** the roof of the car.
over	- above/across - on the surface of	- The plane flew **over** the Atlantic. - She put a sheet **over** the furniture.
round	in a circular movement	The earth moves **round** the sun.
through	from one side to the other	The Seine flows **through** Paris.
throughout	in every part of	The virus spread **throughout** the country.
to	in the direct of / towards	On the way **to** the station.
towards	in the direction of	The child ran **towards** her father.
under	beneath, below	Water flows **under** the bridge.
up	towards or in a higher position	She walked **up** the stairs.

3. CONJUCTIONS

What is a conjunction?
It's a word that joins words, phrases, or clauses.

Okay, read this question and then try to answer it with your eyes closed. What is a conjunction?
How did you do? Did you remember that they join words, phrases, and clauses? Let's look at a few more examples.

- **Words**: silver **and** gold
- **Phrases**: over the river **and** through the woods
- **Clauses**: Marianne planted a flower, **and** she watched it grow.

In all of those examples, we used the conjunction and, one of the most common conjunctions.
But, of course, there are many more conjunctions out there.
All conjunctions perform the same basic function (gluing things together), but there are three types of conjunctions, and you'll be learning about them next!

1. Coordinating Conjunctions
2. Subordinating Conjunctions
3. Correlative Conjunctions

Types of Conjunctions	Definition	Conjunctions
Coordinating Conjunctions	Joins words, phrases and independent clauses Example: The park is green, but it is greener in the winter.	for, and, nor, but, or, yet, so
Subordinating Conjunctions	Introduces a dependent clause and ties it to an independent clause Example: If you leave, I will be lonely.	after, although, as, as if, as long as, as much as, as soon as, as though because, before, even, even if, even though, if, if

		only, if when, if then, inasmuch, in order that, just as, lest, now, now since, now that, now when, once, provided, provided that, rather than, since, so that, supposing, than, that, though, till, unless, until, when, whenever, where, whereas, where if, wherever, whether, which, while, who, whoever, why
Correlative Conjunctions	A pair of conjunctions that must be used together Example: I either want ice cream or yogurt.	both / and, not only / but also, either / or, neither / nor, whether / or, as / as such / that, scarcely / when, as many / as, no sooner / than, rather / than

4. GERUNDS

WHAT ARE GERUNDS? (WITH EXAMPLES)

A gerund is a noun formed from a verb. All gerunds end *-ing*. For example:

- swimming
- running
- drinking

Even though a gerund is a noun, a gerund can still take a direct object (like a verb). This is known as a gerund complement. For example:

- swimming **the lake**
- running **a mile**
- drinking **a beer**

MORE EXAMPLES OF GERUNDS

Below are some more examples of gerunds (shaded) with their roles as nouns explained:

- Acting is fun.
  (Gerund as the subject of a sentence)
- Playing football is fun.
  (Here, *football* is the gerund complement of the gerund *playing*.)
- Acting is merely the art of keeping a large group of people from coughing. (Sir Ralph Richardson, 1902-1983)
  (*Acting* is a gerund as a subject. The gerunds *keeping* and *coughing* are objects of prepositions.)
  (In this example, *a large group of people* is the gerund complement of *keeping*.)
  - Generosity is giving more than you can, and pride is taking less than you need. (Kahlil Gibran, 1883-1931)

(Two gerunds, both subject complements)

- I love acting. It is so much more real than life. (Oscar Wilde, 1854-1900)
  (A gerund as the direct object of the verb *love*)
- You can tell a lot about a fellow's character by his way of eating jellybeans. (Ronald Reagan, 1911-2004)
  (A gerund as the object of a preposition)
- I like to play blackjack. I'm not addicted to gambling, I'm addicted to sitting in a semi-circle. (Mitch Hedberg, 1968-2005)
  (Two gerunds, both objects of prepositions)

GERUND PHRASES

A gerund will often appear in a gerund phrase. A gerund phrase consists of a gerund, its object, and all modifiers. For example (gerund phrase shaded):

- Picking rotten apples from the floor is a great way to get stung by a wasp.

Read more about gerund phrases.

ALL GERUNDS END -*ING* BUT SO DO PRESENT PARTICIPLES

Even though all gerunds end with the suffix -*ing*, not every word which ends -*ing* is a gerund. The other common type of word which ends -*ing* is the present participle. Like gerunds, present participles are also formed from verbs (making them verbals), but they are not used as nouns. They are used as adjectives or when forming verbs in a progressive tense. For example:

- Running the tap will clear the air pocket.
  (This is a gerund.)
- Can you fix the **running** tap?
  (This is a present participle as an adjective.)
- The tap **was running** for an hour.
  (This is a present participle used to form the past progressive tense.)

5. WORD FORMATION

Use the words in capitals to form a word that fits in the space in the same line!

Detroit is renowned for the _____ of cars.	**PRODUCE**
If you make a good _____ at the interview, you will get the job	**IMPRESS**
Teaching and medicine are more than _____, they're professions.	**OCCUPY**
My history teacher has a vast _____ of past events.	**KNOW**
You are never too old to go to college and gain some _____	**QUALIFY**
My greatest _____ was graduating from the university.	**ACHIEVE**

The weatherman said there is a strong _____ of rain today.	**POSSIBLE**
Some old laws are no longer _____.	**EFFECT**
Athens is _____ for its ancient buildings.	**FAME**
He was caught shoplifting so now he has a _____ record.	**CRIME**
Despite her severe _____, she fulfilled her goals in life.	**DISABLE**
Being _____ is the worst thing that can happen to someone.	**EMPLOY**
If you buy presents in the summer your _____ can be very high.	**SAVE**
Due to the pilot's _____, the copilot managed to land safely.	**GUIDE**
It's important to also see the less _____ sides of the job.	**DESIRE**
I was surprised at his _____ to give up.	**REFUSE**
Children are by nature _____ of danger.	**AWARE**
She is always _____ towards her parents.	**RESPECT**
the hospital has the best medical _____ and fast ambulances.	**EQUIP**
You can relax in the comfortable _____ of the hotel.	**SURROUND**
The _____ looked dark and there were hardly any other guests.	**ENTER**

UNIT – III

1. REFERENCE WORDS

Reference Words and How to Use Them It/they/them	Used to replace ideas that have been mentioned previously	□ Opening an office in Brno seems sensible. **It** would take advantage of the growing market there. □ They've sent us ten different reports. It's going to take time to go through **them**.

		There are new customers joining up all the time. If we are to benefit from **them** we need to make sure we provide useful content.
This/these	Used to refer to objects or ideas that are near in place or time	You suggested last week that we should open an office in Brno. **This** seems to be a good idea if we want to benefit from the market there. Thank you for sending me sample materials from your new course. **These** have given me a good idea of the high quality training that your company delivers.
That/those	Used to refer to objects that are more remote in place or time	The minister was accused of having lied to the journalist about the fraud cases. **That** is not exactly what happened though. He was misquoted. Staff who wish to take advantage of this new benefit are able to apply immediately. **Those** who do not apply before the deadline will have another opportunity in three months time.

2. DISCOURSE

List of Discourse Markers *(Penanda Wacana)*

ADDING (menambah)
And *(dan)*
Also *(juga)*
as well as *(serta)*
moreover *(tambahan pula)*
too *(sangat / juga)*
furthermore *(tambahan pula)*
additionally *(tambahan itu)*

TO EXPRESS (menunjukkan pendapat)
frankly *(secara terus terang)*
I think (saya rasa/ *fikir)*
in my opinion *(pada pendapat saya)*
actually *(sebenarnya)*
I believe that.. *(saya percaya bahawa..)*
unfortunately *(malangnya)*
luckily *(nasib baik)*
honestly *(secara jujurnya)*

SEQUENCING (susunan)
once upon a time *(pada suatu hari)*
from this point *(dari sini)*
at the beginning *(pada awalnya)*
first, second, third... *(pertama, kedua, ketiga...)*
finally *(akhirnya)*
next *(seterusnya)*
meanwhile *(sementara itu)*
after *(selepas)*
after that *(selepas itu)*
then *(kemudian)*
subsequently (*seterusnya / kemudiannya)*
in conclusion *(kesimpulannya)*
in the end *(pada akhirnya)*
at last *(akhirnya)*
final finally *(akhir sekali)*

ILLUSTRATING (menggambarkan)
for example *(sebagai contoh)*
such as *(seperti)*
like *(seperti)*
for instance (misalnya)
in the case of *(dalam kes)*
as revealed by... *(seperti yang didedahkan oleh..)*
illustrated by *(digambarkan oleh)*

CAUSE and EFFECT (sebab dan akibat)
because *(sebab / kerana)*
so *(jadi)*
therefore *(oleh itu)*
thus *(dengan itu / oleh itu)*
consequently *(akibatnya)*
hence *(justeru / seterusnya / dengan itu)*
as a result *(sebagai kesannya)*

COMPARING (membandingkan)
similarly *(sama seperti / begitu juga)*
likewise *(begitu juga / demikian juga)*
as with (seperti dengan)
like *(seperti)*
equally *(sama / dengan sama rata)*
in the same way.. *(melalui cara yang sama)*

QUALIFYING (melayakkan)
But *(tetapi)*
However *(walau bagaimanapun)*
Although *(walaupun)*
Unless *(melainkan jika)*
Except *(kecuali)*
apart from *(selain daripada..)*
as long as *(selagi)*
if *(jika)*
even so *(walaupun begitu)*
even though *(walaupun)*
in another way *(dengan cara lain)*
on the other hand *(dengan kata lain)*
for all that *(untuk semua itu)*

-MEENOZ'12-

MARKERS

3. JUMBLED SENTENCES

1. Eminent
S1:In the middle of one side of the square sits the Chairman of the committee, the most important person in the room.
P :For a committee is not just a mere collection of individuals.
Q :On him rests much of the responsibility for the success or failure of the committee.
R :While this is happening we have an opportunity to get the 'feel' of this committee.
S :As the meeting opens, he runs briskly through a number of formalities.
S6:From the moment its members meet, it begins to have a sort nebulous life of its own.
The Proper sequence should be:
A. RSQP B. PQRS C. SQPR D. **QSRP**

2.
S1:A force of exists between everybody in the universe.
P :Normally it is very small but when the one of the bodies is a planet, like earth, the force is considerable.
Q :It has been investigated by many scientists including Galileo and Newton.
R :Everything on or near the surface of the earth is attracted by the mass of earth.
S :This gravitational force depends on the mass of the bodies involved.
S6:The greater the mass, the greater is the earth's force of attraction on it. We can call this force of attraction gravity.
The Proper sequence should be:
A. PRQS B. PRSQ C. QSRP D. **QSPR**

3.
S1:Calcutta unlike other cities kepts its trams.
P :As a result there horrendous congestion.
Q :It was going to be the first in South Asia.
R :They run down the centre of the road
S :To ease in the city decided to build an underground railway line.
S6:The foundation stone was laid in 1972.
The Proper sequence should be:
A. PRSQ B. PSQR C. SQRP D. **RPSQ**

4.
S1:For some time in his youth Abraham Lincoln was manager for a shop.
P :Then a chance Customer would come.
Q :Young Lincoln way of keeping shop was entirely unlike anyone else's
R :Lincoln would jump up and attend to his needs and then revert to his reading.
S :He used to lie full length on the counter of the shop eagerly reading a book.

S6:Never before had Lincoln had so much time for reading as had then.
The Proper sequence should be:
A. SRQP B. **QSPR** C. SQRP D. QPSR

5.
S1:All the land was covered by the ocean.
P :The leading god fought the monster, killed it and chopped its body in to two halves.
Q :A terrible monster prevented the gods from separating the land from the water.
R :The god made the sky out of the upper part of the body and ornamented it with stars.
S :The god created the earth from the lower part, grew plants on it and populated it with animals.
S6:The god moulded the first people out of clay according to his own image and mind.
The Proper sequence should be:
A. PQRS B. PQSR C. QPSR D. **QPRS**

4. DEGREES OF COMPARISON

Comparison can be made using the three forms of the adjective.
Adjective is a word and it qualifies a noun. It gives more information about the noun.
eg. The lion is a ***strong*** animal. Rita is a ***beautiful*** girl.
Adjectives are of three degrees. (1) Positive (2) Comparative (3) Superlative
eg. David is a ***rich*** man.
eg. Cancel is ***more dreadful*** *than cholera*.
eg. Mount Everest is ***the highest*** *peak in the world*.
Study the following models.

Positive	Comparative	Superlative
Very few boys in the class are as ***tall*** as John.	John is ***taller*** than any other boys in the class.	John is the tallest boy in the class.
	John is ***taller*** than most other boys in the class.	John is ***one of the tallest*** boy in the class.

Adjective and Adverbs have three degrees, Positive, Comparative and Superlative.
The Pasitive Degree is used to denote the mere existence of quality.
The Positive Degree of an adjective in comparison is the adjective in its simple form. It is used to denote the mere existence of some quality of what we speak about. It is used when ***no comparison is made***.

- It is a *tall* building.
- Apple is *sweet* to taste.

The Comparative Degree is used to compare the qualities of two persons or things.

The Comparative Degree denotes the existence of a higher degree of the quality than the positive. It is used ***when two things*** (or two sets of things) are compared.

- This building is *taller* than any other building.
- Apple is *sweeter* than pear.
- **The Superlative Degree** denotes the existence of the highest degree of the quality. It is used when more than two things are compared.
- This is the *tallest* building.
- Apple is the *sweetest* fruit.

The Superlative Degree is used when more than two nouns or things are compared.

- Johnsy is kind (*Positive Degree*)
- Johnsy is kinder than Rosy (*Comparative Degree*)
- Johnsy is the kindest of all (*Superlative Degree*)

Comparative	Positive
The Blue Mountain Express runs faster than the Green Express.	The Green Express does not run so fast as the Blue Mountain Express.
1. Very few books are read _____ Harry Potter by children. (a) as much as (b) more than (c) the most	
2. Life in the desert is _____ than life in the mountains. (a) as dreary as (b) the dreariest (c) drearier	
3. Honey is _____ medicine. (a) as good as (b) better than (c) the best	
4. School days are _____ than college days. (a) as enjoyable as (b) more enjoyable (c) the most enjoyable	
5. The coconut tree has _____ any other tree. (a) as much uses as (b) more uses than (c) the most uses of	
6. Dogs are _____ as dolphins. (a) as friendly (b) more friendly (c) the most friendly	
7. No other country sells _____ number of cellphones as India after America. (a) as many (b) more (c) the most	

8. Nursing is _____ many other professions. (a) as noble as (b) nobler than (c) the most noble	
9. Very few channels are watched by children _____ the Cartoon Channel. (a) as much as (b) more than (c) the most	
10. Black is the _____ colour. (a) as dark as (b) darker than (c) darkest	

5. PRONOUNS

WHAT IS A PRONOUN?

In grammar, a pronoun is defined as a word or phrase that may be substituted for a **noun** or noun phrase, which once replaced, is known as the pronoun's antecedent. How is this possible? In a nutshell, it's because pronouns can do everything that nouns can do. A pronoun can act as a subject, direct object, indirect object, object of the **preposition**, and more.

Without pronouns, we'd have to keep on repeating nouns, and that would make our speech and writing repetitive, not to mention cumbersome. Most pronouns are very short words. Examples include:

- **Indefinite pronouns** – those referring to one or more unspecified objects, beings, or places
- **Personal pronouns** – those associated with a certain person, thing, or group; all except you have distinct forms that indicate singular or plural number
- **Reflexive pronouns** – those preceded by the adverb, adjective, pronoun, or noun to which they refer, and ending in **–SELF** or **–SELVES**
- **Demonstrative pronouns** – those used to point to something specific within a sentence
- **Possessive pronouns** – those designating possession or ownership
- **Relative pronouns** – those which refer to nouns mentioned previously, acting to introduce an adjective (relative) clause
- **Interrogative pronouns** – those which introduce a question
- **Reciprocal pronouns** – those expressing mutual actions or relationship; i.e. one another
- **Intensive pronouns** – those ending in **–SELF** or **–SELVES** and that serve to emphasize their antecedents

PRONOUN RULES

There are a few important rules for using pronouns. As you read through these rules and the examples in the next section, notice how the pronoun rules are followed. Soon you'll see that pronouns are easy to work with.

- Subject pronouns may be used to begin sentences. For example: We did a great job.
- Subject pronouns may also be used to rename the subject. For example: It was she who decided we should go to Hawaii.
- Indefinite pronouns don't have antecedents. They are capable of standing on their own. For example: No one likes the sound of fingernails on a chalkboard.

- Object pronouns are used as direct objects, indirect objects, and objects of prepositions. These include: you, me, him, her, us, them, and it. For example: David talked to her about the mistake.
- Possessive pronouns show ownership. They do not need apostrophes. For example: The cat washed **ITS** whiskers.

EXAMPLES OF PRONOUNS

In the following examples, the pronouns are italicized.

1. **WE** are going on vacation.
2. Don't tell **ME** that **YOU** can't go with **US**.
3. **ANYBODY** who says **IT** won't be fun has no clue **WHAT THEY** are talking about.
4. **THESE** are terribly steep stairs.
5. **WE** ran into **EACH OTHER** at the mall.
6. I'm not sure **WHICH** is worse: rain or snow.
7. **IT** is one of the nicest Italian restaurants in town.
8. Richard stared at **HIMSELF** in the mirror.
9. The laundry isn't going to do **ITSELF**.
10. **SOMEONE** spilled orange juice all over the countertop!

PRONOUN EXERCISES

The following exercises will help you gain greater understanding about how pronouns work. Choose the best answer to complete each sentence.

1. This is _____ speaking.
   A. John
   B. He
   C. He john
   D. Am
2. Greg is as smart as _____ is.
   A. I
   B. me
   C. she
   D. we
3. The dog chewed on _____ favorite toy.
   A. it's
   B. it is
   C. its'
   D. its
4. It could have been _____ .
   A. Jerry
   B. anyone
   C. better
   D. more difficult
5. Terry is taller than _____ am.
   A. I
   B. me
   C. she

D. we

Answers

1. B. This is he speaking.
2. C. Greg is as smart as she is.
3. D. The dog chewed on its favorite toy.
4. B. It could have been anyone.
5. A. Terry is taller than I am.

List of Pronouns

As you read through this list of pronouns, remember that each one of these pronouns is a word that can be used to take the place of a noun. Think about ways to use the pronouns on this list in sentences, as this will increase your understanding.

I, We, Me, Us, You, She, He, Her, Him, They, Them, It, That, Which, Who, Whom, Whose, Whichever, Whoever, Whomever, This, These, That, Those, Anybody, Anyone, Anything, Each, Either, Everyone, Everybody, Everything, Nobody, Neither, No one, Nothing, Somebody, One, Someone, Something, Few, Many, Both, Several, Any, All, Some, Most, None, Myself, Yourself, Ourselves, Yourselves, Herself, Himself, Themselves, Itself, Who, What, Which, Whose, Whom

6. ONE WORD SUBSTITUTION WITH MEANING

1. Audience – a number of people listening to a lecture
2. Altruist – one, who considers the happiness and well-being of others first
3. Atheist – a person who does not believe in God
4. Anthropologist – one, who studies the evolution of mankind
5. Autocracy – government by one person
6. Autobiography – the life history of a person written by himself
7. Amputate – to cut off a part of a person's body which is infected
8. Arsenal – a place for ammunition and weapons
9. Archives – a place where government or public records are kept
10. Amateur – a man who does a thing for pleasure and not as a profession
11. Aristocracy – government by the nobles
12. Aquatic – animals/plants ere which live in water
13. Amphibian – animals which live both on land and sea
14. Ambidexter – one, who can use either hand with ease
15. Alimony – allowance paid to wife on legal separation
16. Anthology – a collection of poems
17. Abdication – voluntary giving up of throne in favour of someone
18. Arbitrator – a person, appointed by two parties to solve a dispute
19. Astronomer – a person, who studies stars, planets and other heavenly bodies
20. Astrologer – a person who studies the influence of heavenly bodies on human beings
21. Anthology – a collection of poems
22. Axiom – a statement which is accepted as true without proof
23. Agenda – a list of headings of the business to be transacted at a meeting
24. Anarchist – one, who is out to destroy all governance, law and order
25. Almanac – an annual calender with positions of stars
26. Bigamy – the practice of having two wives or husbands at a time

27. Bibliophile – a lover and collector of books
28. Bouquet – a collection of flowers
29. Bureaucracy – government by the officials
30. Belligerent – a person, nation that is involved in war
31. Biennial – an event which happens once in two years
32. Blasphemy – the act of speaking disrespectfully about sacred things
33. Creche – a nursery where children are cared for while their parents are at work
34. Cosmopolitan – a person who regards whole world as his country
35. Chauffeur – one, who is employed to drive a motor car
36. Curator – a person incharge of a museum
37. Carnivorous – one, who lives on flesh
38. Cannibal – one, who feeds on human flesh
39. Contemporaries – belonging to or living at the same time
40. Cloak room – a place for luggage at railway station
41. Cynosure – centre of attraction
42. Connoisseur – a critical judge of any art and craft
43. Crusade – a religious war
44. Choreographer – one, who teaches dancing
45. Cacographist – a person, who is bad in spellings
46. Calligraphist – a person, who writes beautiful handwriting
47. Cynic – one, who sneers at the aims and beliefs of his fellow men
48. Convalescent – one, who is recovering health
49. Cavalry – soldiers, who fight on horse back
50. Cardiologist – a person, who is specialist in heart diseases

7. ADVERBS

Find the adjective in the first sentence and fill the gap with the adverb.

1. Joanne is happy. She smiles _____.
2. The boy is loud. He shouts _____.
3. Her English is fluent. She speaks English_____.
4. Our mum was angry. She spoke to us_____.
5. My neighbor is a careless driver. He drives _____.
6. The painter is awful. He paints _____.
7. Jim is a wonderful piano player. He plays the piano_____.
8. This girl is very quiet. She often sneaks out of the house_____.
9. She is a good dancer. She dances really_____.
10. This exercise is simple. You_____ have to put one word in each space.
11. Please don't walk too _____.
12. I had to open the door_____ to get the package inside.
13. In his bathrobe, John stepped _____to _____ retrieve the morning paper.
14. He is _____ tired from the journey.
15. I walked_____.

Answer Key:-

2. happily	6. awfully	11. fast
3. loudly	7 wonderfully	12. wide

4. fluently	8. quietly	13. outside, quickly
5. angrily	9. well	14. completely
6. carelessly	10. simply	15.downstairs

Rewrite the complete sentence using the adverbs.

1. They go to the movies.(often)
2. She listens to classical music.(rarely)
3. He reads the newspaper.(sometimes)
4. She complains about her husband.(always)
5. I drink coffee.(sometimes)
6. Frank is ill.(often)
7. He feels terrible(usually)
8. I go jogging in the morning.(always)
9. She helps her daughter with her homework.(never)
10. we watch television in the evening.(always)
11. I smoke.(never)
12. I eat meat.(seldom)
13. I eat vegetables and frits.(always)
14. She listens to classical music.(rarely)
15. Sara smiles.(never)

Answer Key:-

1. They often go to the movies.
2. She rarely listens to classical music.
3. He sometimes reads the newspaper.
4. She always complains about her husband.
5. I sometimes drink coffee.
6. Frank often is ill.
7. He usually feels terrible.
8. I always go jogging in the morning.
9. She never helps her daughter with her homework.
10. We always watch television in the evening.
11. I never smoke.
12. I seldom eat meat.
13. I always eat vegetables and frits.
14. She rarely listens to classical music.
15. Sara never smiles.

UNIT- IV

1. COMPARE & CONTRAST PARAGRAPH WRITING

The Comparison paragraph compares two subjects and discusses how they are alike and lists a few examples. In the contrast paragraph, two subjects are discussed as how they are different, again, listing a few examples.

Transitional expressions used in comparison:

- in the same way
- and, also, in addition
- as well as
- both, neither
- each of

- just as...so
- similarly
- like
- too
- the same

Transitional expressions used in contrast:

- although
- whereas
- but
- however
- conversely
- on the other hand
- in contrast
- while
- yet
- unlike

Here are some examples of a compare-contrast paragraph:

Acoustic and Electric Guitars

As broad as their sounds are, there are several very distinct similarities and differences between electric and acoustic guitars. For instance, both utilize the use of a body for the neck to attach to and a neck with frets for finger placement. The strings attach to the lower end of the body and go all the way to the head, or the top of the neck. They both use strings that vary in gauge, or size, which are vital to produce sound when they are picked, hammered on, or strummed as a group. Similarly, each is tuned in the same manner to produce the proper tone desired. An acoustic guitar needs no amplifier to make its sound loud enough to be heard. An acoustic guitar uses the body of the guitar as its amplifier. Because the body is very thick and hollow it is able to project its own natural sound loudly. This makes it very portable and capable of being played virtually anywhere. An acoustic guitar doesn't need any foot pedals, volume and tone knobs, or any other hardware like that to produce the sound it makes. An electric guitar is very hard to hear without an amplifier. An electric guitar requires the use of an amplifier to transport the sound though pickups that are secured in the body. These sounds are transferred through a cable connected to the guitar. The cable then goes to the amplifier which produces the sounds out of the speakers. Volume and tone knobs on the electric guitar can make it louder or change the sound of the strings being played. Additionally, foot pedals can be added to produce even more different sounds so that the musical capabilities of the electric guitar are almost limitless. There's not a lot of music that I listen to that doesn't have some sort of electric or acoustic guitar in the mix, either as the main instrument or as small as a fill in for a certain sound. The genre of the music frequently dictates which type of guitar should be used.

2. WRITING PERSONAL LETTERS

Typical Purposes for Personal Letters

- to inform
- to keep in touch
- to share
- to persuade

Typical Formats for Personal Letters

- casual, conversational wording
- correct spelling
- casual use of punctuation, capitalization, and grammar
- personal and interesting details
- handwritten or typed format
- traditional or electronic distribution
- indented paragraphs

Typical Audience Expectations for Personal Letters

- typing or handwriting is easy to read
- messages is easy to understand
- references are familiar

Typical Components of Personal Letters

- date
- salutation
- introduction
- body
- conclusion
- closing
- signature

1. Write a personal letter within a text box in a word processing program. Then label the components of the letter.
2. Write a one-page personal letter. Exchange letters with a partner, and use the criteria in this section to evaluate your partner's personal letter.
3. How would you commemorate the following events in the life of a close friend who lives on the other side of the country? Would you send that friend a personal letter, an e-mail, or a text message or post a message on the wall of her social networking page? Discuss the implications of your choices of medium.
   a. the birth of her first child
   b. her wedding
   c. the death of her grandmother
   d. her big promotion at work
   e. her graduation from college

4. TENSES

tense	Affirmative/Negative/Question	Use	Signal Words
Simple	**A:** He speaks.	action in the present taking	always, every …,

Present	**N:** He does not speak. **Q:** Does he speak?	place **regularly, never or several times** facts actions taking place one after another action set by a timetable or schedule	never, normally, often, seldom, sometimes, usually if sentences type I (*If I **talk**, ...*)
Present Progressive	**A:** He is speaking. **N:** He is not speaking. **Q:** Is he speaking?	**action taking place in the moment of speaking** action taking place only for a limited period of time action arranged for the future	at the moment, just, just now, Listen!, Look!, now, right now
Simple Past	**A:** He spoke. **N:** He did not speak. **Q:** Did he speak?	action in the past taking place **once, never or several times** actions taking place one after another action taking place in the middle of another action	yesterday, 2 minutes ago, in 1990, the other day, last Friday if sentence type II (*If I **talked**, ...*)
Past Progressive	**A:** He was speaking. **N:** He was not speaking. **Q:** Was he speaking?	action **going on** at a certain time in the past actions taking place at the same time action in the past that is interrupted by another action	while, as long as
Present Perfect Simple	**A:** He has spoken. **N:** He has not spoken. **Q:** Has he spoken?	putting emphasis on the **result** action that is still going on action that stopped recently finished action that has an influence on the present action that has taken place once, never or several times before the moment of speaking	already, ever, just, never, not yet, so far, till now, up to now
Present Perfect Progressive	**A:** He has been speaking. **N:** He has not been speaking. **Q:** Has he been speaking?	putting emphasis on the **course or duration** (not the result) action that recently stopped or is still going on finished action that influenced the present	all day, for 4 years, since 1993, how long?, the whole week
Past Perfect Simple	**A:** He had spoken. **N:** He had not spoken. **Q:** Had he spoken?	action taking place before a certain time in the past sometimes interchangeable	already, just, never, not yet, once, until that

		with past perfect progressive putting emphasis only on the **fact** (not the duration)	day if sentence type III (*If I **had talked**, …*)
Past Perfect Progressive	**A:** He had been speaking. **N:** He had not been speaking. **Q:** Had he been speaking?	action taking place before a certain time in the past sometimes interchangeable with past perfect simple putting emphasis on the **duration or course** of an action	for, since, the whole day, all day
Future I Simple	**A:** He will speak. **N:** He will not speak. **Q:** Will he speak?	action in the future that cannot be influenced **spontaneous** decision assumption with regard to the future	in a year, next …, tomorrow If-Satz Typ I (*If you ask her, she **will help** you.*) *assumption*: I think, probably, perhaps
Future I Simple (going to)	**A:** He is going to speak. **N:** He is not going to speak. **Q:** Is he going to speak?	**decision** made for the future conclusion with regard to the future	in one year, next week, tomorrow
Future I Progressive	**A:** He will be speaking. **N:** He will not be speaking. **Q:** Will he be speaking?	action that is **going on**at a certain time in the future action that is sure to happen in the near future	in one year, next week, tomorrow
Future II Simple	**A:** He will have spoken. **N:** He will not have spoken. **Q:** Will he have spoken?	action that will be **finished** at a certain time in the future	by Monday, in a week
Future II Progressive	**A:** He will have been speaking. **N:** He will not have been speaking. **Q:** Will he have been speaking?	action taking place before a certain time in the future putting emphasis on the **course** of an action	for …, the last couple of hours, all day long
Conditional I Simple	**A:** He would speak. **N:** He would not speak. **Q:** Would he speak?	action that **might** take place	if sentences type II (*If I were you, I **would go** home.*)
Conditional I Progressive	**A:** He would be speaking. **N:** He would not be speaking. **Q:** Would he be speaking?	action that might take place putting emphasis on the **course** / **duration**of the	

		action	
Conditional II Simple	**A:** He would have spoken. **N:** He would not have spoken. **Q:** Would he have spoken?	action that **might**have taken place in the past	if sentences type III (*If I had seen that, I **would have helped***.)
Conditional II Progressive	**A:** He would have been speaking. **N:** He would not have been speaking. **Q:** Would he have been speaking?	action that might have taken place in the past puts emphasis on the **course** / **duration** of the action	

1. Fill up the blanks with correct form of verbs:

1. We _____ (watch) TV when it started to rain.
2. I _____ (want) to visit you yesterday, but you were not at home.
3. Look! It _____ (rain), so we can't _____ (go)to the beach.
4. There are a lot of clouds! It_____ (go) to rain soon.
5. The sun _____ (rise) in the East.
6. Since 2011 they _____ (visit) their son every year.
7. While the doctor _____ (examine) Mr Jones, his son was waiting outside this morning.
8. I _____ (wait) have been waiting for my girlfriend for two hours.
9. After Larry _____ (see) the film on TV, he decided to buy the book.
10. Wait a minute, I _____ (carry) this box for you.

2. Choose the correct verb tense to complete each of the following sentences:

11. When I arrived, Anthony _____ with his Nintendo.
    a) was playing b) has been playing c) had been played
12. The train wasn't there. It _____ already
    a) had been left b) had left c) has left
13. Tara is exhausted. She _____.
    a) has been running b) had been running c) has running
14. How long _____ each other when they got married?
    a) had they known b) have they known c) would they know
15. I didn't have much to eat at lunch because _____ a lot at breakfast.
    a) I would have eaten b) I'd eaten c) I was eating
16. I _____ long when the alarm went off.
    a) had been sleeping b) hadn't been sleeping c) will have been sleeping
17. She _____ to speak English before she arrived in America.
    a) had learned already b) will have already learned c) had already learned
18. Ram _____ good French because he had been studying for several years.
    a) spoke b) had spoken c) has spoken

19. Shakespeare would be famous even if he _____ "To be or not to be..."
a) had not written b) had not been writing c) has not written
20. I _____ from him for ten years when I received his last letter.
a) didn't hear b) hadn't been hearing c) hadn't heard
21. I _____ for seven hours last night.
a) was slept b) slept c) has slept
22. I _____ that you were here.
a) wasn't knowing b) didn't knew c) didn't know
23. When I came to this city, I _____ anyone.
a) haven't known b) didn't know c) wasn't knowing
24. You know where I live - I _____ in the same place for the past three years.
a) have been living b) had lived c) was living
25. You _____ my brother, have you?
a) haven't seen b) didn't see c) didn't saw

Answers:

1.were watching 2.wanted 3.is raining, go 4.is going 5.rises 6.have visited 7.was examining 8. have been waiting 9. had seen 10. will carry 11.was playing 12.had left 13. has been running 14.had they known 15.I'd eaten 16. hadn't been sleeping 17. had already learned 18.Spoke 19. had not written 20. hadn't heard 21.slept 22.didn't know 23.didn't know 24. have been living 25. haven't seen

5. DIFFERENT FORMS OF WORDS

Noun	Adjective	Verb
1. Comparison	Comparable	Compare
2. Defending	Defensive	Defend
3. Prediction	Predictable	Predict
4. Mystery	Mysterious	Mystify
5. Reaction	Reactive	React
6. Correction	Corrective	Correct
7. Familiarity	Familiar	Familiarize
8. Life	Livable	Live
9. Movement	Movable	Move
10. Disposal	Disposable	Dispose
11. Expense	Expensive	Expend
12. Extraction	Extractable	Extract
13. Imagination	Imaginative	Imagine

14. Manufacture	Manufacturing	Manufacture
15. Profession	Professional	Profess
16. Invention	Inventive	Invent
17. Irrigation	Irrigative	Irrigate
18. Generation	Generable	Generate
19. Education	Educatable	Educate
20. Advise	Advice	Advisory
21. Contribute	Contribution	Contributory
22. Classify	Classification	Classifiable
23. Impress	Impression	Impressive
24Develop	Development	Developable
25. Govern	Government	Governing
26. Decide	Decision	Decisive
27. Magnetize	Magnetism	Magnetic
28. Electrify	Electrification	Electric
29. Introduce	Introduction	Introductory
30. Grow	Growth	Growing.

6. PHRASAL VERBS

phrasal verb	**meaning**	**example sentence**
ask*somebody***out**	invite on a date	Brian **asked** Judy **out** to dinner and a movie.
ask around	ask many people the same question	I **asked around** but nobody has seen my wallet.
add up to*something*	equal	Your purchases **add up to** $205.32.
back*something***up**	reverse	You'll have to **back up** your car so that I can get out.
back*somebody***up**	support	My wife **backed** me **up** over my decision to quit my job.
blow up	explode	The racing car **blew up** after it crashed into the fence.

phrasal verb	meaning	example sentence
blow*something***up**	add air	We have to **blow** 50 balloons **up**for the party.
break down	stop functioning (vehicle, machine)	Our car **broke down** at the side of the highway in the snowstorm.
break down	get upset	The woman **broke down** when the police told her that her son had died.
break*something***down**	divide into smaller parts	Our teacher **broke** the final project **down** into three separate parts.
break in	force entry to a building	Somebody **broke in** last night and stole our stereo.
break into*something*	enter forcibly	The firemen had to **break into** the room to rescue the children.
break*something***in**	wear something a few times so that it doesn't look/feel new	I need to **break** these shoes **in**before we run next week.
break in	interrupt	The TV station **broke in** to report the news of the president's death.
catch up	get to the same point as somebody else	You'll have to run faster than that if you want to **catch up** with Marty.
check in	arrive and register at a hotel or airport	We will get the hotel keys when we **check in**.

UNIT – V

1. WRITING ESSAYS

1.Write an essay about Cyber crime
2.Draft an essay on Science.
3.Write an essay on “the impacts of atomic radiation”.
4.Sketch on an essay on “The Problems and Benefits of Technology ”
5.Sort out “the impacts of atomic radiation” in an essay of 250 words.
6.Write an essay stating the role of electronic media in imparting education
7.Draft an essay on the natural resources of Energy.
8.Describe on the good and evil effects of Friendship in an essay of 250 Words.

9.Sort out the evil effects of Television in an essay of 250 Words.
10. The Advantage and Disadvantages of the Internet- describe in an essay of 250 Word.
11. Technology and science.
12. Engineers in Nation Building
13. Unemployment in India.
14. India in 2020.
15. Nuclear power.
16. Alternate Sources of energy.
17. Are computer better than human Brain.
18. Global warming.
19. English in Today's world.
20. Rainwater Harvesting.
21. Write an Essay in about 300 words describing Internet and its uses.
22. Write an Essay in about 300 words on explaining the wisdom of investing money in articles of gold.
23. Write an Essay in about 300 words describing Computer and its uses.
24. Describe on the Advantages of using Solar and Wind Power as Alternative sources of Energy in India.
25. Write an Essay in about 300 words explaining water as invaluable natural resources', which has to be spent carefully.
26. Write an Essay in about 300 words explaining the importance of indoor games and its types.
27. Write an Essay in about 300 words comparing Newspaper and Television as a media of mass communication.
28. Write an Essay in about 300 words on Spoken English and its uses.
29. Write an Essay about India in the development of science and Technology.
30. Write an essay about jobs and unemployment in India.
31. Attempt an essay on Power conservation.
32. Attempt an essay on Business English.
33. Write an essay about hygiene
34. Write an essay about good health
35. Write an essay about the merits and demerits of working abroad.
36. Describe your favorite vacation spot. Use at least three of the five senses (taste, touch, sound, sight, smell) in your description.
37. Write a story based on this line : "By evening, she was running a high fever ... "
38. School life is the most difficult time but it also keeps many joys for us. Do you agree?
39. Examination day often makes many pupils worry. Do you agree?
40. "Money is the root of all evil". Give your opinion.
41. How should people protect and preserve the natural resources?
42. Describe a pet which you have kept.
43. Consider yourself as a Car and write an autobiography for you.
44. Talk about rainy season.
45. "The luxuries of today are the necessities of tomorrow". Discuss.

2. MODAL VERBS

What are modal verbs?

Modals (also called **modal verbs, modal auxiliary verb**s, **modal auxiliaries**) are special verbs which behave irregularly in English. They are different from normal verbs like "work, play, visit..." They give additional information about the ***function*** of the main verb that follows it. They have a great variety of ***communicative functions***.

Here are some characteristics of modal verbs:

- They never change their form. You can't add "s", "ed", "ing"...
- They are always followed by an infinitive without "to" (e.i. the bare infinitive.)
- They are used to indicate modality allow speakers to express certainty, possibility, willingness, obligation, necessity, ability

List of modal verbs

Here is a list of modal verbs:

can, could, may, might, will, would, shall, should, must

The verbs or expressions DARE, OUGHT TO, HAD BETTER, and NEED NOT behave like modal auxiliaries to a large extent and my be added to the above list

Use of modal verbs:

Modal verbs are used to express functions such as:

1. Permission
2. Ability
3. Obligation
4. Prohibition
5. Lack of necessity
6. Advice
7. possibility
8. probability

Examples of modal verbs

Here is a list of modals with examples:

Modal Verb	Expressing	Example
must	Strong obligation	You must stop when the traffic lights turn red.
	logical conclusion / Certainty	He must be very tired. He's been working all day long.
must not	prohibition	You must not smoke in the hospital.
can	ability	I can swim.
	permission	Can I use your phone please?
	possibility	Smoking can cause cancer.
could	ability in the past	When I was younger I could run fast.
	polite permission	Excuse me, could I just say something?
	possibility	It could rain tomorrow!
may	permission	May I use your phone please?
	possibility, probability	It may rain tomorrow!
might	polite permission	Might I suggest an idea?
	possibility, probability	I might go on holiday to Australia next year.
need not	lack of necessity/absence of obligation	I need not buy tomatoes. There are plenty of tomatoes in the fridge.
should/ought to	50 % obligation	I should / ought to see a doctor. I have a terrible headache.
	advice	You should / ought to revise your lessons
	logical conclusion	He should / ought to be very tired. He's been working all day long.
had better	advice	You 'd better revise your lessons

Remember
Modal verbs are followed by an infinitive without "to", also called the bare infinitive.

*Examples**:***

- You **must <u>stop</u>** when the traffic lights turn red.
- You **should <u>see</u>** to the doctor.
- There are a lot of tomatoes in the fridge. You **need not <u>buy</u>** any.

<u>3. THE MOST COMMON ENGLISH IDIOMS</u>

These English idioms are extremely common in everyday conversation in the United States. You will hear them in movies and TV shows and can use them to make your English sound more like that of a native speaker.

Idiom	Meaning	Usage
A blessing in disguise	a good thing that seemed bad at first	as part of a sentence
A dime a dozen	Something common	as part of a sentence
Beat around the bush	Avoid saying what you mean, usually because it is uncomfortable	as part of a sentence
Better late than never	Better to arrive late than not to come at all	by itself
Bite the bullet	To get something over with because it is inevitable	as part of a sentence
Break a leg	Good luck	by itself
Call it a day	Stop working on something	as part of a sentence
Cut somebody some slack	Don't be so critical	as part of a sentence
Cutting corners	Doing something poorly in order to save time or money	as part of a sentence
Easy does it	Slow down	by itself
Get out of hand	Get out of control	as part of a sentence
Get something out of your system	Do the thing you've been wanting to do so you can move on	as part of a sentence
Get your act together	Work better or leave	by itself
Give someone the benefit of the doubt	Trust what someone says	as part of a sentence
Go back to the drawing board	Start over	as part of a sentence
Hang in there	Don't give up	by itself
Hit the sack	Go to sleep	as part of a sentence

Idiom	Meaning	Usage
It's not rocket science	It's not complicated	by itself
Let someone off the hook	To not hold someone responsible for something	as part of a sentence
Make a long story short	Tell something briefly	as part of a sentence
Miss the boat	It's too late	as part of a sentence
No pain, no gain	You have to work for what you want	by itself
On the ball	Doing a good job	as part of a sentence
Pull someone's leg	To joke with someone	as part of a sentence
Pull yourself together	Calm down	by itself
So far so good	Things are going well so far	by itself
Speak of the devil	The person we were just talking about showed up!	by itself
That's the last straw	My patience has run out	by itself
The best of both worlds	An ideal situation	as part of a sentence
Time flies when you're having fun	You don't notice how long something lasts when it's fun	by itself
To get bent out of shape	To get upset	as part of a sentence
To make matters worse	Make a problem worse	as part of a sentence
Under the weather	Sick	as part of a sentence
We'll cross that bridge when we come to it	Let's not talk about that problem right now	by itself
Wrap your head around something	Understand something complicated	as part of a

Idiom	Meaning	Usage
		sentence
You can say that again	That's true, I agree	by itself
Your guess is as good as mine	I have no idea	by itself

COMMON ENGLISH IDIOMS & EXPRESSIONS

These English idioms are used quite regularly in the United States. You may not hear them every day, but they will be very familiar to any native English speaker. You can be confident using any of them when the context is appropriate.

Idiom	Meaning	Usage
A bird in the hand is worth two in the bush	What you have is worth more than what you might have later	by itself
A penny for your thoughts	Tell me what you're thinking	by itself
A penny saved is a penny earned	Money you save today you can spend later	by itself
A perfect storm	the worst possible situation	as part of a sentence
A picture is worth 1000 words	Better to show than tell	by itself
Actions speak louder than words	Believe what people do and not what they say	by itself
Add insult to injury	To make a bad situation worse	as part of a sentence
Barking up the wrong tree	To be mistaken, to be looking for solutions in the wrong place	as part of a sentence
Birds of a feather flock together	People who are alike are often friends (usually used negatively)	by itself
Bite off more than you can chew	Take on a project that you cannot finish	as part of a sentence
Break the ice	Make people feel more comfortable	as part of a sentence
By the skin of your teeth	Just barely	as part of a sentence
Comparing apples to oranges	Comparing two things that cannot be compared	as part of a

Idiom	Meaning	Usage
		sentence
Costs an arm and a leg	Very expensive	as part of a sentence
Do something at the drop of a hat	Do something without having planned beforehand	as part of a sentence
Do unto others as you would have them do unto you	Treat people fairly. Also known as "The Golden Rule"	by itself
Don't count your chickens before they hatch	Don't count on something good happening until it's happened.	by itself
Don't cry over spilt milk	There's no reason to complain about something that can't be fixed	by itself
Don't give up your day job	You're not very good at this	by itself
Don't put all your eggs in one basket	What you're doing is too risky	by itself
Every cloud has a silver lining	Good things come after bad things	by itself
Get a taste of your own medicine	Get treated the way you've been treating others (negative)	as part of a sentence
Give someone the cold shoulder	Ignore someone	as part of a sentence
Go on a wild goose chase	To do something pointless	as part of a sentence
Good things come to those who wait	Be patient	by itself
He has bigger fish to fry	He has bigger things to take care of than what we are talking about now	by itself
He's a chip off the old block	The son is like the father	by itself
Hit the nail on the head	Get something exactly right	by itself
Ignorance is bliss	You're better off not knowing	by itself
It ain't over till the fat lady sings	This isn't over yet	by itself

Idiom	Meaning	Usage
It takes one to know one	You're just as bad as I am	by itself
It's a piece of cake	It's easy	by itself
It's raining cats and dogs	It's raining hard	by itself
Kill two birds with one stone	Get two things done with a single action	by itself
Let the cat out of the bag	Give away a secret	as part of a sentence
Live and learn	I made a mistake	by itself
Look before you leap	Take only calculated risks	by itself
On thin ice	On probation. If you make another mistake, there will be trouble.	as part of a sentence
Once in a blue moon	Rarely	as part of a sentence
Play devil's advocate	To argue the opposite, just for the sake of argument	as part of a sentence
Put something on ice	Put a project on hold	as part of a sentence
Rain on someone's parade	To spoil something	as part of a sentence
Saving for a rainy day	Saving money for later	as part of a sentence
Slow and steady wins the race	Reliability is more important than speed	by itself
Spill the beans	Give away a secret	as part of a sentence
Take a rain check	Postpone a plan	as part of a sentence
Take it with a grain of salt	Don’t take it too seriously	as part of a sentence
The ball is in your court	It's your decision	by itself
The best thing since sliced bread	A really good invention	as part of a

Idiom	Meaning	Usage
		sentence
The devil is in the details	It looks good from a distance, but when you look closer, there are problems	by itself
The early bird gets the worm	The first people who arrive will get the best stuff	by itself
The elephant in the room	The big issue, the problem people are avoiding	as part of a sentence
The whole nine yards	Everything, all the way.	as part of a sentence
There are other fish in the sea	It's ok to miss this opportunity. Others will arise.	by itself
There's a method to his madness	He seems crazy but actually he's clever	by itself
There's no such thing as a free lunch	Nothing is entirely free	by itself
Throw caution to the wind	Take a risk	as part of a sentence
You can't have your cake and eat it too	You can't have everything	by itself
You can't judge a book by its cover	This person or thing may look bad, but it's good inside	by itself

V.S.B. ENGINEERING COLLEGE, KARUR
DEPARTMENT OF MATHEMATICS

Matrices and Calculus

UNIT -1
DIFFERENTIAL CALCULUS
PART – A

1.Find the domain of a function y = $\sqrt{x+4}$

Solution: The square root of negative number is not defined, the domain of f consistsof all values x such that x+4 ≥ 0 . ie x ≥ -4

The domain is the interval [-4,∞)

2.Find the domain of the range of the function defined by the co ordinates {(-4,1),(-2,2.5),(2,-1),(3,-2)}

Solution : The domain is simply the x values given X = {-4,-2,2,3}

The range consists of f(x) values f(x) = {-1,1,2.5,2}

Therefore f(x) is discontinuous function

3. Find $\lim_{x\to o} \frac{1-\cos x}{x}$

Solution: Let $\lim_{x\to o} \frac{1-\cos x}{x} = \lim_{x\to o} \frac{2\sin^2\frac{x}{2}}{2}$

$$= \lim_{x\to o}\left\{\frac{\sin\frac{x}{2}}{\frac{x}{2}}\right\} . \lim_{x\to o}\frac{x}{2} = 1.0 = 0$$

4. Find $\lim_{x\to o} \frac{e^x-1}{x}$

Solution: put $e^x - 1 = y$, therefore $y+1 = e^x$

$x = \log_e(y+1)$ as $x\to 0$ $y\to 0$

Therefore $\lim_{x\to o} \frac{e^x-1}{x}$ =

$\lim_{y\to o} \frac{y}{\log_e(y+1)}$ $= \lim_{y\to o} \frac{1}{\log_e(y+1)^{\frac{1}{y}}}$

$= \frac{1}{\log_e e} = 1$

5. Evaluate the following $\lim_{x\to o} \frac{x^3-8}{x-2}$

Solution: $\lim_{x\to a} \frac{x^n-a^n}{x-a} = n\, a^{n-1}$ for all relational values of n.

Given $\lim_{x\to 2} \frac{x^3-8}{x-2} = \lim_{x\to 2} \frac{x^3-2^3}{x-2} = 3.2^2 = 12$

6. $\lim_{x\to o} \frac{x^4-1}{x-1}$

Solution: $\lim_{x\to a} \frac{x^n-a^n}{x-a} = n\, a^{n-1}$

$\Longrightarrow \lim_{x\to 1} \frac{x^4-1}{x-1} = 4.1 = 4$

7. Find $\lim_{x\to o} \frac{(1+x)^3-1-3x}{x^2}$

Solution: Given $\lim_{x\to o} \frac{(1+x)^3-1-3x}{x^2} = \lim_{x\to o} \frac{1+x^3+3x^2+3x-1-3x}{x^2}$

$= \lim_{x\to o} \frac{3x^2+x^3}{x^2} = \lim_{x\to o} x+3 = 3$

8. Show that $\lim_{x\to o} \frac{(1+x)^n-1}{x} = n$

Solution: put $(1+x)^n - 1 = y \Longrightarrow (1+x)^n = 1+y$

Taking log on both sides we get , n log (1+x)= log (1+y)

$\lim_{x\to o} \frac{(1+x)^n-1}{x} = \lim_{x\to o} \left\{\frac{((1+x)^n-1)\log(1+x)}{\log(1+x).x}\right\}$

$= \lim_{y\to o} \frac{y}{\frac{1}{n}\log(1+y)} \cdot \lim_{x\to o} \log(1+x)^{\frac{1}{x}}$

$= \lim_{y\to o} n.\frac{1}{\log(y+1)^{\frac{1}{y}}} \lim_{x\to o} \log(1+x)^{\frac{1}{x}}$

$= \frac{n}{\log e} \log e = n$

9. Where are each of the following functions discontinuous f (x) = $\frac{x^2-x-2}{x-2}$

Solution: If x=2, then the denominator of f(x)=0

Therefore f(2) is not defined. So f is discontinuous at 2.

10. Find the discontinuous function f(x) = $\begin{cases} \frac{1}{x^2} \text{ if } x \neq 0 \\ 1 \text{ if } x = 0 \end{cases}$

Solution: Here f(0)=1 is defined. But $\lim_{x\to 0} f(x) = \lim_{x\to 0} \frac{1}{x^2} = \infty$ does not exists

So f is discontinuous at 0.

11. $\lim_{x\to -2} \frac{x^3+2x^2-1}{5-3x}$

Solution: Let f(x)= $\lim_{x\to -2} \frac{x^3+2x^2-1}{5-3x}$ is rational and is continuous on its domain.

$\lim_{x\to -2} \frac{x^3+2x^2-1}{5-3x} = \lim_{x\to -2} f(x) = f(-2)$

$= \frac{-2^3+2(-2)^2-1}{5-3(-2)}$

$\lim_{x\to -2} \frac{x^3+2x^2-1}{5-3x} = \frac{-1}{11}$

12. Evaluate the following limits and justify each steps $\lim_{x\to 5} 2x^2 - 3x + 4$

Solution: $\lim_{x\to 5} 2x^2 - 3x + 4 = \lim_{x\to 5} 2x^2 - \lim_{x\to 5} 3x + \lim_{x\to 5} 4$

$= 2(5)^2 - 3(5)+4 = 39$

13. If f and g are continuous functions with f(3) = 5 and $\lim_{x\to 3} (2f(x) - g(x) = 4)$ **find g(3).**

Solution: Given f(3) = 5

$\lim_{x\to 3} [2f(x) - g(x)] = \lim_{x\to 3} 2f(x) - \lim_{x\to 3} g(x)$

$= 2 f(3) - g(3)$

Given $\lim_{x\to 3} 2f(x) - g(x) = 4$

2f(3)- g(3)=4

$2\times 5 - g(3) = 4 \Rightarrow 10 - g(3) = 4 \Rightarrow g(3) = 6$

14. Define : Derivatives

Solution: The derivative of afunction f at a number a , denoted f ´(a) , is

f ´(a) = $\lim_{h \to o} \frac{f(a+h)-f(a)}{h}$ if this limit exists

if x = a+h then h = x – a and h⟶0 iff x⟶ a. therefore an equivalent way of slating the definition of the derivatives, as the finding tangent limit is

f ´(a) = $\lim_{x \to a} \frac{f(x)-f(a)}{x-a}$

15. Find the derivative of the function f(x) = $x^2 - 8x - 9$ at the number a.

Solution: f ´(a) = $\lim_{h \to o} \frac{f(a+h)-f(a)}{h}$

$= \lim_{h \to o} \frac{(a+h)^2-8(a+h)+9-(a^2-8a+9)}{h}$

$= \lim_{h \to o} \frac{2ah+h^2-8h}{h} \Longrightarrow \lim_{h \to o} 2a + h - 8$

= 2a+0-8 = 2a-8

Hence f(x) = $x^2 - 8x - 9$ at a is 2a-8

16.Prove that the function f(x) = $x^3 - 3x^2 + 6$ is positive for all values of x = 2.

Solution: Given f(x)=$x^3 - 3x^2 + 6$

f ´(x)= $3x^2 - 6x$= 3x (x-2)

Therefore if x>2 then f ´(x) is positive

f(x) is an increasing function for x>2 but f(2)=2

(i.e) positive , therefore x>2 f(x) is positive.

17. Find an equation of the tangent line to the parabola y = $x^2 - 8x + 9$ at the point (3,- 6).

Solution : f(x)= $x^2 - 8x + 9$

f ´(x)= 2x-8

At x=a ⟹ f ´(a)=2a-8

Therefore , the slope of the tangent line at (-3,6) is,

f ´(3)= 2(3)-8= -2, thus an equation of the tangent line is ,

y-(-6)= -2x.

18. If f(x) = xe^xfind f ´(x).

Solution: f ´(x)=$\frac{d}{dx}(xe^x) = x\frac{d}{dx}(e^x)+e^x\frac{d}{dx}(x)$

$= xe^x + e^x.1$

$=(x+1).e^x$.

19.Find f´(x) if f(x) = sin(cos (tan x)).

Solution: Let f(x) = sin(cos (tan x)).

f ´(x).= cos(cos (tan x))$\frac{d}{dx}\cos(\tan x)$

= cos (cos (tan x)) [sin (tan x)] $\frac{d}{dx}(\tan x)$

= - cos(cos(tan x)) sin (tan x) $sec^2 x$.

20. Find y´´ if $x^4 + y^4 = 16$.

Solution: differentiating the equation implicity with respect to x, we get , $4x^3 + 4y^3y'$=0

$\Longrightarrow$ y´=$\frac{x^3}{y^3}$

Y ´´= $\frac{d}{dx}(\frac{x^3}{y^3})$,

$= -\frac{y^3\frac{d}{dx}(x^3)-x^3\frac{d}{dx}y^3}{y^{3^2}}$

$=\frac{3x^2y^3-3x^3y^2\left(-\frac{x^3}{y^3}\right)}{y}$

$= -\frac{3x^2(x^4+y^4)}{y^7}$ $= -\frac{48x^2}{y^7}$

21. Find the domain of the function g(x) = $\frac{1}{x^2-x}$

Solution: Given g(x) = $\frac{1}{x^2-x} = \frac{1}{x(x-1)}$

When x = 0 or x=1, thus the domain of g is

{x / x ≠ 0, x ≠ 1}

therefore the interval of the domain is, (- ∞,0) ∪ (0,1) ∪ (1,∞).

22.prove that $\frac{d}{dx}(sinh^{-1}x) = \frac{1}{\sqrt{1+x^2}}$

Solution: Let y= $sinh^{-1}x$

Sinh y=x

Differentiate with respect to y ,cosh y $\frac{dy}{dx} = 1$

Since -sin$h^2y + cosh^2y = 1$ and cosh y ≥ 0.

Cos h y =$\sqrt{1+sinh^2y} \Rightarrow \frac{dy}{dx} = \frac{1}{\cosh y} \Rightarrow \frac{1}{\sqrt{1+sinh^2y}}$

$\Rightarrow \frac{dy}{dx} = \frac{1}{\sqrt{1+x^2}}$

23. Find the maxima and minima of the function $2x^3 - 3x^2 - 36x + 10$.

Solution:

Let f(x)= $2x^3 - 3x^2 - 36x + 10.$

At the maximum or minimum f ´(x)=0

f ´(x)=$6x^2 - 6x - 36$=6(*x*-3)(*x*+*2)*

x=3 and x= -2 give maximum or minimum

f ´´(x)=6(2x-1), when x=3, f ´´(x)= 6(6-1)=30, positive

when x= -2, f ´´(x) = 6(-4-1) = -30, negative

Therefore x= -2 gives the maximum and x=3 gives the minimum.

Maximum value=f(-2)= 54

Minimum value=f(3)= -71.

24. Differentiate f(x)=$\log_{10}(2 + \sin x)$

Solution: $\frac{d}{dx}(\log_a x) = \frac{1}{x\,In\,a}$

Using this formula with a=10, we have

f´(x)=$\frac{d}{dx}\log_{10}(2+\sin x) = \frac{1}{(2+\sin x)\,In\,10}\frac{d}{dx}(2+\sin x)$

=$\frac{\cos x}{(2+\sin x)\,In\,10}$

25. Find the absolute maximum and minimum values of the function

f(x)=x^3-3x^2+1. $\frac{-1}{2} \le x \le 4$

solution: since f is continuous on $[\frac{-1}{2},4]$

f(x)=x^3-3x^2+1 $\Longrightarrow$ f´(x) = 3x^2-6x = 3x(x-2)

f´(x) exists for all x .

when f ´´ (x) = 0 (i.e) x=0 or x=2

f(0)=1 f(2)=-3

the values of f at the end points of the interval are

f($\frac{-1}{2}$) = $\frac{1}{8}$ f(4)=17

maximum value f(4)=17

minimum value f(2)= -3

PART – B

1. Find the domain and range of the function f(x) = $\frac{\sqrt{x+2}}{x^2-9}$ without using a graph
2. Find the domain and range for the function h = 20t – 4.9t^2
3. Find the domain and range of function y = $\frac{1}{x+3}$ - 5
4. Show that the function f(x)=1-$\sqrt{1-x^2}$ is continuous on the interval [-1,1].
5. The position of a partile is given by the equation of motion s = f(t) = $\frac{1}{(1+t)}$, where t is measured in seconds and s in meters. Find the velocity and the speed after 2 seconds.
6. Find f ´(a) , if f(x) =3 – 2x + 4x^3
7. Find f ´(x) if f(x) = $\frac{1-x}{2+x}$
8. Assuming that the petrol burnt in driving a motor varies as the cube of its velocity, show that the most economical speed when going against a current of c miles per hour is $\frac{3}{2}$ c miles per hour.
9. Compute $\lim_{n\to\infty}(\sqrt{x^2}+1-x)$
10. If f(x)=$\sqrt{x-1}$, find the derivative of f. state the domain of f ´.

11. Find an equation of the tangent line to the curve y = $\frac{e^x}{1+x^2}$ at the point (1,$\frac{e}{2}$).

12. Differentiate f(x) = $\frac{\sec x}{1+\tan x}$ for what value of x does the graph of f have a horizontal tangent?

13. .a) If x $(1+y)^{\frac{1}{2}}$ + y $(1+x)^{\frac{1}{2}}$ = 0 prove that $\frac{dy}{dx}$ = - $\frac{1}{(1+x)^2}$

14. If y = $\sqrt{(\ \sin x+\sqrt{\sin x+\sqrt{\sin x}}\ \ \ldots.to\ infinity\ )}$ find $\frac{dy}{dx}$.

15. Find the nth differential co efficient of $\cos^5\theta \sin^7\theta$

16. The greatest value of ax+by where a and b are positive and $x^2+xy+y^2=3k^2$ is 2k$\sqrt{a^2-ab+b^2}$

17. Find the maxima and minima of the function $2x^3-3x^2-36x+10.$

18. Find the maximum value of $\frac{\log x}{x}$ for positive values

UNIT – 2
FUNCTIONS OF SEVERAL VARIABLES
PART -A

1.If $u=\frac{y}{z}+\frac{z}{x}$, **find the value of** $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+z\frac{\partial u}{\partial z}$.

Solution: Given $u=\frac{y}{z}+\frac{z}{x}$,

$\frac{\partial u}{\partial x}=\frac{-z}{x^2}$, $\frac{\partial u}{\partial y}=\frac{1}{z}$, $\frac{\partial u}{\partial z}=\frac{-y^2}{z^2}+\frac{1}{x}$.$=> x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+z\frac{\partial u}{\partial z}=-\frac{z}{x}+\frac{y}{z}-\frac{y}{z}+\frac{z}{x}=0$.

2.If $x+y+z=u, y+z=uv, z=uvw$ **find** $\frac{\partial(x,y,z)}{\partial(u,v,w)}$

Solution: Given

$u = x + y + z,$

$x = x - (y + z),$

$= uv - z$

$y = uv - uvw$

Now, $J = \begin{vmatrix} \frac{\partial x}{\partial} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix} = \begin{vmatrix} 1-v & -u & 0 \\ v-vw & u-uw & -uv \\ vw & uw & uv \end{vmatrix} = \begin{vmatrix} 1-v & -u & 0 \\ vw & uw & uv \\ v & u & o \end{vmatrix} R_3 \to R_3 + R_2 = u^2 v$

3. Find the Jacobian of the transformation $u = \frac{2x-y}{2}, v = \frac{y}{2}$

Solution:

$\frac{\partial(u,v)}{\partial(x,y)} = J = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$ Now, $\frac{\partial u}{\partial x} = 1, \frac{\partial u}{\partial y} = \frac{-1}{2}, \frac{\partial v}{\partial y} = 0, \frac{\partial v}{\partial y} = \frac{1}{2} = \begin{vmatrix} 1 & -\frac{1}{2} \\ 0 & \frac{1}{2} \end{vmatrix} = \frac{1}{2}$

4.Find $\frac{du}{dt}$ if u= $\frac{x}{y}$ where x=e^t, y=log t

Solution:

$\frac{du}{dt} = \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial y}\frac{dy}{dt} \to (1)$

now $\frac{\partial u}{\partial x} = \frac{1}{y}; \frac{\partial u}{\partial y} = -\frac{x}{y^2}; \frac{dx}{dt} = e^t; \frac{dy}{dt} = \frac{1}{t} \to (2)$

Substituting equ (1) in (2) we get, $\frac{du}{dt} = \frac{1}{y}e^t.e^t - \frac{x}{y^2}.\frac{1}{t}$

5.If u= $\frac{x^2+y^2}{\sqrt{x+y}}$ **Prove that** $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{3}{2}u$

Solution: Given that u(x,y)= $\frac{x^2+y^2}{\sqrt{x+y}}$

u(tx,ty) = $\frac{(tx)^2+(ty)^2}{\sqrt{tx+ty}} = \frac{t^2(x^2+y^2)}{t^{1/2}\sqrt{x+y}} = \frac{t^{3/2}(x^2+y^2)}{\sqrt{x+y}} = r^{3/2}u$

u is homogenous of degree 3/2. $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{3}{2}u$ ∵ By Euler theorem.

6.Finddy/dx when $x^3+y^3 = 3axy$.

Solution: let f(x,y)= $x^3+y^3 - 3axy$

$$\frac{dy}{dx} = -\frac{\left(\frac{\partial f}{\partial x}\right)}{\left(\frac{\partial f}{\partial y}\right)} = \frac{3x^2 - 3ay}{3y^2 - 3ax} = -\frac{x^2 - ay}{y^2 - ax}$$

7. If u= $x^2+y^2+z^2$ and x= e^t ,y=e^t sin t, z= e^tcos t, find du/dt without actual substitution.

Solution: $\frac{du}{dt} = \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial y}\frac{dy}{dt} + \frac{\partial u}{\partial z}\frac{dz}{dt}$

= 2x e^t+2y(e^t sint +cos t)+2z(e^tcos t- e^tsin t)=2 e^t [e^t+ $e^t$$\sin^2 t$ + e^tsintcos t+ e^t $\cos^2$ t- e^tsintcos t]

=2 e^t [e^t + e^t ($\sin^2$ t+$\cos^2$ t)] =(2 e^t) (2 e^t)=4 e^{2t}

8. If $(\cos x)^y = (\sin y)^x$ find dy/dx.

Solution: Taking log on both sides we get, f(x,y)= y log cos x-x log sin y=0

f_x = -y tan x-log sin y, f_y= logcos x-x cot y $\therefore \frac{dy}{dx} = -\frac{f_x}{f_y} = \frac{y\tan x + \log\sin y}{\log\cos x - x\cot y}$

9.Finddy/dx given $ax^2+2hxy+by^2+2gx+2fy+c=0$

Solution: Let $\phi(x, y) = ax^2+2hxy+by^2+2gx+2fy+c=0$

Then $\frac{dy}{dx} = -\frac{\left(\frac{\partial \phi}{\partial x}\right)}{\left(\frac{\partial \phi}{\partial y}\right)} = -\frac{2(ax + hy + g)}{2(hx + by + f)} = -\frac{(ax + hy + g)}{(hx + by + f)}$

10.Properties of Jacobians.

Solution: 1.If u and v are the functions of x and y then $\frac{\partial(u,v)}{\partial(x,y)} \times \frac{\partial(x,y)}{\partial(u,v)} = 1$(inverse property of thejacobians)

2. If u, v are functions of x,y and x,y are themselves function of r,s then

$\frac{\partial(u,v)}{\partial(x,y)} \frac{\partial(x,y)}{\partial(r,s)} = \frac{\partial(u,v)}{\partial(r,s)}$

11.If x= r cosθ ,y= rsinθ find $\frac{\partial(x,y)}{\partial(r,\theta)}$

Solution: $\frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r[\cos^2\theta + \sin^2\theta] = 1$

12.If u= $\frac{y^2}{2x}, v = \frac{x^2 + y^2}{2x}$ find $\frac{\partial(u,v)}{\partial(x,y)}$

Solution: $\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{-y^2}{2x^2} & \frac{y}{x} \\ \frac{x^2-y^2}{2x^2} & \frac{y}{x} \end{vmatrix}$

$= -\frac{y^3}{2x^3} - \frac{y(x^2-y^2)}{2x^3} = -\frac{y}{2x}$

13.Write Taylors series expansion.

Solution: $f(x,y) = f(a,b) + [h f_x(a,b) + k f_y(a,b)] + (1/2!)[h^2 f_{xx}(a,b) + 2hk f_{xy}(a,b) + k^2_{yy}(a,b)] + \ldots$

14.Expand $f(x,y) = e^{xy}$ in Taylors series at (1,1) upto second degree.

Solution:

Function	Value at (1,1)
$f(x,y)=e^{xy}$	$f=e$
$f_x = ye^{xy}$	$f_x=e$
$f_y=xe^{xy}$	$f_y=e$
$f_{xx}= y^2e^{xy}$	$f_{xx}=e$
$f_{xy}=y^2e^{xy}$	$f_{xy}=2e$
$f_{yy}=x^2e^{xy}$	$f_{yy}=e$

$\therefore e^{xy} = e + \frac{(x-1)e + (y-1)e}{1!} + \frac{(x-1)^2 + 2(x-1)(y-1)2e}{2!} + (y-1)^2 e + \ldots$

$= 2\left[1 + \frac{(x-1)+(y-1)}{1!} + \frac{(x-1)^2 + 4(x-1)(y-1) + (y-1)^2}{2!}\right]$

15. Define stationary value

Solution: A function f(x,y) is said to be stationary at (a,b) or f(a,b) is said to bea stationary value of f(x,y) if $f_x(a,b)=0$and $f_y(a,b) =0$.

16. A flat circular plate is heated so that the temperature at any point (x,y) is $u(x,y)=x^{-2}+2y^2-x$. Find the coldest point on the plate.

Solution: $u= x^{-2}+2y^2-x$

$u_x= 2x-1, u_x=0, u_y=0, u_y=4y, 2x-1=0, 4y=0, x=1/2, y=0$, $A= u_{xx}=2$; $C=u_{yy}=4$; $B=u_{xy} =0$

$A=AC-B^2>0$

u is maximum at (1/2,0) and its minimum value is -1/4

17. State Euler's theorem on homogeneous function .

Solution: Let u=f(x,y) be a homogeneous functions of degree n then $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu$

18. Define Saddle point

Solution: The points at which $rt-s^2$ is negative there is no maxima and minima (such points are called saddle points)

19. Expand $e^x \cos y$ in powers of x and y upto terms of first degree

solution: let $f(x,y)= e^x \cos y$

The Taylor series in power of x and y is

$f(x,y)=f(0,0)+xf_x(0,0)+yf_y(0,0)+\ldots$

$f(0,0)=e \cos 0=1$, $f_x=e^0 \cos 0=1$, $f_y(0,0)= - e^0 \sin 0 =0$

f(x,y)=1+x(1)+y(0)+...=1+x

20. Evaluate $\int_0^1 \int_x^1 dydx$

Solution: Given $\int_0^1 \int_x^1 dydx = \int_0^1 [y]_0^x dx = \int_0^1 (x-0)dx$

$$= \int_0^1 xdx = \left(\frac{x^2}{2}\right)_0^1 = 1/2$$

21. Using Euler's theorem, given u(x,y) is a homogeneous function of degree n, prove that $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = n(n-1)u$ **.**

Solution: By Euler's theorem $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu$ …….. (1)

Differentiating (1) partially w.r.to x, we get

$$x\frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x} + y\frac{\partial^2 u}{\partial x \partial y} = n\frac{\partial u}{\partial x}$$

$$x\frac{\partial^2 u}{\partial x^2} + y\frac{\partial^2 u}{\partial x \partial y} = (n-1)\frac{\partial u}{\partial x} \text{.....(2)}$$

Differentiating (1) partially w. r.to y we get

$$x\frac{\partial^2 u}{\partial x \partial y}\, y\frac{\partial^2 u}{\partial y^2} + \frac{\partial u}{\partial y} = n\frac{\partial u}{\partial y}$$

$$y\frac{\partial^2 u}{\partial y^2} + x\frac{\partial^2 u}{\partial y^2} = (n-1)\frac{\partial u}{\partial y} \text{.....(3)}$$

$(2)\times x + (3)\times y \Rightarrow$

$$x^2\frac{\partial^2 u}{\partial x^2} + xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} + xy\frac{\partial^2 u}{\partial x \partial y} = (n-1)x\frac{\partial u}{\partial x} + .(n-1)y\frac{\partial u}{\partial y} \text{.....(2)}$$

$$= (n-1) = (n-1)(nu)$$

$$but \quad \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial x \partial y}$$

$$weget\ x^2\frac{\partial^2 u}{\partial x^2} + xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} = n(n-1)u$$

22.Using the definition of total derivative, find the value of $\frac{du}{dt}$ **given**

$u = y^2 - 4ax; x = at^2; y = 2at$

$$\frac{du}{dt}=\frac{\partial u}{\partial x}\frac{dx}{dt}+\frac{\partial u}{\partial y}\frac{dy}{dt}\text{.....(1)}$$

$$u=y^2-4ax\Rightarrow\frac{\partial u}{\partial x}=-4a\ \&\ \frac{\partial u}{\partial y}=2y=4at$$

Solution:

$x=at^2$ y=2at

$\frac{dx}{dt}=2at$ $\frac{dx}{dt}=2a$

$$\frac{du}{dt}=(-4a)(2at)+(4at)(2a)=0$$

23.If $u=x^3y^2+x^2y^3$ **where** $x=at^2$ **and y=2at then find** $\frac{du}{dt}$ **?**

Solution:

$u=x^3y^2+x^2y^3$

$\frac{\partial u}{\partial x}=3x^2y^2+2xy^3=xy^2\left[3x+2y\right]$

$=\left(at^2\right)^2\left(2at\right)^2\left[3at^2+4at\right]=4a^4t^5\left[3t+4\right]$

$\frac{\partial u}{\partial y}=2x^2+3x^2y^2=x^2y[2x+3y]$

$=(at^2)^2(2at)[2at^2+6at]$

$=a^2t^4\,2at(2at)[t+3]$

$=4a^4t^6[t+3]$

$$x=at^2\Rightarrow\frac{dx}{dt}=2at, y=2at\Rightarrow\frac{dy}{dt}=2a$$

$$(1)\Rightarrow\frac{du}{dt}=[4a^4t^5(3t+4)](2at)+[4a^4t^6(t+3)](2a)$$

$$=8a^5t^6(3t+4)+8a^5t^6(t+3)$$

$$=8a^5t^6(4t+7)$$

24.If u=2xy, v=x^2-y^2, x=rcosθ,y=rsinθ then compute $\frac{\partial(u,v)}{\partial(r,\theta)}$ **=?**

Solution:u=2xy v=x^2-y^2 x=rcosθ y=rsinθ

$u = 2xy \quad v = x^2 - y^2 \quad x = r\cos\theta \quad y = r\sin\theta$

$\frac{\partial u}{\partial x} = 2y \quad \frac{\partial v}{\partial x} = 2x \quad \frac{\partial x}{\partial r} = \cos\theta \quad \frac{\partial y}{\partial x} = \sin\theta$

$\frac{\partial u}{\partial y} = 2x \quad \frac{\partial v}{\partial y} = -2y \quad \frac{\partial x}{\partial \theta} = -r\sin\theta \quad \frac{\partial y}{\partial \theta} = r\cos\theta$

$$\frac{\partial(u,v)}{\partial(r,\theta)} = \frac{\partial(u,v)}{\partial(x,y)}\frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial u}{\partial y} \end{vmatrix} \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} 2y & 2x \\ 2x & -2y \end{vmatrix} \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix}$$

$$= (-4y^2 - 4x^2)(r\cos^2\theta + r\sin^2\theta)$$

$$= -4(x^2 + y^2)r$$

$$\therefore \frac{\partial(u,v)}{\partial(r,\theta)} = -4r(r^2) = -4r^3.$$

25.Find $\frac{du}{dt}$ if $u = \sin\left(\frac{x}{y}\right)$.where x=e^t ,y=t^2.

Solution:

$$\frac{du}{dt} = \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial y}\frac{dy}{dt}$$

$$= \frac{1}{y}\cos\left(\frac{x}{y}\right)e^t + \left(\frac{-x}{y^2}\right)\cos\frac{x}{y}2t = \frac{1}{y}e^t\cos\left(\frac{x}{y}\right) - \left(\frac{x}{y^2}\right)2t\cos\frac{x}{y}$$

$$= \frac{e^t}{t^2}\cos\left(\frac{e^t}{t^2}\right) - 2t\frac{e^t}{(t^2)^2}\cos\frac{e^t}{t^2} = \frac{e^t}{t^2}\cos\left(\frac{e^t}{t^2}\right)\left[1 - \frac{2}{t}\right]$$

26.If $u = \frac{y^2}{2x}, v = \frac{x^2 + y^2}{2x}$, Find $\frac{\partial(u,v)}{\partial(x,y)}$

Solution:

: $$\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} \text{.....(1)}$$

$u = \frac{y^2}{2x} \qquad v = \frac{x^2 + y^2}{2x}$

$$\frac{\partial u}{\partial x} = \frac{y^2}{2}\left[\frac{-1}{x^2}\right] = \frac{-y^2}{2x^2}, \frac{\partial v}{\partial x} = \frac{(2x)(2x) - (x^2 + y^2)(2)}{(2x)^2} = \frac{x^2 - y^2}{2x^2}$$

$$\frac{\partial u}{\partial x} = \frac{2y}{2x} = \frac{y}{x}, \frac{\partial v}{\partial x} = \frac{1}{2x}[2y] = \frac{y}{x} \therefore (1) \Rightarrow \frac{\partial u}{\partial x} = \begin{vmatrix} \frac{-y^2}{2x^2} & \frac{y}{x} \\ \frac{x^2 - y^2}{2x^2} & \frac{y}{x} \end{vmatrix} = \frac{-y^2}{2x^3} - \frac{y(x^2 - y^2)}{2x^3} = \frac{-y}{2x}$$

27.Given $u(x,y) = x^2\tan^{-1}\left(\frac{y}{x}\right)$, find the value of $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy}$.

Solution: Given:

$$u(x,y)=x^2\tan^{-1}\left(\frac{y}{x}\right)\Rightarrow u(tx,ty)=(tx)^2\tan^{-1}\left(\frac{ty}{tx}\right)\Rightarrow=t^2x^2\tan^{-1}\frac{y}{x}=t^2u(x,y)$$

$\Rightarrow$u is a homogeneous function of degree 2 and it is differentiable partially twice.

$$x^2\frac{\partial^2u}{\partial x^2}+2xy\frac{\partial^2u}{\partial x\partial y}+y^2\frac{\partial^2u}{\partial y^2}=n(n-1)u=2(2-1)u=2u$$

28.Write the sufficient condition for f(x,y) to have a maximum value at (a,b).

Solution: If f(x,y) has amaximum or minimum value at (a,b), then its partial derivatives vanish

i.e., $f_x(a,b)=0,\ f_y(a,b)=0$

$(i)\ f_{xx}<0 \quad \& \quad f_{xx}f_{yy}-f^2xy>0\ at(a,b)\Rightarrow Maximum$

$(ii)\ f_{xx}>0 \quad \& \quad f_{xx}f_{yy}-f^2xy>0\ at(a,b)\Rightarrow Minimum$

29.Find $\frac{du}{dx}$**,if** $u=\tan^{-1}\left(\frac{y}{x}\right)$ **where** $y=\tan^2 x$**.**

Solution: Given:

$$\frac{du}{dx}=\frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}\frac{dy}{dx}\text{.....(1)}$$

$$u=\tan^{-1}(y/x)$$

$$\frac{\partial u}{\partial x}=\frac{1}{1+\left(\frac{y}{x}\right)^2}\left[\frac{-y}{x^2}\right]=\frac{1}{1+\frac{y^2}{x^2}}\left[\frac{-y}{x^2}\right]=\frac{-y}{x^2+y^2},\frac{\partial u}{\partial y}=\frac{1}{1+\left(\frac{y}{x}\right)^2}\left[\frac{1}{x}\right]=\frac{x}{x^2+y^2}$$

$$y=\tan^2 x\ \Rightarrow\ \frac{dy}{dx}=2\tan x[\sec^2 x]$$

$$\therefore\frac{du}{dx}=\frac{-y}{x^2+y^2}+\frac{x}{x^2+y^2}2\tan x\sec^2 x=\frac{1}{x^2+y^2}+[-y+2x\tan x\sec^2 x$$

$$=\frac{1}{x^2+y^2}+[-\tan^2 x+2x\tan x\sec^2 x$$

$$=\frac{\tan x}{x^2+y^2}[-\tan x+2x\sec^2 x]=\frac{\tan x}{x^2+y^2}[-\tan x+2x(1+\tan^2 x)]=\frac{-y+2x\sqrt{y}(1+y)}{x^2+y^2}$$

30.If $u=\frac{y^2}{x}\ \&\ v=\frac{x^2}{y}$**, Find** $\frac{\partial(x,y)}{\partial(u,v)}$**.**

Solution: $\frac{\partial(u,v)}{\partial(x,y)}=\begin{vmatrix}\frac{\partial u}{\partial x} & \frac{\partial u}{\partial y}\\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y}\end{vmatrix}=\begin{vmatrix}\frac{-y^2}{x^2} & \frac{2y}{x}\\ \frac{2x}{y} & \frac{-x^2}{y^2}\end{vmatrix}=-3,Hence\ \frac{\partial(x,y)}{\partial(u,v)}=\frac{-1}{3}$

31. If $u=\sin^{-1}\frac{\sqrt{x}-\sqrt{y}}{\sqrt{x}+\sqrt{y}}$, find $x\frac{\partial u}{\partial x}+y\frac{\partial v}{\partial y}$.

$u = \sin^{-1}\dfrac{\sqrt{x}-\sqrt{y}}{\sqrt{x}+\sqrt{y}}, \Rightarrow \sin u = \dfrac{\sqrt{x}-\sqrt{y}}{\sqrt{x}+\sqrt{y}}$

Let $f(x,y) = \sin u = \dfrac{\sqrt{x}-\sqrt{y}}{\sqrt{x}+\sqrt{y}}$

$$f(tx,ty) = \sin u = \frac{\sqrt{t}\sqrt{x}-\sqrt{t}\sqrt{y}}{\sqrt{t}\sqrt{x}+\sqrt{t}\sqrt{y}} = \frac{\sqrt{x}-\sqrt{y}}{\sqrt{x}+\sqrt{y}} = f(x,y)$$

f is a homogeneous function of degree 0 in x and y .

Hence using Euler's theorem

$$x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} = (0)(f) = 0 \Rightarrow i.e., x\frac{\partial(\sin u)}{\partial x} + y\frac{\partial(\sin u)}{\partial y} = 0 \Rightarrow x\cos u\frac{\partial u}{\partial x} + y\cos u\frac{\partial u}{\partial y} = 0 \qquad i.e., x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 0$$

32.If x=rcosθ, y=rsinθ find ∂r⁄∂x

Solution:

$x = r\cos\theta \quad , \quad y = r\sin\theta$

Given: $x^2 = r^2\cos^2\theta \quad , \quad y^2 = r^2\sin^2\theta$

$$x^2 + y^2 = r^2\cos^2\theta + r^2\sin^2\theta = r^2[\cos^2\theta + \sin^2\theta = r^2$$

$$i.e., r = \sqrt{x^2+y^2} \Rightarrow \frac{\partial r}{\partial x} = \frac{1}{2\sqrt{x^2+y^2}}|2x| = \frac{x}{\sqrt{x^2+y^2}} = \frac{x}{r}$$

33.Find the stationary points of $z=x^2-xy+y^2-2x+y$.

Solution:

$$z = x^2 - xy + y^2 - 2x + y$$

$$i.e., z = f(x,y) = x^2 - xy + y^2 - 2x + y$$

$$= f_x(x,y) = 2x - y - 2$$

$$= f_y(x,y) = -x + 2y + 1$$

To find the stationary points

Put $f_x = 0, \quad f_y = 0$

$f_x = 0 \Rightarrow \quad 2x - y - 2 = 0$

$2x - y = 2$ .....(1)

$f_y = 0 \Rightarrow -x + 2y + 1 = 0$

$-x + 2y = -1$ .....(2)

$(1)\times 2 \Rightarrow \quad 4x - 2y = -1$ .....(3)

$(2) \Rightarrow \quad -x + 2y = -1$ .....(4)

$(3) + (4) \Rightarrow \quad 3x = 3 \Rightarrow x = 1$

Put $x = 1$ *in* (1), *weget* $2 - y = 2 \Rightarrow y = 0$

$\therefore$ The stationary value is (1,0).

34.If $u = \sin^{-1}\frac{x}{y} + \tan^{-1}\frac{y}{x}$, *Find* $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$.

Solution:

$u = f(x,y) = \sin^{-1}\frac{x}{y} + \tan^{-1}\frac{y}{x}$ $f(tx,ty) = \sin^{-1}\frac{tx}{ty} + \tan^{-1}\frac{ty}{tx} = \sin^{-1}\frac{x}{y} + \tan^{-1}\frac{y}{x} = f(x,y)$

$\therefore$ u is a homogeneous function of degree 0 in x and y.

$\therefore$ By Eulers theorem, $x\frac{\partial u}{\partial x} + y\frac{\partial v}{\partial y} = 0 \times u = 0$

34.If $z=u^2+v^2$, $u=r\cos\theta, v=r\sin\theta$, Find $\frac{\partial z}{\partial \theta}$.

Solution: Given: $z=u^2+v^2$, , $u=r\cos\theta$, $v=r\sin\theta$, i.e., $z=r^2\cos^2\theta+r^2\sin^2\theta= r^2[\cos^2\theta+\sin^2\theta$, $z = r^2$ $\Rightarrow \left(\frac{\partial z}{\partial \theta}\right)_r = 0.$

35.Expande^{xy} in power of x-1and y-1 upto first degree terms.

Solution;

Function	*Valueat*(1,1)
$f(x,y) = e^{xy}$	$f = e$
$f_x = ye^{xy}$	$f_x = e$
$f_y = xe^{xy}$	$f_y = e$
$f_{xx} = y^2e^{xy}$	$f_{xx} = e$
$f_{xy} = xye^{xy} + e^{xy}$	$f_{xy} = e + e = 2e$
$f_{yy} = x^2e^{xy}$	$f_{yy} = e$

By Taylor's series expansion,

$\therefore e^{xy} = \frac{e + (x-1)e + (y-1)e}{1!} + \frac{(x-1)^2e + 2(x-1)(y-1)2e + (y-1)^2e}{2!} + ...$

$= e\left[\frac{1 + (x-1) + (y-1)}{1!} + \frac{(x-1)^2 + 2(x-1)(y-1)2 + (y-1)^2}{2!} + ...\right]$

36. $u = \sin^{-1}\left(\frac{x^3 - y^3}{x+y}\right)$ $P.T$ $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 2\tan u$

Solution:

Given: $\sin u = (x^3-y^3) / (x+y)$

Sin u is a homogeneous function of degree 2.

$\therefore \Rightarrow x\frac{\partial}{\partial x}(\sin u) + y\frac{\partial}{\partial y}(\cos u) = 2\sin u \Rightarrow x\cos u\frac{\partial u}{\partial x} + y\cos u\frac{\partial u}{\partial y} = 2\sin u x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 2\tan u$

37.If $u = \frac{y}{z} + \frac{z}{x} + \frac{x}{y}$, **gives the value of** $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z}$.

Solution:

$Given: \quad u = \frac{y}{z} + \frac{z}{x} + \frac{x}{y}$

$\frac{\partial u}{\partial x} = \frac{1}{y} - \frac{z}{x^2} \Rightarrow \quad x\frac{\partial u}{\partial x} = \frac{x}{y} - \frac{z}{x}$.....(1)

$\frac{\partial u}{\partial y} = \frac{-x}{y^2} + \frac{1}{z} \Rightarrow \quad y\frac{\partial u}{\partial y} = \frac{-x}{y} + \frac{y}{z}$.....(2)

$\frac{\partial u}{\partial z} = \frac{-y}{z^2} + \frac{1}{x} \Rightarrow \quad z\frac{\partial u}{\partial z} = \frac{-y}{z} + \frac{z}{x}$.....(3) $\therefore x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0$

38. Find the taylor's series of x^y near the point (1,1) upto the first degree term.

Solution:

Function	*valueat* (1,1)
$f(x,y) = x^y$	$f = 1$
$f_x = yx^{y-1}$	$f_x = 1$
$f_y = x^y \log x$	$f_y = 0$

By Taylor's theorem.,

$f(x,y) = f(a,b) + [hf_x(a,b) + kf_y(a,b) + ...]$]

$Here\ a = 1, b = 1,\ h = x - a = x - 1,\ k = y - b = y - 1$

$x^y = 1 + [(x-1)(1) + (y-1)(0)] + ... = 1 + (x-1) + ...$

39.If $u=\log(x^2+xy+y^2)$., Prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 2$.

Solution: Given: $u=\log(x^2+xy+y^2)$.,

Here u is not a homogeneous function

$\therefore e^u = x^2+xy+y^2$, e^u is a homogeneous function of degree 2 in x,y.

$\therefore$ By Euler's theorem ,we have

$$x\frac{\partial}{\partial x}(e^u) + y\frac{\partial}{\partial y}(e^u) = 2e^u \Rightarrow xe^u\frac{\partial}{\partial x} + ye^u\frac{\partial}{\partial y} = 2e^u \Rightarrow \quad x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y} = 2.$$

40. Find $\frac{du}{dt}$, if $u=x^2+y^2$, $x= at^2$, $y=2at$.

Solution: $\frac{du}{dt} = \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial y}\frac{dy}{dt}$ .....(1)

$u = x^2 + y^2 \qquad x = at^2 \qquad y = 2at$

$\frac{\partial u}{\partial x} = 2x \qquad \frac{dx}{dt} = 2at \qquad \frac{dy}{dt} = 2a$

$\frac{\partial u}{\partial y} = 2y = 4at \quad \therefore (1) \Rightarrow \frac{du}{dt} = (2at^2)(2at) + (4at)(2a) = 4a^2t^3 + 8a^2t = 4a^2t(t^2+2)$

41. Find $\frac{dy}{dx}$ when ysinx=xcosy.

Solution: Given: f(x,y)=ysinx-xcosy

P= $\partial f/\partial x$ =ycosx-cosy and Q= $\partial f/\partial y$ =sinx+xsinydy/dx = p/q = (ycosx-cosy) / (sinx+xsiny).

42.Find stationary points of the function $f(x,y)=x^3-y^3-3xy$.

Solution: Given: $f(x,y)= x^3-y^3-3xy$

$f_x = 3x^2 - 3y,\ f_y = -3y^2 - 3x$

For extremum, $f_x = 0$ & $f_y = 0$

$\therefore 3x^2 - 3y = 0 \quad \& \quad -3y^2 - 3x = 0$

$x^2 = y \quad \& \quad y^2 = -x$

Eliminating y we get,

$$(x^2)^2 = -x \Rightarrow x^4 + x = 0 \Rightarrow x(x^3+1) = 0$$

$\therefore \quad x = 0, y = 0$ *&when* $x = -1, y = 1$

Therefore the stationary points are (0,0) and (-1,1)

43.Find the jacobian $\frac{\partial(x,y)}{\partial(u,v)}$, if x=u(1+v) and y=v(1+v).

Solution:

$$\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 1+v & u \\ v & 1+u \end{vmatrix} = 1+u+v.$$

44.If $u = y^2 / x$, $v = x^2 / y$, find $\frac{\partial(u,v)}{\partial(x,y)}$.

Solution:

$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} \Rightarrow u = \frac{y^2}{x}, v = \frac{x^2}{y}$$

$$\frac{\partial u}{\partial x} = -\frac{y^2}{x^2}, \frac{\partial v}{\partial x} = \frac{2x}{y}$$

$$\frac{\partial u}{\partial y} = \frac{2y}{x}, \frac{\partial v}{\partial y} = -\frac{x^2}{y^2}, \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} -\frac{y^2}{x^2} & \frac{2y}{x} \\ \frac{2x}{y} & \frac{-x^2}{y^2} \end{vmatrix} = \frac{x^2y^2}{x^2y^2} - \frac{4xy}{xy} = 1-4 = -3$$

45.Expand e^x cosy near (1, π/2) by Taylor's series upto first degree terms only.

Solution:

Function	*valueat* $(1,\frac{\pi}{4})$
$f(x,y)=e^x\cos y$	$f(a,b)=e(\frac{1}{\sqrt{2}})=\frac{e}{\sqrt{2}}$
$f_x=e^x\cos y$	$f_x=\frac{e}{\sqrt{2}}$
$f_y=e^x\sin y$	$f_y=-\frac{e}{\sqrt{2}}$

Taylor's series upto first degree.

$$f(x,y)=f(a,b)+\frac{1}{1!}[hf_x(a,b)+kf_y(a,b)]$$

Here $a=1,b=\frac{\pi}{4},h=x-a=x-1,k=y-b=y=\frac{\pi}{4}$

$$\therefore f(x,y)=\frac{e}{\sqrt{2}}+\frac{1}{1!}\left[(x-1)\frac{e}{\sqrt{2}}-(y-\frac{\pi}{4})\frac{e}{\sqrt{2}}\right]=\frac{e}{\sqrt{2}}\left[1+(x-1)-(y-\frac{\pi}{4})\right]$$

46.Prove that $xu_x+yu_y=\frac{5}{2}\tan u$ **if** $u=\sin^{-1}\left(\frac{x^3+y^3}{\sqrt{x}+\sqrt{y}}\right)$**.**

Solution: Given: $u=\sin^{-1}\left(\frac{x^3+y^3}{\sqrt{x}+\sqrt{y}}\right)$

Sinu is a homogeneous function of degree 5/2.

$$\therefore x\frac{\partial}{\partial x}(\sin u)+y\frac{\partial}{\partial y}(\sin u)=\frac{5}{2}\sin u\Rightarrow x\cos u\frac{\partial u}{\partial x}+y\cos u\frac{\partial u}{\partial y}=\frac{5}{2}\sin u$$

$$\div\cos u\Rightarrow\quad x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\frac{5}{2}tanu$$

47.Find the minimum point of f((x,y)=$x^2+y^2+6x+12$.

Solution

Given: $f(x,y)=x^2+y^2+6x+12$

$$f_x=2x+6,\ f_y=2y,\ f_{xx}=2,\ f_{yy}=2,\ f_{xy}=0$$

The stationary points are given by $f_x=0$ and $f_y=0$

$$f_x=0\Rightarrow 2x+6=0\Rightarrow 2x=6\Rightarrow x=-3,\qquad f_y=0\Rightarrow 2y=0\Rightarrow y=0$$

$\therefore$ (-3,0) is the extreme point of f(x,y)

$$A=f_{xx}=2,\qquad B=f_{xy}=0,\qquad c=f_{yy}=2\Rightarrow AC-B^2=4>0$$

$\therefore(-3,0)$ *is a* min*imum po*int.

48. *If* $u=\tan^{-1}\left(\frac{x^2+y^2}{x+y}\right)$ *Find* $\frac{\partial u}{\partial x}$.

Solution:

$Given: u = \tan^{-1}\left(\frac{x^2+y^2}{x+y}\right)$

$$\frac{\partial u}{\partial x} = \frac{1}{1+\left[\frac{x^2+y^2}{x+y}\right]}\left[\frac{(x+y)(2x)-(x^2+y^2)(1)}{(x+y)^2}\right]$$

$$= \frac{(x+y)^2}{(x+y)^2+(x^2+y^2)} = \frac{x^2+2xy-y^2}{x^4+y^4+2x^2y^2+x^2+y^2+2xy}.$$

PART -B

1. (i) Expand $f(x,y)=4x^2+xy+6y^2+x-20y+21$ as Taylor series about (-1,1)

   (ii) If $u= 4x^2+6xy$, $v= 2y^2xy$, $x=r\cos\theta$, $y= r\sin\theta$ evaluate $\frac{\partial(u,v)}{\partial(r,\theta)}$

2. (i) Evaluate $\int_0^\infty \frac{e^{-ax}\sin x}{x}dx$, a>0 and deduce $\int_0^\infty \frac{\sin x}{x}dx = \frac{\pi}{2}$

   (ii) Find the minimum value of xy^2z^2 subject to x+y+z=24 using Lagrange's multiplier.

3. (i) Find the maximum and minimum values of x^2-xy+y^2-2x+y.

   (ii) Expand $x^2y+3y-2$ in powers of (x-1) and (y+2) using Taylor's expansion.

4. Determine the maximum value of the nth root of a product of n positive members $x_1, x_2, \ldots, x_n$ provided that their sum is equal to a given number 'a'.

5. (i) If $u= \log(x^2+y^2+z^2-3xyz)$, show that $\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)^2 u = \frac{-9}{(x+y+z)^2}$

   (ii) Locate the stationary point of the function $f(x,y)=x^4+y^4-2x^2+4xy-2y^2$

6. If $u= \log(x^3+y^3+z^3-3xyz)$ prove that $\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)^2 u = \frac{9}{(x+y+z)^2}$

7. Find the minimum value of $x^2+y^2+z^2$ with the constraint $xy+yz+zx=3a^2$

8. (i) If $u=\frac{y^2}{2x}, v=\frac{x^2+y^2}{2x}$ find $\frac{\partial(u,v)}{\partial(x,y)}$

   (ii) Prove that rectangular solid of maximum volume which can be inscribed in sphere is cube.

9. (i) If $u=\sin^{-1}\left(\frac{x^2+y^2}{x+y}\right)$ prove that $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y} = \tan u$

   (ii) Investigate the maxima and minima if any of the function $y^2+4xy+3x^2+x^3$

10. Find the volume of greatest rectangular parallelepiped that can be inscribed in the ellipsoid $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$

11. (i) If $u = x^y$., Show that $u_{xxy}=u_{xyx}$.

    (ii) If $u=\log(x^2+y^2)+\tan^{-1}(y/x)$. Prove that $u_{xx}+u_{yy}=0$.

(iii) Find the jacobian $\frac{\partial(x,y,z)}{\partial(r,\theta,\varphi)}$ of the transformation x=rsinθcosφ, y=rsinθsinφ and z=rcosθ.

12. (i) Find the maximum value of $x^m y^n z^p$ subject to the condition x+y+z=a.

(ii) find the Taylor's series expansion of e^xsiny at the point (-1,π/4) upto third degree terms.

13.(i)If u= $r\left[\frac{x}{y},\frac{y}{z},\frac{z}{x}\right]$, Prove that $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+z\frac{\partial u}{\partial z}=0$.

14. (i) Examine the function $f(x,y)=x^3y^2(12-x-y)$ for extreme values.

(ii) Find the volume of the greatest rectangle parallelopiped inscribed in the ellipsoid whose equation is $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$

15. (i) If z=f(x,y). where $x=u^2-v^2$, y=2uv, prove that $\frac{\partial^2 z}{\partial u^2}+\frac{\partial^2 z}{\partial v^2}=4(u^2+v^2)\left(\frac{\partial^2 z}{\partial x^2}+\frac{\partial^2 z}{\partial y^2}\right)$

(ii) Find the Taylor's series expansion of $x^2y^2+2x^2y+3xy^2$ in powers of (x+2) and (y-1) upto third degree terms.

16. (i) If x+y+z=u , y+z=uv , z=uvw , Prove that $\frac{\partial(x,y,z)}{\partial(u,v,w)}=u^2v$

(ii) Find the extreme values of the function $f(x,y)=x^3+y^3-3x-12y+20$

17. (i)If u=f(x,y) where x=rcosθ ,y=rsinθ, Prove that $\left(\frac{\partial u}{\partial x}\right)^2+\left(\frac{\partial u}{\partial y}\right)^2=\left(\frac{\partial u}{\partial r}\right)^2+\frac{1}{r^2}\left(\frac{\partial u}{\partial \theta}\right)^2$

18. (i) Find the maximum and minimum values of x^2-xy+y^2-2x+y.

(ii) A rectangle box open at the top is to have a volume of 32 cc. Find the dimensions of the box that requires the least material for its construction.

19. (i) u= $\sin^{-1}\left(\frac{x^2+y^2}{x+y}\right)$, Prove that 1) $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\tan u$ 2) $x^2\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\partial y}+y^2\frac{\partial^2 u}{\partial y^2}=\tan^3 u$

(ii) Test for an extrema of the function $f(x,y)=x^4+y^4-x^2-y^2-1$.

20 (i) If x=ucosα - vsinα , y=usinα +vcosα and z=f(x,y), show that $\frac{\partial^2 z}{\partial x^2}+\frac{\partial^2 z}{\partial y^2}=\frac{\partial^2 z}{\partial u^2}+\frac{\partial^2 z}{\partial v^2}$

21. (i) If g(x,y)=ψ(u,v) where $u=x^2-y^2$, v=2xy, prove that $\frac{\partial^2 g}{\partial x^2}+\frac{\partial^2 g}{\partial y^2}=4(x^2+y^2)\left[\frac{\partial^2 \Psi}{\partial u^2}+\frac{\partial^2 \Psi}{\partial v^2}\right]$

(ii) Find the shortest and longest distances from the point (1,2,-1) to the sphere $x^2+y^2+z^2=24$, using the method of lagrange multipliers.

22. (i)Expand e^{-x} logy as a Taylor's series in powers of x and (y-1) upto third degree terms.

(ii) Find the jacobian of y_1 , y_2 , y_3 with respect to x_1 , x_2 , x_3 if $y_1=\frac{x_2x_3}{x_1}$, $y_2=\frac{x_3x_1}{x_2}$, $y_3=\frac{x_1x_2}{x_3}$

23. If $u=\log(x^3+y^3+z^3-3xyz)$ Prove that $\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)^2 u=\frac{-9}{(x+y+z)^2}$.

24. Find the minimum value of $x^2+y^2+z^2$ with the constraint $xy+yz+zx=3a^2$.

25. (i)If $u=(x^2+y^2+z^2)^{-1/2}$ prove that $\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2}=0$

(ii)If u= x(1-y) ,v=xy(1-z),w=xyz , Prove that $\frac{\partial(u,v,w)}{\partial(x,y,z)}=x^2y$.

UNIT – IV

MULTIPLE INTEGRALS

PART-A

1.Evaluate where A is the region bounded by x-axis coordinate x=2a and the curve $x^2=4ay$.

Solution: The point A is (2a,a).

The limits of y : y=0 to y= $\frac{x^2}{4a}$

The limits of x: x=0 to x=2a

Therefore $\iint_A xydxdy = \int_0^{2a}\int_0^{\frac{x^2}{4a}} xydydx = \int_0^{2a} x\left[\frac{y^2}{2}\right]_0^{\frac{x^2}{4a}}dx = \frac{1}{2}\int_0^{2a}\frac{x^5}{16a^2}dx$

$$= \frac{1}{32a^2}\int_0^{2a} x^5dx = \frac{1}{32a^2}\left[\frac{x^6}{6}\right]_0^2.a = \frac{64a^6}{32.6.a^2} = \frac{a^4}{3}$$

2.Change the order of integration in $I=\int_0^1\int_{x^2}^{2-x} xydydx$ and hence evaluate the same.

Solution: $I=\int_0^1\int_{x^2}^{2-x} xydydx$

Here first integrate with respect to y and then with respect to x

By changing the order of integration, integrate first w.r.t x, then y.

Region of integration is $Y=x^2 \rightarrow (1)$

$y = 2 - x,\ x + y = 2 \rightarrow (2)$ x=0 , x=1

Solving (1) and (2) the point of integration is (x,y)=(1,1)

The region of integration is (x,y)=(1,1).The region is divided in two parts:

$$I = \int_0^1 \int_0^{\sqrt{y}} xydxdy + \int_1^2 \int_1^{2-x} xydxdy$$

$$= \frac{1}{2}\left[x^2 y\right]_0^{\sqrt{y}} dy + \frac{1}{2}\int_1^2 \left(x^2 y\right)_0^{2-y} dy = \frac{1}{2}\int_0^1 y^2 dy + \frac{1}{2}\int_1^2 y(2-y)^2 dy$$

$$= \frac{1}{2}\left[\frac{y^3}{3}\right]_0^1 + \frac{1}{2}\left[2y^2 - \frac{4}{3}y^3 + \frac{y^4}{4}\right]_1^2 = \frac{1}{6} + \frac{1}{2}\left[10 - \frac{28}{3} - \frac{1}{4}\right] = \frac{1}{6} + \frac{5}{24} = \frac{3}{8}$$

3.Evaluate by changing the order of integration in $\int_0^4 \int_{\frac{x^2}{4}}^{2\sqrt{x}} dydx$

Solution: From the problem the region is divided is to vertical strips, first integrate w.r.t y then w.r.t x

Given curves $y = \frac{x^2}{4} \Rightarrow x^2 = 4y \Rightarrow y = 2\sqrt{x} \Rightarrow y^2 = 4x \Rightarrow x = 0, x = 4$

The point of intersection is (4,4).For changing the order of integration , divide region into horizontal strips.

Strips: $x = \frac{y^2}{4} to 2\sqrt{y}$

$y = 0 to 4 \Rightarrow \int_0^4 \int_{\frac{y^2}{4}}^{2\sqrt{y}} dxdy = \frac{16}{3}$

4. Find the area inside the circle r=a $\sin\theta$ **and outside the cardiod r=a** $(1-\cos\theta)$

Solution: Given curves are r=a $\sin\theta \rightarrow (1)$ and r=a $(1-\cos\theta) \rightarrow (2)$

Solving equations 1 and 2 , find the point of intersection

$$a\sin\theta = a(1-\cos\theta) \Rightarrow \sin\theta + \cos\theta = 1 \Rightarrow \frac{1}{\sqrt{2}}\sin\theta + \frac{1}{\sqrt{2}}\cos\theta = \frac{1}{\sqrt{2}}$$

$$\sin\left(\frac{\pi}{4} + \theta\right) = \sin\frac{\pi}{4} \Rightarrow 0 + \frac{\pi}{4} = \frac{\pi}{4} or \pi - \frac{\pi}{4} \Rightarrow \theta = 0 or \theta = \frac{\pi}{2}$$

$$\text{Area} \int_0^{\frac{\pi}{2}} \int_{a(1-\cos\theta)}^{a\sin\theta} rdrd\theta = \int_0^{\pi/2} \left[\frac{r^2}{2}\right]_{a(1-\cos\theta)}^{a\sin\theta} d\theta = \frac{a^2}{2}\int_0^{\frac{\pi}{2}} \left(\sin^2\theta - \cos^2\theta - 1 - 2\cos\theta\right)d\theta$$

$$Area = \frac{(4-\pi)a^2}{4}$$

5. Compute the area of the surface of the cone $x^2+y^2=z^2$ **which cut out by the cylinder** $(x-a)^2+y^2=a^2$

Solution: $\frac{\partial z}{\partial x}=\frac{x}{z},\frac{\partial z}{\partial y}=\frac{y}{z}$

$$S=\iint_R\sqrt{\left(\frac{\partial z}{\partial x}\right)^2+\left(\frac{\partial z}{\partial y}\right)^2+1}dydx=2\int_0^{2a}\int_0^{\sqrt{2ax-x^2}}\sqrt{2}dydx=2\sqrt{2}\int_0^{2a}\sqrt{2ax-x^2}dx$$

Substitute $x=2a\sin\theta \,\&\, dx=2a\cos\theta d\theta$

$$S=2\sqrt{2}\int_0^{\frac{\pi}{2}}4a^2\cos^2\theta d\theta=2\sqrt{2}4a^2\int_0^{\pi/2}\left(\frac{1+\cos\theta}{2}\right)d\theta\left[\because\cos2\theta=2\cos^2-1\right]=2\sqrt{2}4a^2.\frac{1}{2}\int_0^{\frac{\pi}{2}}(1+\cos\theta)d\theta=2\sqrt{2}\pi a^2$$

6.Compute entire area bounded by $r^2=a^2\cos2\theta$

Solution: Area = 4 times area in the first quadrant.

$$Area=4\int_0^{\pi/2}\int_0^{a\sqrt{\cos\theta}}rdrd\theta=4\int_0^{\pi/4}\frac{a^2}{2}\cos2\theta d\theta=\frac{2a^2}{2}=a^2$$

7.Express the region $x\geq0,y\geq0,z\geq0,x^2+y^2+z^2\leq1$ **by triple integration.**

Solution: $=\int_0^1\int_0^{\sqrt{1-x^2}}\int_{\sqrt{1-x^2}}^{\sqrt{1-x^2-y^2}}dzdydx=\int_0^1\int_0^{\sqrt{1-x^2}}(z)_0^{\sqrt{1-x^2-y^2}}dydx=\int_0^1\int_0^{\sqrt{1-x^2}}\left(\sqrt{1-x^2-y^2}\right)_0dydx$

Put

$r^2=1-x^2$

$$=\int_0^1\int_0^{\sqrt{1-x^2}}\sqrt{r^2-y^2}dydx=\int_0^1\left[\frac{r^2}{2}\sin^{-1}\frac{y}{r}+\frac{y}{2}\sqrt{r^2-y^2}\right]_0^{\sqrt{1-x^2}}dx=\int_0^1\frac{r^2}{2}\sin^{-1}dx=\frac{\pi}{2}\int_0^1\frac{1-x^2}{2}dx=\frac{\pi}{4}\left[x-\frac{x^3}{3}\right]_0^1=\frac{\pi}{6}$$ **8.**

Evaluate $\iiint_v dxdydz$ **where V is the volume enclosed by the cylinder and planes x. z=0 and z=2**

Solution: Limits of z : z=0 and z=2**,** Limits of y : o to $\sqrt{1-y^2}$, Limits of x :-1 to 1

$$\therefore\iiint_v dxdydz=2\int_{-1}^1\int_0^{\sqrt{1-y}}\int_0^{2-x}dzdxdy\ \therefore\iiint_v dxdydz=2\int_{-1}^1\int_0^{\sqrt{1-y}}\int_0^{2-x}dxdydz$$

$$= 2\int_{-1}^{1}\int_{0}^{\sqrt{1-y^2}} [z]_0^{2-x}\,dydx = 2\int_{-1}^{1}\int_{0}^{\sqrt{1-y^2}} (2-x)\,dxdy = 2\int_{-1}^{1}\left[2x - \frac{x^2}{2}\right]_0^{\sqrt{1-y}} dy = 4\int_{-1}^{1}\sqrt{1-y^2}\,dy - \int_{-1}^{1}(1-y^2)dy$$

$$= 4\left[\frac{y}{2}\sqrt{1-y^2} + \frac{1}{2}\sin^{-1}\left(\frac{y}{2}\right)\right]_{-1}^{1} - \left[y - \frac{y^3}{3}\right]_{-1}^{1} = 2\left[\frac{\pi}{2} + \frac{\pi}{2}\right] - \left[2 - \frac{2}{3}\right] = 2\pi - \frac{4}{3}$$

9.Evalulate $\int_0^1\int_0^{1-x}\int_0^{\sqrt{x+y}} zdxdydz$

Solution: Given integral is evaluated in the order dzdydx

Hence the integral= $\int_0^1\int_0^{1-x}\int_0^{\sqrt{x+y}} zdzdydx$

$$\frac{1}{2}\int_0^1\int_0^{1-x}\left[z^2\right]_0^{\sqrt{x+y}} dydx = \frac{1}{2}\int_0^1\int_0^{1-x}(x+y)dydx = \frac{1}{2}\int_0^1\left[xy + \frac{y^2}{2}\right]_0^{1-x} dx = \frac{1}{2}\int_0^1\left[x(1-x) + \frac{(1-x)^2}{2}\right]dx$$

$$= \frac{1}{2}\left[\frac{x^2}{2} - \frac{x^3}{3} - \frac{[1-x]^3}{6}\right]_0^1 = \frac{1}{2}\left[\frac{1}{2} - \frac{1}{3} + \frac{1}{6}\right] = \frac{1}{2}\left[\frac{2}{6}\right] = \frac{1}{6}$$

10. Find the volume of the region bounded by the surface $y = x^2, x = y^2$ **and the planes z=0 ,z=2**

Solution: Required volume = $\int_0^1\int_{x^2}^{\sqrt{x}}\int_0^2 dxdydz$

$$= \int_0^1\int_{x^2}^{\sqrt{x}} (z)_0^2 dydx = 2\int_0^1 [y]_{x^2}^{\sqrt{x}} dx = 2\int_0^1 \left(\sqrt{x} - x^2\right)dx = 2\left[\frac{2}{3} - \frac{1}{3}\right] = \frac{2}{3} \therefore volume = 2/3$$

11.*Evaluate* $\int_0^1\int_1^2 x(x+y)dydx$

Solution: $\int_0^1\int_1^2 x(x+y)dydx = \int_0^1\int_1^2 (x^2+xy)dydx$

$$= \int_0^1\left[x^2 + \frac{xy^2}{2}\right]_{y=1}^{y=2} dx = \int_0^1\left[(2x^2+2x) - (x^2 + \frac{x}{2})\right]dx = \int_0^1\left[2x^2 + 2x - x^2 - \frac{x}{2}\right]dx = \int_0^1\left[x^2 + \frac{3x}{2}\right]dx$$

$$= \left[x^3 + \frac{3x^2}{4}\right]_0^1 = \left(\frac{1}{3} + \frac{3}{4}\right) - (0+0) = \frac{13}{12}$$

12.Evaluate $\int_0^4 \int_1^{x^2} e^{y/x} dydx$

Solution: Let I= $\int_0^4 \int_1^{x^2} e^{y/x} dydx$

$$\int_0^4 \left[\frac{e^{y/x}}{\frac{1}{x}} \right]_{y=0}^{y=x^2} dx = \int_0^4 \left[\frac{e^x}{1/x} - \frac{1}{1/x} \right] dx = \int_0^4 \left[xe^x - x \right] dx = \int_0^4 x(e^x - 1) dx = \left[x(e^x - x) - (1)(e^x - \frac{x^2}{2}) \right]_0^4$$

$$= \left[4(e^4 - 4) - (e^4 - 8) - (0 - 1) \right] = [(4e^4 - 16 - e^4 + 8) + 1] = 3e^4 - 8 + 1 = 3e^4 - 7$$

13.Change the order of integration $\int_0^\infty \int_1^\infty \frac{e^{-y}}{y} dydx$ **and evaluate it**

Solution:

$$HINT: \int_0^\infty \int_1^\infty \frac{e^{-y}}{y} dydx = \int_0^\infty \int_0^Y \frac{e^{-y}}{y} dxdy = \int_0^\infty \left[\frac{e^{-y}}{y} x \right]_{x=0}^{x=y} dy = \int_0^\infty (e^{-y} - 0) dy = \left[\frac{e^{-y}}{-1} \right]_0^\infty = -[e^{-y}]_0^\infty = -[0 - 1] = 1$$

14. Find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Solution: Area of ellipse =4 X area of quadrant.

Divide the area into horizontal strips of width δy.

x varies from x =0 to x = $\frac{a}{b}\sqrt{b^2 - y^2}$, y varies from y= 0 to y= b

The required area=$4\int_0^b \int_1^{\frac{a}{b}\sqrt{b^2-y^2}} dxdy = 4\int_0^b [x]_0^{\frac{a}{b}\sqrt{b^2-y^2}} dy = 4\int_0^b \left[\frac{a}{b}\sqrt{b^2 - y^2} - 0 \right] dy$

$$= \frac{4a}{b}\int_0^b \sqrt{b^2 - y^2} dy = \frac{4a}{b}\left[\frac{b^2}{2} \sin^{-1}\frac{y}{b} + \frac{y}{2}\sqrt{b^2 - y^2} \right]_0^b = \frac{4a}{b}\left[\left(\frac{b^2}{2}\frac{\pi}{2} + 0 \right) - (0 + 0) \right] = \frac{4a}{b}\frac{b^2}{2}\frac{\pi}{2} = \pi ab$$

15. Evaluate $\int_C [(3xy^2 + y^3)dx + (x^3 + 3xy^2)dy]$ **where C is the parabola y^2=4ax from the point (0,0) to the point (a,2a).**

Solution: The given integral I= $\int_C [(3xy^2 + y^3)dx + (x^3 + 3xy^2)dy]$ The line integral is evaluated along the parabola y^2=4ax. The resulting definite integral is evaluated betweenthe concerned limits, got from the end points of

$put\ x = \frac{y^2}{4a},\ \mathrm{dx} = \frac{1}{4a}2ydy = \frac{1}{2a}ydy$

$$I = \int_0^{2a}\left[\left(3.\frac{1}{4a}y^2.y^2 + y^3\right)\frac{y}{2a}dy + \left(\frac{1}{64a^3}y^6 + 3.\frac{1}{4a}y^2y^2\right)dy\right]$$

$$= \int_0^{2a}\left(\frac{5}{4a}y^4 + \frac{3}{8a^2}y^5 + \frac{1}{64a^3}y^6\right)dy = \left(\frac{1}{4a}y^5 + \frac{1}{16a^2}y^6 + \frac{1}{448a^3}y^7\right)_0^{2a} = \frac{86}{7}a^4$$

16. Evaluate $\iiint (x^2+y^2+z^2)\,\mathrm{dxdydz}$ taken over the volume closed by the sphere $x^2+y^2+z^2=a^2$ by transforming into spherical coordinates.

Solution: I= $\iiint_V (x^2+y^2+z^2)dxdydz$ where V is the volume of the sphere $x^2+y^2+z^2=a^2$

To change into the spherical polar coordinates. Put x=r cos θ sin ϕ , y= r sin θ sin ϕ, z= r cos ϕ and $x^2+y^2+z^2=r^2$Here r varies from 0 to a, θ varies from 0 to 2π , ϕ varies from 0 to π.

$$\mathrm{I} = \int_0^{\pi}\int_0^{2\pi}\int_0^{1} r^2.r^2\sin\phi dr\ d\theta\ d\phi$$

$$= \left(\int_0^{1} r^4dr\right)\left(\int_0^{\pi}\sin\phi d\phi\right)\left(\int_0^{2\pi}d\theta\right) = \left(\frac{1}{5}\right)(2)(2\pi) = \frac{4\pi}{5}$$

17. Evaluate $\int_0^1\int_0^2\int_0^3 xyz\ dzdydx$

Solution: Let $\mathrm{I} = \left[\int_0^1 xdx\right]\left[\int_0^2 ydy\right]\left[\int_0^3 zdz\right] = \left[\frac{x^2}{2}\right]_0^1\left[\frac{y^2}{2}\right]_0^1\left[\frac{z^2}{2}\right]_0^3 = \frac{9}{2}$

18. Evaluate $\int_0^1\int_0^{1-x}\int_0^{x+y} e^z\,dxdydz$

Solution: *Let* $\mathrm{I} = \int_0^1\int_0^{1-x}\int_0^{x+y} e^z\,dxdydz$

$$= \int_0^1\int_0^{1-x}\int_0^{x+y} e^z dzdydx = \int_0^1\int_1^{1-x}[e^z]_0^{x+y}dydx = \int_0^1\int_1^{1-x}[e^{x+y}-e^0]dydx = \int_0^1\int_1^{1-x}[e^{x+y}-1]dydx = \int_0^1\int_1^{1-x}[e^xe^y - y]_{y=0}^{y=1-x}dx$$

$$= \int_0^1\left[(e^xe^{1-x} - (1-x)) - e^x\right]dx = \int_0^1[e^1 - (1-x) - e^x]dx$$

$$= \int_0^1(e-1+x-e^x)dx = \left[e^x - x + \frac{x^2}{2} - e^x\right]_0^1 = \left[\left(e-1+\frac{1}{2}-e\right) - (0-0+0-1)\right] = \frac{-1}{2}+1 = \frac{1}{2}$$

19. Express $= \int_0^a \int_y^a \frac{x^2 dxdy}{(x^2+y^2)^{3/2}}$ **in polar coordinates and then evaluate it.**

Solution: The region of integration is bounded by y=0, y=a, x=y, x=a

Let us transform this integral in polar coordinates by taking x=r cos θ, y= r sin θ, dxdy=r dr d θ

$$I = \int_0^{\pi/4} \int_0^{a\sec\theta} \frac{(r\cos\theta)^2 \, r \, drd\theta}{[(r\cos\theta)^2+(r\sin\theta)^2]^{3/2}} = \int_0^{\pi/4} \int_0^{a\sec\theta} \cos^2\theta drd\theta = \int_0^{\pi/4} \cos^2\theta [r]_0^{a\sec\theta} d\theta$$

$$= \int_0^{\pi/4} \cos^2\theta a\sec\theta d\theta = \int_0^{\pi/4} a \cos\theta d\theta = a\int_0^{\pi/4} \cos\theta d\theta = a[\sin\theta]_0^{\pi/4} = a\left(\frac{1}{\sqrt{2}} - 0\right) = \frac{a}{\sqrt{2}}$$

20. Transform the integral into polar coordinates and hence evaluate

$\int_0^a \int_0^{\sqrt{a^2-x^2}} \sqrt{x^2+y^2}\, dydx$.

Solution:

The region of integration is y=0,y= $\sqrt{a^2-x^2}$, x=0and x=a

i.e., y=0, $x^2+y^2=a^2$, x=0 and x=a i.e., the given region is a quadrant of the circle $x^2+y^2=a^2$

In polar coordinates the given region of integration is bounded by the curves r=0,r=a, θ =0 and $\theta = \pi/2$

$\therefore \int_0^a \int_0^{\sqrt{a^2-x^2}} \sqrt{x^2+y^2}\, dydx = \int_0^{\pi/2} \int_0^a r \, r \, dr \, d\theta$ [$\because$ In polar coordinates $r^2=x^2+y^2$ and dx dy=r drdθ]

$$= \int_0^{\pi/2} \left(\frac{r^3}{3}\right)_0^a d\theta = \frac{a^3}{3}\int_0^{\pi/2} d\theta = \frac{a^3}{3}[\theta]_0^{\pi/3} = \frac{a^3\pi}{6}$$

21. Evaluate $\int_0^1 \int_x^{\sqrt{x}} xy(x+y)dxdy$

Solution: $\int_0^1\int_x^{\sqrt{x}} xy(x+y)dxdy = \int_0^1\int_x^{\sqrt{x}} xy(x+y)dydx = \int_0^1\int_x^{\sqrt{x}} (x^2y+xy^2)dxdy = \int_0^1\left[x^2\frac{y^2}{2}+\frac{xy^3}{3}\right]_{y=x}^{y=\sqrt{x}}dx$

$= \int_0^1\left[\left(x^2\frac{x}{2}+x\frac{x^{\frac{3}{2}}}{3}\right)-\left(x^2\frac{x^2}{2}+x\frac{x^3}{3}\right)\right]dx = \int_0^1\left[\frac{x^3}{3}+\frac{x^{\frac{5}{2}}}{3}-\frac{5}{6}x^4\right]dx$

$= \left[\frac{x^8}{8}+\frac{x^{\frac{7}{2}}}{\left(\frac{7}{2}\right)(3)}-\frac{5}{6}\frac{x^5}{5}\right]_0^1 = \left(\frac{1}{8}+\frac{2}{21}-\frac{1}{6}\right)-(0+0-0) = \frac{3}{56}$

22. Evaluate $\int_c\left[x^2dy+y^2dx\right]$ **where C is the path y=x from (0, 0) to (1, 1)**

Solution: $\int_c x^2dy+y^2dx$ Here y=x; dy=dx= $\int_c x^2(dx)+x^2dx = \int_c 2x^2dx = \int_0^1 2x^2dx = \left[2\frac{x^3}{3}\right]_0^1 = \frac{2}{3}-0 = \frac{2}{3}$

23. Write down the double integral, to find the area between the circles $r = 2\sin\theta$ **and** $r = 4\sin\theta$

Solution: $r = 2\sin\theta$ is the circle with diameter 2 and $r = 4\sin\theta$ is the circle with diameter 4.

Initial line is the diameter and pole lies on circle. ∴ Thus the required integral is

$I = \int_0^{\pi}\int_{2\sin\theta}^{4\sin\theta} rdrd\theta = \int_0^{\pi}\left[\frac{r^2}{2}\right]_{2\sin\theta}^{4\sin\theta}d\theta = \int_0^{\pi}6\sin^2\theta d\theta = 6\int_0^{\pi}\left[\frac{1-\cos 2\theta}{2}\right]d\theta = 3\int_0^{\pi}(1-\cos 2\theta)d\theta$

$= 3\left[\theta-\frac{\sin 2\theta}{2}\right]_0^{\pi} = 3\left[(\pi-0)-(0-0)\right] = 3\pi$

24. Change of order of integration in $I = \int_0^1\int_{x^2}^{2-x} f(x,y)dxdy$

Solution: Given $I = \int_0^1\int_{x^2}^{2-x} f(x,y)dxdy = \int_0^1\int_{x^2}^{2-x} f(x,y)dydx$ (c page 7)

The region of integration is bounded by x=0, x=1, $y = x^2$ and y=2-x

Here x varies from x=0 to x=1 and varies from $y = x^2$ and y=2-x

$I = I_1 + I_2 = I = \int_0^1\int_{x^2}^{1} f(x,y)dydx + \int_0^1\int_0^{2-x} f(x,y)dydx$

Changing the order we get

$I = \int_0^1\int_0^{\sqrt{y}} f(x,y)dxdy + \int_1^2\int_1^{2-y} f(x,y)dydx$

25. Evaluate $\int_0^{\pi}\int_0^{\sin\theta} rdrd\theta.$

Solution: Given: $\int_0^{\pi}\int_0^{\sin\theta} r dr d\theta.$ =

$$\int_0^{\pi}\left[\frac{r^2}{2}\right]_{r=0}^{r=\sin\theta} d\theta = \int_0^{\pi}\left[\frac{\sin^2\theta}{2}-0\right]d\theta = \frac{1}{2}\int_0^{\pi}\sin^2\theta d\theta = \frac{1}{2}\int_0^{\pi}\left[\frac{1-\cos 2\theta}{2}\right]d\theta = \frac{1}{4}\left[\theta-\frac{\sin 2\theta}{2}\right]_0^{\pi} = \frac{\pi}{4}$$

26. Change the order of integration for the double integral $\int_0^1\int_0^x f(x,y)dxdy$.

Solution: Given: $I = \int_0^1\int_0^x f(x,y)dxdy = \int_0^1\int_0^x f(x,y)dydx$

The region of integration is bounded by x=0,x=1,y=0 and y=x.Here x varies from x=0 to x=1and y varies from y=0 to y=x

Changing the order we get $\int_0^1\int_y^1 f(x,y)dxdy$

27. Evaluate $\int\int_R dxdy,$ where R is the shaded region in the figure.

Solution:

$$\int\int_R dxdy = \int_0^2\int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} dxdy = \int_0^2[x]_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} dy = \int_0^2\left[\left(\sqrt{4-y^2}\right)-\left(-\sqrt{4-y^2}\right)\right]dy = 2\left[\frac{y}{2}\sqrt{4-y^2}+\frac{4}{2}\sin^{-1}\frac{y}{2}\right]_0^2 = 2\pi$$

28. Find the area enclosed by the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

Solution: Area of ellipse= $4\times$ area of quadrant.

Divide the area into horizontal strips of width δy.

X varies from x=0 to $x=\frac{a}{b}\sqrt{b^2-y^2}$ and Y varies from y=0 to y=b $\therefore$ The required

$$\text{area} = \int_0^2\int_0^{\frac{a}{b}\sqrt{b^2-y^2}} dxdy = 4\int_0^b[x]_0^{\frac{a}{b}\sqrt{b^2-y^2}} dy = 4\int_0^b\left[\frac{a}{b}\sqrt{b^2-y^2}\right]dy = \frac{4a}{b}\left[\frac{b^2}{2}\sin^{-1}\frac{y}{b}+\frac{y}{2}\sqrt{b^2-y^2}\right]_0^b = \frac{4a}{b}\frac{b^2}{2}\frac{\pi}{2} = \pi ab$$

29. Evaluate $\int_a^b\int_c^d\int_f^g e^{x+y+z}dzdydx$

Solution: let $I = \int_a^b\int_c^d\int_f^g e^{x+y+z}dzdydx = \int_a^b\int_c^d\int_f^g e^x e^y e^z dzdydx = \left[\int_a^b e^x dx\right]\left[\int_c^d e^y dy\right]\left[\int_f^g e^z dz\right] = \left[e^x\right]_a^b\left[e^y\right]_c^d\left[e^z\right]_f^g$

$= \left(e^b-e^a\right)\left(e^d-e^c\right)\left(e^g-e^f\right)$

30. Change the order of integration $\int_0^{\infty}\int_0^{\infty} f(x,y)dydx$.

Solution: The region of integration is bounded by x=0,x= ∞ ,y=0 and y= ∞

Here x varies from x=0 to x= ∞ and y varies from y=0 to y= ∞

Here x=0 to x= ∞ represents vertical path and y=0 to y= ∞ represents vertical strip

Changing the order of integration is nothing but to change the vertical path into horizontal path and to change the vertical strip to horizontal strip.Hence by changing the order we get

$$\int_0^{\infty}\int_0^{y} f(x,y)dydx$$

31. Transforms into polar coordinates the integral $\int_0^{a}\int_y^{a} f(x,y)dxdy$

Solution: The region of integration is bounded by x=x, y=a, y=0 and y=a

Let us transform thus integral in polar coordinates by taking

$x = r\cos\theta, y = r\sin\theta, dxdy = rd\theta dr$

$$I = \int_0^{\frac{\pi}{4}}\int_0^{a\sec\theta} f(r\cos\theta, r\sin\theta)rd\theta dr$$

32. Why do we change the order of integration in multiple integrals? Justify your Solution with an example.

Some of the problems connected with double integrals, which seen to be complicated, can be made easy to handle by a change in the order of integration.

Example: $\int_0^{\infty}\int_x^{\infty}\frac{e^{-y}}{y}dydx$ = difficult to solve

Changing the order we get

$$= \int_0^{\infty}\int_x^{\infty}\frac{e^{-y}}{y}dydx = \int_0^{\infty}\frac{e^{-y}}{y}[x]_0^{y}dy = \int_0^{\infty}\frac{e^{-y}}{y}ydy = \int_0^{\infty}e^{-y}dy = \left[\frac{e^{-y}}{y}\right]_0^{\infty} = 1-\left(\frac{1}{-1}\right) = 1$$

33. Sketch roughly the region of integration for the following double integral $\int_0^{a}\int_0^{\sqrt{a^2-x^2}} f(x,y)dxdy$

Solution: $\int_0^{a}\int_0^{\sqrt{a^2-x^2}} f(x,y)dxdy$ is correct form

Here x varies from x=0 to x=a

And y varies from y=0 to y= $\sqrt{a^2-x^2}$

$y^2 = a^2 - x^2 = x^2 + y^2$

34. Express the volume bounded by $x \geq 0, y \geq 0, z \geq 0$ **and** $x^2 + y^2 + z^2 \leq 1$ **in triple integration.**

Solution: $v=\int_0^1\int_0^{\sqrt{1-x^2}}\int_0^{\sqrt{1-x^2-y^2}} dzdydx$

35.Express the area enclosed by $x, y \geq 0$ **and** $x^2+y^2=4$ **as double integral.**

Solution:

$A=\int_0^2\int_0^{\sqrt{4-y^2}} dxdy$

36. Shade the region of the integration $\int_0^2\int_0^y dxdy$

Solution: Given y varies from y=0 to y=2 and x varies from x=0 to x=y

37. Evaluate $\int_0^1\int_1^2 \left(y^2+x^2\right)dxdy$

Solution: Given:

$$\int_0^1\int_1^2 \left(y^2+x^2\right)dxdy=\int_0^1\left[y^2x+\frac{x^3}{3}\right]_{x=1}^{x=2}dy=\int_0^1\left[\left(2y^2+\frac{8}{3}\right)-\left(y^2+\frac{1}{3}\right)\right]dy=\int_0^1\left(2y^2+\frac{8}{3}-y^2-\frac{1}{3}\right)dy=\int_0^1\left(y^2+\frac{7}{3}\right)dy=\frac{8}{3}$$

38. If $v=\int_0^1\int_0^{\sqrt{1-x^2}}\int_0^{\sqrt{1-x^2-y^2}} dzdydx$**. Point out the region of integration.**

Solution: Given Here x varies from x=0 to x=1

And y varies from y=0 to y=$\sqrt{1-x^2}$ $y^2=1-x^2 \Rightarrow x^2+y^2=1$

Z varies from z=0 to $z=\sqrt{1-x^2-y^2} \Rightarrow z^2=1-x^2-y^2 \Rightarrow x^2+y^2+z^2=1$

39. Evaluate $\int_0^1\int_0^x dxdy$

Solution: Given: $\int_0^1\int_0^x dxdy=\int_0^1[y]_{y=0}^{y=x}dx=\int_0^1[x-0]dx=\int_0^1 xdx\left[\frac{x^2}{2}\right]_0^1=\left[\frac{1}{2}-0\right]=\frac{1}{2}$

40. Change the order of integration $\int_0^\infty\int_x^\infty \frac{e^{-y}}{y}dydx$**. (c-195)**

Solution: The region of integration is bounded by x=0,x=∞ ,y=0 and y=∞

Here x varies from x=0 to x= ∞

And y varies from y=0 to y= ∞

Change the order of integration we get,

$$=\int_0^{\infty}\int_x^{\infty}\frac{e^{-y}}{y}dxdy$$

41. Evaluate $\int_a^b\int_c^d\int_f^g xy^2 zdzdydx$

Solution: Let $I=\int_a^b\int_c^d\int_f^g xy^2 zdzdydx=\left[\int_0^2 xdx\right]\left[\int_1^3 y^2dy\right]\left[\int_1^2 zdz\right]=\left[\frac{x^2}{2}\right]_0^2\left[\frac{y^3}{3}\right]_1^3\left[\frac{z^2}{2}\right]_1^2=(2)\left(\frac{26}{3}\right)\left(\frac{3}{2}\right)=26$

42. Evaluate $\int_1^2\int_3^4 dydx$

Solution:

$$\int_1^2\int_3^4 dydx=\int_1^2 x\left[\frac{y^2}{2}\right]_{y=3}^{y=4}dx=\int_1^2 x\left[8-\frac{9}{2}\right]dx=\int_1^2 x\left(\frac{7}{2}\right)dx=\frac{7}{2}\left[\frac{x^2}{2}\right]_1^2=\frac{21}{4}$$

43. Evaluate $\int_1^3\int_3^4\int_1^4 xyzdzdydx$

Solution:

$$\int_1^3\int_3^4\int_1^4 xyzdzdydx=\left[\int_1^3 xdx\right]\left[\int_3^4 ydy\right]\left[\int_1^4 zdz\right]=\left[\frac{x^2}{2}\right]_1^3\left[\frac{y^2}{2}\right]_3^4\left[\frac{z^2}{2}\right]_1^4=\frac{1}{8}\left[x^2\right]_1^3\left[y^2\right]_3^4\left[z^2\right]_1^4=\frac{1}{8}[9-1][16-9][16-1]=105$$ **44.**

Evaluate $\int_0^3\int_0^2 xy(x+y)dydx$

Solution:

$$\int_0^3\int_0^2 xy(x+y)dydx=\int_0^3\int_0^2\left[x^2y+xy^2\right]dydx=\int_0^3\left[\frac{x^3}{3}y+\frac{x^2}{2}y^2\right]_{x=0}^{x=2}dy=\int_0^3\left[\frac{8}{3}y+2y^2\right]dy$$

$$=\left[\frac{8}{3}\frac{y^2}{2}+\frac{2y^3}{3}\right]_0^3=12+18=30$$

45. Change the order of integration in $\int_0^1\int_0^{2\sqrt{x}} x^2dxdy$ **(c-320)**

Solution: Given $\int_0^1 \int_0^{2\sqrt{x}} x^2 dxdy = \int_0^1 \int_0^{2\sqrt{x}} x^2 dydx$

Here x varies from x=0 to x=1

And y varies from y=0 to $y=2\sqrt{x}$ => $y^2=4x$

Changing the order of integration is nothing but to change the vertical path into horizontal path and then to change the vertical strip PQ into horizontal strip RS.

Here

$$\int_0^1 \int_0^{2\sqrt{x}} x^2 dydx = \int_0^2 \int_{\frac{y^2}{4}}^1 x^2 dxdy$$

46. Transform the integration $\int_0^\infty \int_0^\infty x^2 dxdy$ to polar coordinates.

Solution: Given: $\int_0^\infty \int_0^\infty x^2 dxdy$

Let us transform this integral in polar coordinates by taking $x = r\cos\theta, y = r\sin\theta dxdy = rd\theta dr$

$$I = \int_0^{\frac{\pi}{2}} \int_0^\infty r^2 \cos^2\theta r d\theta dr = \int_0^{\frac{\pi}{2}} \int_0^\infty r^3 \cos^2\theta d\theta dr$$

47. Evaluate $\int_0^1 \int_0^3 \int_0^4 ydzdydx$

Solution: $\int_0^1 \int_0^3 \int_0^4 ydzdydx = \left[\int_0^1 dz\right]\left[\int_0^3 ydy\right]\left[\int_0^4 dx\right] = [z]_0^1 \left[\frac{y^2}{2}\right]_0^3 [dx]_0^4 = [1-0]\left[\frac{9}{2}-0\right][4-0] = 18$

48. Evaluate $\int_0^\pi \int_0^{\cos\theta} rdrd\theta.$

Solution: Given: $\int_0^\pi \int_0^{\cos\theta} rdrd\theta. =$

$$\int_0^\pi \left[\frac{r^2}{2}\right]_{r=0}^{r=\cos\theta} d\theta = \int_0^\pi \left[\frac{\cos^2\theta}{2}-0\right] d\theta = \frac{1}{2}\int_0^\pi \cos^2\theta d\theta = \frac{1}{2}\int_0^\pi \left[\frac{1+\cos 2\theta}{2}\right] d\theta = \frac{1}{4}\left[\theta + \frac{\sin 2\theta}{2}\right]_0^\pi = \frac{\pi}{4}$$

49. Find the area of circle of radius 'a' by double integration in polar coordinates.

Solution: The equation of circle with pole on the circle and diameter through the point as initial line is $r = 2a\cos\theta$ Area= $2 \times upper$ area$=2\int_0^{\frac{\pi}{2}} \int_0^{2a\cos\theta} rdrd\theta = \int_0^{\frac{\pi}{2}} \left(r^2\right)_0^{2a\cos\theta} d\theta = 4a^2 \int_0^{\frac{\pi}{2}} \cos^2\theta d\theta = 4a^2 \frac{1}{2}\frac{\pi}{2} = \pi a^2$

50. Evaluate $\int_0^1 \int_x^1 \sin\left(y^2\right) dydx$

Solution: Given: I= $\int_0^1 \int_x^1 \sin\left(y^2\right) dydx$ Changing the order we get

$$I = \int_0^1 \int_x^1 \sin\left(y^2\right) dydx = \int_0^1 \left(\sin^2 y\right)[x]_{x=0}^{x=y} dy = \int_0^1 \left(\sin^2 y\right) ydy = \int_0^1 \sin t \frac{dt}{2} = \frac{1}{2}[-\cos t]_0^1 = -\frac{1}{2}[\cos t]_0^1 = \frac{1-\cos 1}{2}$$

PART -B

1.Change the order of integration and evaluate $\int_1^2 \int_0^{4-x^2} (x+y)dydx.$

2.Evaluate $\iint xy(x+y)dxdy$ over the area between $y=x, y=x^2.$

3.Evaluate $\int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^{\sqrt{a^2-y^2-x^2}} xdxdydz$, transforming the variable.

4.Change the order of integration and evaluate $\int_0^1 \int_x^{\sqrt{2-x^2}} \frac{x}{\sqrt{x^2+y^2}} dxdy.$

5.Change the order of integration and evaluate $\int_0^a \int_x^a (x^2+y^2)dxdy.$

6. Change the order of integration and evaluate $\int_0^{4a} \int_{x^2/4a}^{2\sqrt{ax}} xydydx.$

7.Evaluate $\iint x^2 ydxdy$ over the first quadrant $x^2+y^2=1$.

8. Find the area of the cardiod $r=a(1+\cos\theta).$

9. Find the volume of the sphere $x^2+y^2+z^2=a^2$ using triple integrals.

10. By changing to polar co-ordinates, evaluate $\int_0^{2a} \int_0^{\sqrt{2ax-x^2}} (x^2+y^2)dxdy.$

11. (i) Find the area inside the circle $r=a\sin\theta$ but lying outside the cardiod $r=a(1-\cos\theta)$

(ii) Evaluate $\int_0^a \int_0^b \int_0^c \left(x^2+y^2+z^2\right) dxdydz$

12. (i) Change to the special polar co- ordinates and hence evaluate $\iiint_V \frac{1}{x^2+y^2+z^2} dxdydz$, where V is the volume of the sphere $x^2+y^2+z^2=a^2$.

(ii)Change the order of integration and hence evaluate $\int_1^3 \int_{y=0}^{\frac{6}{x}} x^2 dydx$.

13. (i) Change the order of integration in $\int_0^a \int_{a-y}^{\sqrt{a^2-x^2}} ydxdy$

(ii) Express $\int_0^a \int_y^a \frac{x^2 dxdy}{\left(x^2+y^2\right)^{\frac{3}{2}}}$ in polar coordinates and then evaluate it.

14.(i)Evaluate $\int_C \left[\left(3xy^2+y^3\right)dx+\left(x^3+3xy^2\right)dy\right]$ where Cis the parabola $y^2=4ax$ from (0,0),(a,2a).

(ii) Evaluate $\int_0^{\log 2} \int_0^x e^{x+y+z} dxdydz$

15. (i) Evaluate $\int_0^\infty \int_0^\infty e^{-\left(x^2+y^2\right)} dxdy$ using the polar coordinates.

(ii) Change the order of integration in the integral $\int_0^a \int_{x^2}^{2-x} xydxdy$

16.(i)Find the double integration, the area enclosed by the curves $y^2=4ax$ and $x^2=4ay$

(ii) Find the volume of the ellipsoid $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$

17.(i) Change the order of integration in the integral $\int_0^a \int_{\frac{x^2}{a}}^{2a-x} xydydx$ and hence evaluate it.

(ii) By transforming into polar coordinates, evaluate $\iint \left(\frac{x^2y^2}{x^2+y^2}\right) dxdy$ over annular region

18. (i) Find the value of $\int\int\int xyzdxdydz$ through the positive spherical octant for which $x^2+y^2+z^2 \le a^2$

(ii) Find the volume of thetetrahedran bounded by the plane $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$ and the coordinate plane x=0, y=0, z=0

19(i) Evaluate $\int\int(x-y)dxdy$ over the region between the line =x and the parabola $y=x^2$.

(ii)Evaluate $\int\int\int x^2yzdxdydz$ taken over the tetrahedron bounded by the planes x=0,y=0,z=0 and $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$

20.(i)Change the order of integration $\int_0^a \int_{a-\sqrt{a^2-y^2}}^{a+\sqrt{a^2+y^2}} xydxdy$ and evaluate it.

(ii) A circular hole of radius b is made centrally through a sphere of radius a. Find the volume of the remainingsphere.

21. Transform into polar coordinates and evaluate $\int_0^2 \int_0^{\sqrt{2x-x^2}} \frac{xdydx}{x^2+y^2}$

22. Change the order of integration and evaluate $\int_0^4 \int_{x^2/4}^{2\sqrt{x}} xydydx.$

23. Evaluate $\int_0^1 \int_0^{1-x} \int_0^{1-x-y} xydxdydz$

24. Find the volume of the tetrahedron bounded by the plane x+y+z=1 the coordinate planes.

25. Express the volume of the sphere $x^2+y^2+z^2=a^2$ as a volume integral and hence evaluate it.

PART-A

1. Find the particular integral of $(D^2+2D+1)y=e^{-x}\cos x$. **(N/D 2009),(M/J 2009)**

Solution:

$$P.I=\frac{1}{(D+1)^2}e^{-x}\cos x=e^{-x}\frac{1}{(D-1+1)^2}\cos x$$

$$=e^{-x}\frac{1}{(D)^2}\cos x=e^{-x}\frac{1}{D}\left[\int\cos x dx\right]$$

2. Solve $(D^2+D+1)y=0$. **(N/D 2009),(M/J 2009)(M/J 2016)**

Soln:

$A.E: m^2+m+1=0$

$$\Rightarrow m=\frac{-1\pm\sqrt{1-4}}{2}$$

$$=\frac{-1\pm\sqrt{-3}}{2}=\frac{-1\pm i\sqrt{3}}{2}=-\frac{1}{2}\pm\frac{\sqrt{3}}{2}i$$

C.F $= e^{-\frac{1}{2}x}\left(A\cos\frac{\sqrt{3}}{2}x+B\sin\frac{\sqrt{3}}{2}x\right)$

$\therefore$ y$= e^{-\frac{1}{2}x}\left(A\cos\frac{\sqrt{3}}{2}x+B\sin\frac{\sqrt{3}}{2}x\right)$.

3. Solve the equation $x^2y''-xy'+y=0$ **(M/J 2009)**

Solution:

Given: $x^2y''-xy'+y=0$ …….(1)

Put $x=e^z$, log x=z, $xD=D'$, $x^2D^2=D'(D'-1)$

(1) $\Rightarrow (D'(D'-1)-D'+1)y=0$

$\Rightarrow (D'^2-2D'+1)y=0$

A.E: $m^2-2m+1=0$ => m=1,1; y= C.F=$(Ax+B)e^x$

4. Find the particular integral of $(D^2+1)y=\sin x$ **(M/J 2010)**

Solution:

P.I=$\frac{1}{D^2+1}\sin x=\frac{1}{-1+1}\sin x$ [Replace D^2 by -1^2, ordinary rule fails]

$$=x\frac{1}{2D}\sin x=\frac{x}{2}\frac{1}{D}\sin x$$

$$=\frac{x}{2}\int\sin x dx=\frac{x}{2}(-\cos x)=-\frac{x\cos x}{2}$$

5. Solve the equation $(D^2-6D+13)y=0$ **(N/D 2009)**

Solution:

Given :$(D^2-6D+13)y=0$

$$A.E: \quad m^2 - 6m + 13 = 0 \quad \Rightarrow m = \frac{6 \pm \sqrt{36 - 4(1)(13)}}{2}$$

$$= \frac{6 \pm \sqrt{36 - 52}}{2} = \frac{6 \pm \sqrt{-16}}{2} = \frac{6 \pm 4i}{2} = 3 \pm 2i$$

C.F = $e^{3x}(A\cos 2x + B\sin 2x)$

6. Transforms the equation $x^2y'' + xy' = x$ into a linear differential equation with constant coefficients(M/J 2010),(A/M 2011),(N/D 2011)

Solution:

Given: $x^2y'' + xy' = x$

$\Rightarrow (x^2D^2 + xD)y = x$ ……..(1)

Put $x = e^z$, log x=z, $xD = D'$, $x^2D^2 = D'(D' - 1)$

(1)=> $[D'(D' - 1) + D']y = e^z$

=> $[D'^2 - D' + D']y = e^z$

=> $D'^2 y = e^z$

7. Find the particular integral of $(D^2 - 2D + 2)y = e^x \cos x$ (N/D 2010)

Solution:

P.I=$\frac{1}{D^2-2D+2} e^x \cos x = e^x \frac{1}{(D+1)^2-2(D+1)+2} \cos x$ [Replace D by D+1]

$= e^x \frac{1}{D^2+2D+1-2D-2+2} \cos x = e^x \frac{1}{D^2+1} \cos x$

$= e^x \frac{1}{-1+1} \cos x$ [Replace $D^2 by - 1^2$]

$= e^x x \frac{1}{2D} \cos x$[Ordinary rule fails]

$= x\frac{e^x}{2} \frac{1}{D} \cos x = x\frac{e^x \sin x}{2}$

8. Reduce the equation $(x^2D^2 + xD + 1)y = \log x$ into an ordinary differential equation with constant coefficients (N/D 2010)

Solution:

Given: $(x^2D^2 + xD + 1)y = x$ ……..(1)

Put $x = e^z$, log x=z, $xD = D'$, $x^2D^2 = D'(D' - 1)$

(1)=> $[D'(D' - 1) + D' + 1]y = z$

=> $[D'^2 - D' + D'^+ + 1]y = z$

=> $[D'^2 + 1]y = z$

9. Find the particular integral of $(D^2 - 4)y = \cosh 2x$ (A/M 2011)

Solution:

Given: $(D^2 - 4)y = \cos hx = \frac{e^{2x} + e^{-2x}}{2} = \frac{1}{2}e^{2x} + \frac{1}{2}e^{-2x}$

P.I1=$\frac{1}{D^2-4}\frac{1}{2}e^{2x} = \frac{1}{2}\frac{1}{D^2-4}e^{2x}$

$=\frac{1}{2}\frac{1}{4-4}e^{2x}$ [Replace D by 2]

$=\frac{1}{2}\frac{1}{0}e^{2x}$ [Ordinary rule fails]

$=\frac{1}{2}x\frac{1}{2D}e^{2x} = \frac{1}{4}x\frac{1}{D}e^{2x}$ $=\frac{x}{4}\frac{e^{2x}}{2} = \frac{x}{8}e^{2x}$ [Replace D by 2]

P.I2=$\frac{1}{D^2-4}\frac{1}{2}e^{-2x} = \frac{1}{2}\frac{1}{D^2-4}e^{-2x}$

$=\frac{1}{2}\frac{1}{4-4}e^{-2x}$ [Replace D by -2]

$=\frac{1}{2}\frac{1}{0}e^{-2x}$ [Ordinary rule fails]

$=\frac{1}{2}x\frac{1}{2D}e^{-2x} = \frac{1}{4}x\frac{1}{D}e^{-2x}$

$=\frac{x}{4}\frac{e^{-2x}}{-2} = \frac{-x}{8}e^{-2x}$ [Replace D by -2]

P.I=P.I1+P.I2=$\frac{x}{8}e^{2x}-\frac{x}{8}e^{-2x}$

10. Find the particular integral of $(D^2+4)y=\sin 2x$ (N/D 2011)

Solution:

$$P.I=\frac{1}{(D)^2+4}\sin 2x$$ [Replace D^2 by -2^2]

$=\frac{1}{-4+4}\sin 2x$ [Ordinary rule fails]

$=x\frac{1}{2D}\sin 2x = \frac{x}{2}\int \sin 2x\,dx$

$=\frac{x}{2}\frac{-\cos 2x}{2}=\frac{-x}{4}\cos 2x$

11. Find the particular integral of $(D-1)^2 y=e^x \sin x$ (M/J2012)

Solution: P.I=$\frac{1}{(D-1)^2}e^x \sin x = e^x\frac{1}{(D+1-1)^2}\sin x$ [Replace D by D+1]

$= e^x\frac{1}{D^2}\sin x = e^x\frac{1}{D}\cos x = -e^x \sin x$

12. Transforms the differential equation $(2x+3)^2\frac{d^2y}{dx^2}-2(2x+3)\frac{dy}{dx}-12y=6x$ to a differential equation with constant Coefficients. (M/J 2012)

Solution:

Given: $(2x+3)^2\frac{d^2y}{dx^2}-2(2x+3)\frac{dy}{dx}-12y=6x$

$[(2x+3)^2D^2-2(2X+3)D-12]y=6x$ .....(1)

$Put \qquad 2x+3=e^z, \qquad \log(2x+3)=z$

=>2x=$e^z-3 => x=\frac{e^z}{2}-\frac{3}{2}$

Let (2x+3)D=$2D'$

$(2x+3)^2D^2=2^2D'(D'-1)$=$4D'(D'-1)$

(1)=> $[4D'(D'-1)-4D'-12]Y=6[\frac{e^z}{2}-\frac{3}{2}]$

$=>[4D'^2 - 4D' - 4D' - 12]y = 3e^z - 9$

$=>[4D'^2 - 8D' - 12]y = 3e^z - 9$

$=>[D'^2 - 2D' - 3]y = \frac{3}{4}e^z - \frac{9}{4}$

13. Find the particular integral of $(D^2 - 4D + 4)y = 2^x$ (N/D 2012)

Solution: $(D-2)^2 y = 2^x = e^{\log 2^x} = e^{x\log 2} = e^{(\log 2)x}$

$P.I = \frac{1}{(D-2)^2} e^{(\log 2)x} = \frac{1}{(\log 2 - 2)^2} e^{(\log 2)x} = \frac{2^X}{(\log 2 - 2)^2}$ [Replace D by log 2]

14. Find the particular integral of $(D^2 - 2D + 1)y = \cosh x$ (M/J 2013)

Solution:

$(D^2 - 2D + 1)y = \cosh x = \frac{e^x + e^{-x}}{2} = \frac{1}{2}e^x + \frac{1}{2}e^{-x}$

P.I1$= \frac{1}{D^2 - 2D + 1}(\frac{1}{2})e^x = \frac{1}{2}\frac{1}{1-2+1}e^x$

$= \frac{1}{2}\frac{1}{0}e^x$ [Ordinary rule fails]

$= \frac{1}{2}x\frac{1}{2D-2}e^x = \frac{1}{2}x\frac{1}{2-2}e^x$ [0rdinary rule fails]

$= \frac{x^2}{2}\frac{1}{2}e^x$

P.I2$= \frac{1}{D^2 - 2D + 1}(\frac{1}{2})e^{-x} = \frac{1}{2}\frac{1}{1+2+1}e^{-x}$

$= \frac{1}{8}e^{-x}$

P.I$= P.I_1 + P.I_2$

$= \frac{x^2}{4}e^x + \frac{1}{8}e^{-x}$

15. Solve $(x^2D^2 + 4xD + 2)y = 0$ (M/J 2013)

Solution:

Given: $(x^2D^2 + 4xD + 2)y = 0$ .....(1)

Put $x = e^z$, log x= z, $xD = D'$, $x^2D^2 = D'(D'-1)$

(1) => $[D'(D'-1) + 4D' + 2]y = 0$

=> $[D'^2 - D' + 4D' + 2]y = 0$

=>$[D'^2 + 3D' + 2]y = 0$

A.E: $m^2 + 3m + 2 = 0$

=>(m+1)(m+2)=0

=>m=-1,m=-2

y= C.F$= Ae^{-z} + Be^{-2z}$

$= Ae^{-\log x} + Be^{-2\log x}$

$= Ax^{-1} + Bx^{-2}$

16. Solve the equation $(D^2 - 4)y = 1$ (N/D 2013)

Solution:

Given:$(D^2-4)y=1e^{0x}$

A.E: $m^2-4=0$ => m=± 2

=>C.F=A$e^{-2x}+Be^{2x}$

P.I=$\frac{1}{D^2-4}1e^{0x}=-\frac{1}{4}e^{0x}$

Solution y=C.F+P.I

=A$e^{-2x}+Be^{2x}-\frac{1}{4}e^{0x}$

17. Transforms the differential equation $\left(3x^2D^2+5xD+7\right)y=\frac{2}{x\log x}$ to a differential

equation with constant Coefficients (N/D 2013)

Solution:

Given:$(3x^2D^2+5xD+7)y=\frac{2}{x\log x}$.....(1)

Put $x=e^z$, log x= z, $xD=D'$, $x^2D^2=D'(D'-1)$

(1) => $[3D'(D'-1)+5D+7]y=\frac{2}{ze^z}$

=> $[3D'^2-3D'+5D+7]y=\frac{2}{ze^z}$

=>$[3D'^2+2D'+7]y=\frac{2}{ze^z}$

18. Find the particular integral of $\left(D^2-9\right)y=e^{-3x}$ (M/J 2014)

Solution:$(D^2-9)y=e^{-3x}$

P.I=$\frac{1}{D^2-9}e^{-3x}$=$\frac{1}{(-3)^2-9}e^{-3x}$ [Replace D by -3]

=$\frac{1}{0}e^{-3x}$ [Ordinary rule fails]

=x$\frac{1}{2D}e^{-3x}$

=$\frac{x}{2}\frac{1}{(-3)}e^{-3x}$ [Replace D by -3]

=$\frac{-x}{6}e^{-3x}$

19. Solve $\frac{d^2y}{dx^2}+2\frac{dy}{dx}+y=0$ (M/J 2014)

Solution:

Given:$(D^2+2D+1)y=0$

A.E: $m^2+2m+1=0$ =>$(m+1)^2=0$

=>m=-1, -1

Solution y=(Ax+B) e^{-x}

20. Find the particular integral of $(D^2+2D+2)y=e^{-x}\sin 2x$.(N/D 2014)

Solution:

P.I=$\frac{1}{D^2+2D+2}e^{-x}\sin 2x$= $e^{-x}\frac{1}{(D-1)^2+2(D-1)+2}\sin 2x$ [Replace D by D-1]

=$e^{-x}\frac{1}{D^2-2D+1+2D-2+2}\sin 2x=e^{-x}\frac{1}{D^2+1}\sin 2x$

$= e^{-x}\frac{1}{-4+1}\sin 2x$ [Replace $D^2 by - 2^2$]

$= \frac{-1}{3}e^{-x}\sin 2x$

21. Solve $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} = 0$ **(N/D 2014)**

Solution:

Given: $(x^2D^2 + xD)y = 0$ ……..(1)

Put $x = e^z$, log x= z, $xD = D'$, $x^2D^2 = D'(D'-1)$

(1) => $[D'(D'-1) + D']y = 0$

=> $[D'^2 - D' + D']y = 0$

=> $[D'^2 y = 0$

A.E: $m^2 = 0$

The general solution is y= C.F=$(Az + B)e^{0z}$

=Az+B=$A\log x + B$

22. Solve $(D^2 - 2D + 5)y = 0$**(M/J 2015)**

Solution:

A.E:$m^2 - 2m + 5$=0

=>m=$\frac{2\pm\sqrt{4-20}}{2}$= $\frac{2\pm\sqrt{-16}}{2} = \frac{2\pm i4}{2}$=1$\pm$2i

y=C.F=$e^x(A\cos 2x + B\sin 2x)$

23. Find the particular integral of$(D^3 + 4D)y = \sin 2x$**(M/J 2015)**

Solution:

P.I=$\frac{1}{D^3+4D}\sin 2x$= $\frac{1}{-4D+4D}\sin 2x$ [Replace D^2 by -4]

=$\frac{1}{0}\sin 2x = x\frac{1}{3D^2+4}\sin 2x$ [Replace D^2 by -4]

=$x\frac{1}{-12+4}\sin 2x$

=$\frac{-x}{8}\sin 2x$

24. If $1 \pm 2i$, $1 \pm 2i$ areb the roots of the auxiliary equation corresponding to a fourth order homogeneous linear differential equation $F(D) = 0$, find its solution.**(M/J 2016)**

Solution:

Given the root are $1 \pm 2i$, $1 \pm 2i$ for $F(D) = 0$

=>$y = e^x(Acos2x + Bsin2x) + e^x(Ccos2x + Dsin2x)$.

25. Find the particular integral of $\left(D^2 - 4D\right)y = e^x x$ **(N/D 2015)**

Solution:

P.I=$\frac{1}{D^2-4D}e^x x$= $e^x\frac{1}{(D+1)^2-4(D+1)}x$ [Replace D by D+1]

$$=e^x \frac{1}{D^2-2D-3} x = e^x \frac{1}{-3(1-\frac{D^2-2D}{3})} x$$

$$=-\frac{e^x}{3}(1-\frac{D^2-2D}{3})^{-1}$$

$$= -\frac{e^x}{3}(x-\frac{2}{3})$$

26. Transform $(x^2y''' - 3xy'' = \frac{\sin(\log x)}{x}$ in to a differential equations with constant coefficients. .

(N/D 2015)

Solution:

Given $(x^2y''' - 3xy'' = \frac{\sin(\log x)}{x}$

Multiply by x,

$(x^3y''' - 3x^2y'' = \sin(\log x)$

Put $x = e^z$, x=log z

$[D'(D'-1)(D'-2) - 3D'(D'-1)]y = \sin z.$

PART-B
ODE WITH CONSTANT COEFFICIENTS:

1. Solve the equation $(D^2 - 3D + 2)y = 2\cos(2x+3) + 2e^x$ **(N/D 2009)**
2. Solve $(D^2 + 16)y = cos^3x$ **(N/D 2010)**
3. Solve $(D^2 - 4D + 3)y = \cos 2x + 2x^2$ **(M/J 2014)**
4. Solve $(D^2 + 3D + 2)y = \sin x + x^2$ **(M/J 2011)**
5. Solve $(D^2 + 5D + 4)y = e^{-x}\sin 2x$ **(A/M 2011),(N/D 2012)**
6. Solve $(D^2 + 4D + 3)y = e^{-x}\sin x$ **(M/J 2010)**
7. Solve $(D^2 - 4D + 3)y = e^x \cos 2x$ **(M/J 2012)**
8. Solve $(D^2 + 4D + 3)y = 6e^{-2x}\sin x \sin 2x$ **(N/D 2011)**
9. Solve $(D^2 - 3D + 2)y = xe^{3x} + \sin 2x$ **(M/J 2015)**
10. Solve $(D^2 + 2D + 5)y = e^{-x}x^2$ **(N/D 2014)**
11. Solve $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = 8xe^x \sin x$ **(N/D 2013)**
12. Solve $(D^2 + 4)y = x^2\cos 2x$ **(M/J 2009),(N/D 2011)**
13. Solve $(D^2 + 2D + 1)y = xe^{-x}cosx$ **(M/J 2016)**
14. Solve $(D^3 - 2D^2 + 4D - 8)y = 6e^{2x} + \sin x \sin 2x$ **(N/D 2015)**

METHOD OF VARIATION OF PARAMETERS

1. Solve $(D^2 + a^2)y =$ tan ax by the method of variation of parameters **(M/J 2009)**

2. Solve $\frac{d^2y}{dx^2} + a^2y = \tan ax$ by the method of variation of parameters **(M/J 2011), (M/J 2014)**
3. Solve $\frac{d^2y}{dx^2} + 4y =$ tan 2x by the method of variation of parameters **(N/D 2014)**
4. Solve $2\frac{d^2y}{dx^2} + 8y = \tan 2x$ by the method of variation of parameters **(N/D 2013)**
5. Solve $(D^2 + 4)y =$ cot 2x by the method of variation of parameters **(N/D 2009), (N/D 2015)**
(N/D 2011)
6. Solve $(D^2 + a^2)y =$ sec ax by the method of variation of parameters **(M/J 2012)**
7. Solve $\frac{d^2y}{dx^2} + y =$ cosec x by the method of variation of parameters **(A/M 2011), (N/D 2012)**
8. Solve $(D^2 + 1)y =$ x sinx by the method of variation of parameters **(M/J 2010)**
9. Using variation of parameters , solve $(2D^2 - D - 3)y = 25e^{-x}$**(N/D 2011)**
10. Solve $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = \frac{e^{-x}}{x^2}$ by the method of variation of parameters **(M/J 2013)**
11. Solveby the method of variation of parameters , $y'' - 2y' + y = e^x \log x$**(M/J 2015)**

CAUCHY AND LEGENDRE EQUATIONS

1. Solve$x^2\frac{d^2y}{dx^2} + 4x\frac{dy}{dx} + 2y = x^2 + \frac{1}{x^2}$**(M/J 2013)**
2. Solve $(x^2D^2 - xD + 1)y = \sin(\log x)$ **(N/D 2014)**

3. Solve $(x^2D^2 + 3xD + 5)y = x\cos(\log x)$**(M/J 2009)**
4. Solve $(x^2D^2 - 3xD + 4)y = x^2\cos(\log x)$**(N/D 2010)**
5. Solve $(x^2D^2 - xD + 4)y = x^2\sin(\log x)$ **(M/J 2012),(N/D 2009)**
6. Solve $(x^2D^2 - 2xD - 4)y = x^2 + 2\log x$**(M/J 2010)**
7. Solve $(x^2D^2 - xD + 1)y = \log x + \pi$**(M/J 2015)**
8. Solve $(x^2D^2 - xD + 1)y = (\frac{\log x}{x})^2$**(M/J 2014)**
9. Solve $\frac{d^2y}{dx^2} + \frac{1}{x}\frac{dy}{dx} + y = \frac{12\log x}{x^2}$ **(N/D 2012)**
10. Solve $x^2\frac{d^2y}{dx^2} - 3x\frac{dy}{dx} + 4y = x^2 \ln x$ **(N/D 2011)**
11. Solve $x^2\frac{d^2y}{dx^2} + 4x\frac{dy}{dx} + y = e^{e^{\log x}}$ **(N/D 2013)**
12. Solve $(1+x)^2\frac{d^2y}{dx^2} + (1+x)\frac{dy}{dx} + y = 2\sin[\log(1+x)]$ **(A/M 2011)**
13. Solve $(1+x)^2\frac{d^2y}{dx^2} + (1+x)\frac{dy}{dx} + y = 2\,4\cos[\log(1+x)]$**(N/D 2011)**
14. Solve $(3x+2)^2\frac{d^2y}{dx^2} + 3(3x+2) - 36y = 3x^2 + 4x + 1$**(M/J 2013)**
15. Solve $x^2y'' - 4xy' + 6y = x^2 + \log x$. **(N/D 2015)**
16. Solve $(x^2D^2 - xD - 2)y = x^2 log x$**(MAY/JUNE-2016)**

SIMULTANEOUS DIFFERENTIAL EQUATIONS

1. Solve $\frac{dx}{dt} + y =$ sin t, x+ $\frac{dy}{dt}$=cos t given x=2 and y=0 at t=0 **(M/J 2009) (N/D 2015)**
2. Solve $\frac{dx}{dt} + 2y =$ sin2t, $\frac{dy}{dt} - 2x$=cos 2t **(M/J 2012),(N/D 2009)**
3. Solve $\frac{dx}{dt} + 2y =$ -sint, $\frac{dy}{dt} - 2x$=cos t **(M/J 2014)**
4. Solve $\frac{dx}{dt} + 2y =$ -sint, $\frac{dy}{dt} - 2x$=cos t given x=1 y=0 at t=0 **(N/D 2010)**
5. Solve $\frac{dx}{dt} - y =$ t, $\frac{dy}{dt} + x = t^2$ **(A/M 2011) (MAY/JUNE-2016)**
6. Solve $\frac{dx}{dt} - y =$ t, $\frac{dy}{dt} + x = t^2$ given x(0)=y(0)=2 **(N/D 2011)**
7. Solve $\frac{dx}{dt} + y = e^t$, $x - \frac{dy}{dt}$=t **(N/D 2012),(N/D 2014)**
8. Solve $\frac{dx}{dt} + 2x + 3y = 2e^{2t}$, $\frac{dy}{dt} + 3x + 2y$=0 **(M/J 2010)**
9. Solve $\frac{dx}{dt} + 5x - 2y =$ t, $\frac{dy}{dt} + 2x + y$=0 **(M/J 2013)**
10. Solve $\frac{dx}{dt} + 2x - 3y =$ t, $\frac{dy}{dt} - 3x + 2y = e^{2t}$ **(N/D 2011)**
11. Solve $\frac{dx}{dt} + 4x + 3y =$ t, $\frac{dy}{dt} + 2x + 5y = e^{2t}$ **(N/D 2013)**
12. Solve the simultaneous equations $\frac{dx}{dt} + \frac{dy}{dt} + 3x =$ sin t, $\frac{dx}{dt} - x + y = \cos t$ **(M/J 2015)**
13. Solve for x from the equations $D^2x + y = 3e^{2t}$, Dx-Dy=$3e^{2t}$ **(M/J 2011)**

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2 MARKS QUESTION WITH ANSWERS

UNIT I- PROPERTIES OF MATTER

1. What is elasticity?

The property of the body to regain its original shape and size, after the removal of deforming force is called elasticity.

2. Define stress and strain.

Stress: The force of resistance per unit area, offered by a body against deformation is known as stress.

Strain: The ratio of change in dimension to the original dimension when subjected to an external load is termed as strain and is denoted by e. It has no unit.

3. Identify tensile stress and tensile strain.

The stress induced in a body, when subjected to two equal and opposite pulls, as a result of which there is an increase in length, is known as tensile stress. The ratio of increase in length to the original length is known as tensile strain.

4. Define compressive stress and compressive strain.

The stress induced in a body, when subjected to two equal and opposite pushes, as a result of which there is a decrease in length, is known as compressive stress. The ratio of increase in length to the original length is known as compressive strain.

5. Define shear stress and shear strain.

The stress induced in a body, when subjected to two equal and opposite forces, which are acting tangentially across the resisting section as a result of which the body tends to shear off across the section is known as shear stress and corresponding strain is known as shear strain.

6. Define Poisson‘s ratio.

The ratio of lateral strain to the linear strain is a constant for a given material, when the material is stressed within the elastic limit. This ratio is Poisson‘s ratio.

7. State Hooke‘s law.

Hooke‘s law is stated as when a material is loaded within elastic limit, the stress is proportional to the strain produced by stress, or Stress/strain=constant. This constant is termed as modulus of elasticity.

8. Define modulus of rigidity.

The ratio of shear stress to the corresponding shear strain when the stress is within the elastic limit is known as modulus of rigidity or shear modulus and is denoted by N.

9. Define modulus of elasticity.

The ratio of tensile stress or compressive stress to the corresponding strain is known as modulus of elasticity or young‘s modulus and is denoted by E.

10. Define Bulk modulus.

When a body is subjected to an uniform direct stress in all the three mutually perpendicular directions, the ratio of the direct stress to the corresponding volumetric strain is found to be a constant is called as the bulk modulus of the material and is denoted by K.

11. Define factor of safety.

Factors of safety, also known as safety factor, is a term describing the load carrying capacity of a system beyond the expected or actual loads. Essentially, the factor of safety is how much stronger the system is than it usually needs to be for an intended load. Safety factors are often calculated using detailed analysis because comprehensive testing is impractical on many projects, such as bridges and buildings, but the structure's ability to carry load must be determined to a reasonable accuracy.

12. What is stability?

The stability may be defined as an ability of a material to withstand high load without deformation.

13. What are the factors that affect the tensile strength?

i) Specimen length ii) Rate of loading and time to break iii) Capacity of machine iv) Effect of humidity and temperature v) Clamping problem

14. What are the factors that affect the elastic modulus?

i) Stress ii) Strain iii) Change of length iv) Elongation

15. What is a twisting moment? Is it same as torque?

Torque is a very simple concept, and has the units of force x distance. For example newton.metre. Torque is therefore a moment. It is typically used to describe mechanical phenomenon (ie mechanisms can have a torque applied or can apply a torque). Such as the moment in a motor shaft or a bolt. Whilst torque is a moment, not all moments are torques. For example a beam is said to undergo a bending moment, but not a torque.

16. Give example for gradually applied load and suddenly applied load.

Gradually applied load

When we lower a body with the help of a crane, the body first touches the platform on which it is to be placed. On further releasing the chain, the platform goes on loading till it is fully loaded by the body. This is the case of gradually applied load.

Suddenly applied load

When we lower a body with the help of a crane, the body is first of all, just above the platform on which it is to be placed. If the chain breaks at once at this moment the whole load of the body begins to act on the platform. This is the case of suddenly applied load.

17. What is resilience?

The strain energy stored by the body within elastic limit, when loaded externally is called resilience.

18. Define strain energy.

Strain energy is the energy absorbed or stored by a member when work is done on it to deform it.

19. Distinguish between suddenly applied and impact load.

When the load is applied all of a sudden and not step wise is called is suddenly applied load. The load which falls from a height or strike and body with certain momentum is called falling or impact load.

20. Define proof resilience?

The maximum strain energy stored in a body up to elastic limit is known as proof resilience.

21. Define strain energy density.

Strain energy density as the maximum strain energy stored in a material within the elastic limit per unit volume. It is also known as modulus of resilience.

22. What are the different types of beams?

1. Cantilever beam: A beam which is fixed at one end and at the other end is known as cantilever beam.
2. Simply supported beam: A beam supported or resting freely on the supports at its both end is known as simply supported beam.
3. Fixed beam: A beam whose both end are fixed or built-in walls is known as fixed beam.
4. Overhanging beam: if the end portion of a beam is extended beyond the support is known as overhanging beam.
5. Continuous beam: A beam which is having more than two supports is known as continuous beam.

23. Name the various types of load.

1. concentrated load or point load
2. Uniformly load
3. Uniformly distributed load

24. Define shear force at a section of a beam.

The algebraic sum of the vertical force at any section of a beam to the right or left of the section is known as shear force.

25. Define bending moment at a section of a beam.

The algebraic sum of the moments of all the force acting to the right or left of the section is known as bending of the beam.

26. What is meant by point of contra flexure?

It is the point where the bending moment is zero where it change sign from positive to negative or vice – versa.

27. State the relationship between the load and shear force.

The rate of the change of shear force is equal to the loading= -W

28. Define clear span and effective span.

The horizontal distance between the supporting walls is called the clear span of the beam. The horizontal distance between the lines of action of end reaction is called effective span.

29. State the theory of simple bending?

If a length of a beam is subjected to a constant bending moment and no share force (i.e. zero shear force) then the stresses will be set up in that length of the beam due to bending moment only and that length of the beam is said to be in pure bending or simple bending. The stresses set up in that length of beam are known as bending stress

30. What are the assumptions made in the theory of simple bending?

1. The material of the beam is perfectly homogeneous and isotropic.
2. The beam material is stressed, within its elastic limit and thus obeys Hooke‘s law.
3. The transverse sections, which were plane before bending, remain plane after bending.
4. Each layer of the beam is free to expand or contract, independently, of the layer, above or below it.

31. Define torsion.

A shaft is said to be in torsion, when equal and opposite torques are applied at the two ends of the shaft. The torque is equal to the product of the force applied (tangentially to the ends of a shaft) and radius of the shaft.

32. What are the assumptions made in the theory of torsion?

(i) The material of the shaft is uniform throughout.(ii) The twist along the shaft is uniform.(iii) Normal cross sections of the shaft, which were plane and circular before twist, remain plane and circular after twist.(iv) All diameters of the normal cross section which were straight before twist, remain straight with their magnitude unchanged, after twist.

33. Why girders are given l shaped?

I beams have very high moment of inertia for the same volume of the given material. So they have high stability in case of bending moments. The two horizontal parts (called flanges) of the I

beam can bear high bending and shearing stress. That means they do not get twisted and tilted easily. That is why they are used in girders, and as rails on the railway tracks.

34. Recall elastic fatigue.

If a body is continuously subjected to stress (or) strain , it gets fatigued (weak) which is called elastic fatigue.

35. How to impurity in a material affect the elasticity of the material? Give an example

The addition of impurities produces variation in the elastic property of the materials. The increase and decrease of elasticity depends on the type of impurity added to it.

- Ex: when potassium is added to gold, the elastic property of gold increase.
- when carbon is added to molten iron, the elastic property of iron decreases, provided the carbon content should be more than 1% in iron.

36. Define moment of a force about a point.

It is defined as a product of a force and perpendicular distance of the line of action of the force from a point.

37. Define center of gravity.

The center of gravity of a body may be defined as the point through which the entire weight of the body is assumed to be concentrated.

38. Define parallel axis theorem

This theorem states that the moment of inertia of a plane area about any axis is the sum of moment of inertia of the area about the passing through the centroidal axis and the product of the area of the plane and square of the perpendicular distance of its centroid from the axis.

$$I_m=I+ah^2$$

39. If the material is in the form of hollow cylinder of inner radius r1 and outer radius r2 then what is the torque per unit twist?

If the material is in the form of hollow cylinder of inner radius r1 and outer radius r2 then the torque per unit twist is $C= n\frac{n\pi}{2L}(r_2 4 - r_1 4)$

40. When a wire is bent back and forth, it becomes hot. Why?

When a wire is bent back and forth, heat is generated due to the area of the elastic hysteresis and frictional force. Hence it becomes hot.

41. What is difference between grain size and particle size?

Particle size is a notation for the comparative dimension of solid particles. But when we are talking about the metal, the crystal contains the internal boundaries. Any of the enclosed part by these boundaries, called as grain and its size is called as a grain size. Generally inside the particle, we can get grains.

42. How size of grains affect the elastic property of the materials?

The elastic nature of material is linked up with its grain size. The metal of smaller grains has better elasticity then the same metal of larger grains.

43. What is meant by thermal stress?

Thermal stress is stress created by any change in temperature to a material. These stresses can lead to fracture or plastic deformation depending on the other variables of heating, which include material types and constraints.

44 . On heating an object, how does thermal stress occur?

As a solid material experiences an increase in temperature, the volume of the structure is ultimately impacted by increasing, a phenomenon known as thermal expansion. This process results from heat's ability to increase a material's kinetic energy.Within solids, molecules are typically located in close proximity to one another, contributing to the defined shape of the structure. As the temperature rises, molecules begin to vibrate at a more rapid speed and push away from one another. This increased separation between the individual atoms causes the solid to expand, thus increasing the volume of the structure.

45. Why an I-section preferred for beams?

I-Section is preferred to rectangular section for resisting Bending Moment (BM) because; in I-Section more than 80% of the bending moment is resisted by the flanges itself. I-Section has higher moment of resistance (MoR) because the areas of flanges is away from the neutral axis (N.A.). In a beam of I-Section, more material is positioned near the outer fibres representing the regions of greatest stress and hence is stronger than a beam of rectangular cross section

46. Write the relation connecting the three modulii of Elasticity.

Young's modules of elasticity Y characterize the ability of the material to resist the change in length.

- Rigidity modulus of elasticity n determines the ability of a material to resist a change in shape while maintaining it volume.
- Bulk modulus K of elasticity characterizes the ability of a material to resist change in its volume not accompanied by a change in shape.
- These elastic constants of the materials are dependent on each other. The expression can be derived by showing the inter-relations between them.
- Three moduli of elasticity Y, n and K are related as $\frac{9}{Y} = \frac{3}{n} + \frac{1}{K}$

47. Mention the factors affecting the Elasticity of Materials.

Effect of stress, Effect of change in temperature, Effect of impurities, Effect of hammering, rolling and annealing, Effect of crystalline nature.

48. What is the importance of safety factor in designing engineering structures?

The good values of safety factor are always adopted to keep the structure for long life.

49. What are the Young's modulus values of the following materials?

Name of the material	GPa Values
Aluminum	69
Brass	100 - 125
Bronze	96 - 120
Copper (Cu)	117
Glass	50 - 90
Rubber	0.01 – 0.1
Steel	200
Wood	11

50. What are the applications of Young's modulus?

- The Young's modulus enables the calculation of the change in the dimension of a bar made of an isotropic elastic material under tensile or compressive loads.
- For instance, it predicts how much a material sample extends under tension or shortens under compression.
- The Young's modulus directly applies to cases of uniaxial stress, that is tensile or compressive stress in one direction and no stress in the other directions.
- Young's modulus is also used in order to predict the deflection that will occur in a statically determinate beam when a load is applied at a point in between the beam's supports.

51. Which is more elastic steel or rubber and why?

Steel. Elasticity is measured as ratio of stress to strain. For a given stress (stretching force per unit area) strain is much smaller in steel than in rubber.

UNIT II- WAVES AND OPTICS

1. Expand LASER.

LASER stands for **L**ight **A**mplification and **S**timulated **E**mission and **R**adiation.

2. Define LASER.

LASER is an artificial light source which emits a powerful,monochromatic Collimated beam of light. The emitted light waves from a LASER source are coherent innature.

3. What are the three process in LASER ?

The three process in LASER are (i) Absorption (ii) Spontanoeous (iii) Stimulated Emission.

4. What is stimulated emission?

The process of induces emissions and photons caused by the incident photons is called as stimulated emission. This process is a key factor to the process of LASER.

5. Define Population Inversion and how it is achieved?

The establishment of condition in which the number of atoms in which the number of atom in higher energy level is more than that in lower energy level is called as population inversion. It is an essential requirement for producing a LASER beam. It is achieved by population inversion.

6. Differentiate between Spontaneous and stimulated emission?

Spontaneous Emission	Stimulated Emission
Emission of light radiation is not triggered by external forces.	Induced Emission of radiations caused by incident Photons.
Emitted Photon travels in random direction	Emitted Photon travels in particular direction
Emitted Photon Cannot be controlled	Emitted Photon Can be controlled.
It is the key factor for light	It is for LASER operation.

7. Define the term upward transition.

The number of stimulated absorptions transistions occurring per unit time is given by Nab = $B_{12}N_1Q$ where B_{12} is a proportionality constant. This process is called as upward transition.

8. Define the term downward transition.

The number of stimulated absorptions transitions occurring per unit time is given by Nsp = $A_{21}N_2$ where A_{21} is a proportionality constant. This process is called as downward transition.

9. What is the usage of pumping action?

The process of creating a population inversion in the Atomic states is known as pumping action, which is essential for producing Laser beam.

10. Define the term population inversion.

It is a situation in which the number of atoms in the higher energy state is more than that of the lower energy states.

11. State the necessary condition for Population inversion.

(i) There must be at least two energy levels.
(ii) There must be a source to supply the energy to the medium.
(iii)The atoms must be continously raised to the excited state.

12. Write the four methods for pumping action.

(i) Optical Pumping.

(ii) Electrical discharge.
(iii)Direct conversion.
(iv)In-Elastic collision between atoms.

13. What are the four characteristics of LASER source?

(i) High directionality.
(ii) High intensity
(iii)Highly mono-chromatic.
(iv)Highly coherent.

14. What are the two types of semi-conductor LASER?

1. Homo-Junction semi-conductor diode LASER.
2. Hetero-Junction semi-conductor diode LASER.

15. What are the applications of Homo-Junction LASER?

1. It is used in fiber–optic communication.
2. It is used in LASER printers and CD players.
3. It is used to heal wounds by infra-red radiations.
4. It is also used as pain killer.

16. What are the applications of Hetero-Junction LASER?

1. It is used in optic communication.
2. It is used in computers and CD ROMs.

17. Name any four industrial applications of LASER.

1. Material Processing
2. LASER Annealing
3. LASER Hardening
4. LASER Cladding

18. Name any four medical applications of LASER. (OR) Mention the medical application of LASER.

1. Opthamology
2. Neuro-surgery
3. Dermotology
4. Gyneocolgy Treatment of detached retina.
5. Performing micro-surgery and blood-less operation.
6. Treatment of human and animal cancer and skin tumours.

19. Define Acceptance angle of the fiber.

The maximum angle at which a ray of light can enter through one end of the fibre and still be totally internally reflected is called as acceptance angle of the fibre.

20. What do you mean by optical fiber optical communication?

For an efficient optical communication system light waves should be transmitted through some guiding mechanism. This type of mechanism is called as optical fiber and the communication through it is called as optical fiber communication.

21. What are the main applications of optical fibre communication?

The main applications of optical fibre communications are to transmit data, voice, video signals, digital data signals using light wave signals.

22. Define numerical aperture.

The sine of the acceptance angle of the fibre is known as numerical aperture (NA). It denotes the light waves gathering capability of the optical fibre. $NA = \sqrt{(n12-n22)}$.

23. What are applications of Step-index Single – mode fibers?

(i) This type of fiber is used in under-sea cable for long distance communication.
(ii) It is used in submarine cable system.

24. Define attenuation.

It is defined as the logarthmic value of ratio of the optical power output (Pout) from a fibre to the power input(Pin). $\alpha = -10\log[Pout/Pin]$.

25. What do you mean by Fiber-Optic sensors?

One of the applications of optical fibres is called as fibre – optic sensors. A sensor is a transducer which converts one form of energy into another form of energy. There are three type of sensors (i) Intrinsic sensors (or) Active sensors (ii) Extrinsic sensors (Passive sensors).

26. What do you meant by free oscillations?

The oscillation of a body or system with its own natural frequency and under no external influence other than the impulse that initiated the motion is called free oscillation.

27. Define damping coefficient.

A damping coefficient is a material property that indicates whether a material will bounce back or return energy to a system. For example, a basketball has a low damping coefficient.

28. How does quantum mechanics differ from Newtonian mechanics?

Newtonian mechanics is valid only for large objects, whereas quantum mechanics is valid only for very small objects. In Newtonian mechanics the position and velocity of an object can be predicted at a particular time, whereas quantum mechanics can only give probabilities.

29. What is the Schrodinger wave function?

The Schrödinger equation is the fundamental equation of physics for describing quantum mechanical behavior. It is also often called the Schrödinger wave equation, and is a partial differential equation that describes how the wavefunction of a physical system evolves over time.

30. What is the potential well?

A potential well is the region surrounding a local minimum of potential energy. Energy captured in a potential well is unable to convert to another type of energy (kinetic energy in the case of a gravitational potential well) because it is captured in the local minimum of a potential well.

31. What is a particle in a box?

In quantum mechanics, the particle in a box model (also known as the infinite potential well or the infinite square well) describes a particle free to move in a small space surrounded by impenetrable barriers. Likewise, it can never have zero energy,

32. How is depletion region formed in pn junction?

Depletion Region. When a p-n junction is formed, some of the free electrons in the n-region diffuse across the junction and combine with holes to form negative ions. In so doing they leave behind positive ions at the donor impurity sites.

33. What is the potential barrier in pn junction?

As more electrons and holes flow in the depletion region the number of positive and negative ions is reduced, causing the depletion region to narrow. The energy loss in overcoming the barrier potential results in a voltage drop across the PN junction equal to the barrier potential. (0.7V for Si, 0.3V for Ge.)

34. Define zero external voltage.

No external voltage potential is applied to the PN junction diode. In the zero bias junction, potential provides higher potential energy to the holes on the P and N side terminals. When the terminals of the junction diode are shorted, few majority charge carriers in the P-side with plenty

energy to overcome the potential barrier to travel across the depletion region. Therefore, with the help of majority charge carriers, the current starts to flow in the diode and it is denoted to as forward current. In the same way, minority charge carriers in the N-side move across the depletion region in reverse direction and it is referred to as reverse current.

35. Define the term of Knee voltage.

Knee voltage is also known as cut in voltage. The minimum amount of voltage threshold inspections required for conducting the diode is known as knee voltage or cut in voltage. And also said as. The forward voltage at which the current through PN junction starts increasing rapidly is known as knee voltage.

36. Define the terms (a) Maximum power current and (b) Peak Inverse Voltage (PIV).

Maximum power current: maximum power point (MPP): The point on a power (I-V) curve that has the highest value of the product of its corresponding voltage and current, or the highest power output. Maximum power point tracker (MPPT): A device that continually finds the MPP of a solar panel or array.

Peak Inverse Voltage (PIV): The maximum value of the reverse voltage that a PN junction or diode can withstand without damaging itself is known as its Peak Inverse Voltage. This rating of Peak Inverse Voltage (PIV) is given and described in the data sheet provided by the manufacturer.

37. Why a junction diode offers very high resistance in reverse biased mode?

Hence, a large amount of electric current is blocked by the depletion region. Thus, reverse biased diode offer large resistance to the electric current. The resistance offered by the reverse biased p-n junction diode is very large compared to the forward biased diode.

38. Define optical pumping

It is the process of establishing population inversion by light energy.

39. Define electrical pumping

It is the process of establishing population inversion by electrical energy.

40. Define chemical pumping

It is the process of establishing population inversion by exothermal chemical reactions.

41. Mention the essential components of laser devices.

Active medium, Pumping source, Optical resonator

42. How can population inversion be achieved in a semiconductor laser?

Population inversion can be achieved in a semiconductor laser by doping, irradiation by light or an electron beam.

43. What is laser communication?

Laser communications systems are wireless connections through the atmosphere. They work similarly to fiber optic links, except the beam is transmitted through free space. The carrier used for the transmission signal is typically generated by a laser diode.

44. What are the performance characteristics of glass fibers?

- Lowest losses
- Smallest intermodal pulse spreading
- Useful at moderately high information rates
- Useful at fairly long range
- Low numerical aperture

45. What are the performance characteristics of plastic fibers?

- High propagation losses
- Limited to very short paths
- Higher coupling effiency
- Large numerical aperture

46. Advantages of fiber optic communication system

- Large information carrying capacity
- Small size and weight
- Electrical isolation
- Immunity to interference and cross talk
- Signal security
- Low transmission loss
- Easy maintenance
- Low cost

47. Types of fiber manufacturing technique.

- Double crucible method (or) crucible-crucible method
- Vapour deposition method

48. What are the conditions for total internal reflections?

Two important conditions for total internal reflection are:

Angle of incidence (i) should be greater than critical angle (ic).

Ray should travel from denser medium to rarer medium.

49. What is the role of a metastable state in lasers?

A large number of excited atoms are accumulated in the metastable state. The population of metastable state can exceed the population at a lower level there by establishing population inversion in a lasing medium. Population inversion could not be created without a metastable state.

50. What are the life times of excited state and metastable state?

Metastable state is an excited state of an atom or other system with a longer lifetime than the other excited states. However, it has a shorter lifetime than the stable ground state. Atoms in the metastable state remain excited for a considerable time in the order of 10^{-6} to 10^{-3}.

51. In Lasers, What does the word 'population' mean?

The number of atoms per unit volume in an energy level is known as population of that energy level.

52. Why laser action is called "inverted absorption"?

If the system is to act as a laser, an incident photon must have a higher probability of causing stimulated emission than of being absorbed i.e. the rate of stimulated emission must exceed that of absorption. In other words, the laser action is possible only when N2 > N1.

53. What are the reasons to believe that light is a wave motion?

Light undergoes interference, diffraction and polarization. These phenomena establish that light is a wave motion.

54. A light wave enters from air into glass. How will the following be affected?

(i) Energy of the wave (ii) Frequency of the wave?

Energy of the wave decreases because a part of the light wave is reflected back into air. (ii) Frequency of the wave remains unchanged.

55. If a wave undergoes refraction, what will be the phase change?

Zero. No phase change occurs during refraction.

56. When monochromatic light travels from one medium to another its wavelength changes but frequency remains the same. Explain.

Frequency is the characteristic of the source while wavelength is the characteristic of the medium. When monochromatic light travels from one medium to another, its speed changes, so its wavelength ($\lambda=c/v$) changes but frequency v remains unchanged.

57. State the essential conditions for two light waves to be coherent.

1. The two waves must be continuous. 2. The two waves should be of same frequency or wavelength. 3. They should have a constant or zero phase difference. 4. Preferably, they should have equal amplitude.

58. Two independent light sources cannot act as coherent sources'. Why?

Two independent sources of light cannot be coherent. This is because light is emitted by individual atoms, when they return to ground state. Even the smallest source of light contains billions of atoms which obviously cannot emit light waves in the same phase.

UNIT III – THERMAL PHYSICS

1. **Differentiate heat and temperature.**

S.No	Heat	Temperature

1.	Heat is energy that is transferred from one body to another as the result of a difference in temperature.	Temperature is a measure of hotness or coldness expressed in terms of any of several arbitrary scales like Celsius and Fahrenheit.
2.	Heat is a measure of how many atoms there are in a substance multiplied by how much energy each atom possesses.	Temperature is related to how fast the atoms within a substance are moving.
3.	Unit: Joules	Kelvin, Celsius or Fahrenheit

2. Define thermal conductivity.

Thermal conductivity refers to the amount/speed of heat transmitted through a material. Heat transfer occurs at a higher rate across materials of high thermal conductivity than those of low thermal conductivity. Materials of high thermal conductivity are widely used in heat sink applications and materials of low thermal conductivity are used as thermal insulation.

3. Define conduction.

Conduction is how heat transfers through direct contact with objects that are touching. Any time that two objects or substances touch, the hotter object passes heat to the cooler object.

4. What is the first law of thermodynamics?

The first law of thermodynamics, also known as Law of Conservation of Energy, states that energy can neither be created nor destroyed; energy can only be transferred or changed from one form to another. For example, turning on a light would seem to produce energy; however, it is electrical energy that is converted.

5. Define convection.

Convection is how heat passes through fluids. A fluid is anything that has loosely moving molecules that can move easily from one place to another. Liquids and gases are fluids. One important property of fluids is that they rise when heated. That's because the molecules spread out and move apart when they get hot. The hot fluid becomes less dense and rises up. Cooler fluid is less dense and so it sinks down. This up-and-down motion creates what are called convection.

6. What is adiabatic condition?

Adiabatic conditions refer to conditions under which overall heat transfer across the boundary between the thermodynamic system and the surroundings is absent.

7. Define radiation.

Radiation is how heat moves through places where there are no molecules. Radiation is actually a form of electromagnetic energy.

8. What are bimetallic strips?

A bimetallic strip is used to convert a temperature change into mechanical displacement. The strip consists of two strips of different metals which expand at different rates as they are heated, usually steel and copper, or in some cases steel and brass.

9. How the circuit breakers are expands for temperature?

Bimetal strips are used in miniature circuit breakers to protect circuits from excess current. A coil of wire is used to heat a bimetal strip, which bends and operates a linkage that unlatches a spring-operated contact. This interrupts the circuit and can be reset when the bimetal strip has cooled down.

10. Can you give any two differences of muffle and reverberatory furnace?

Muffle Furnace	Reverberatory Furnace
A muffle furnace (sometimes retort furnace in historical usage) is a furnace in which the subject material is isolated from the fuel and all of the products of combustion, including gases and flying ash	In this the material to be heated is insulated from the fuel but not with its combustion gases. To make it easier consider this example.
200◦C -1200◦C	Above 1500◦C

11. What is solar power?

Solar power is the conversion of energy from sunlight into electricity, either directly using photovoltaic (PV), indirectly using concentrated solar power, or a combination. Concentrated solar power systems use lenses or mirrors and tracking systems to focus a large area of sunlight into a small beam. Photovoltaic cells convert light into an electric current using the photovoltaic effect.

12. What are heat exchangers?

A heat exchanger is a device used to transfer heat between a solid object and a fluid, or between two or more fluids. The fluids may be separated by a solid wall to prevent mixing or they may be in direct contact. They are widely used in space heating, refrigeration, air conditioning, power stations, chemical plants,petrochemical plants, petroleum refineries, natural-gas processing, and sewage treatment.

13. Define green house effect.

The greenhouse effect is the process by which radiation from a planet's atmosphere warms the planet's surface to a temperature above what it would be without its atmosphere. If a planet's atmosphere contains radioactively active gases they will radiate energy in all directions.

14. Give the principle of solar water heaters?

The working of solar water heaters is very simple to understand. The solar water heaters use two common principles for its functioning. a black surface heats up when left in the sun, by

absorption of solar radiation; The good absorption property of black surfaces is used to improve solar energy absorption in a solar heater. The inside of car/ bus parked in sun for a long time becomes hot. This is because solar radiation can pass through the glass windows of the bus but cannot come out. It is trapped inside and thus heats up the bus. Similarly water passing through insulated pipes kept in the sun becomes hot.

15. How the heat conducts through solid?

Generally conduct heat better than other solids do. In metals, some of the electrons (often one per atom) are not stuck to individual atoms but flow freely among the atoms. Of course, that's why metals are such good conductors of electricity. Now if one end of a bar is hot, and the other is cold, the electrons on the hot end have a little more thermal energy- random jiggling- than the ones on the cold end. So as the electrons wander around, they carry energy from the hot end to the cold end, which is another way of saying they conduct heat.

16. Mention the properties of thermal insulation materials should have?

1. Thermal conductivity of the material should be low.
2. Specific heat capacity should be low.
3. They should have high fire resistance.

17. Write the thermal coefficient equation for the bad conductor using lee's disc method.

The thermal coefficient equation for the bad conductor using lee's disc method is,

$$K=\frac{msrd(r+2h)}{A(\theta_1-\theta_2)(2r+2h)}$$

18. How do solar water heaters work?

The working of solar water heaters is very simple to understand. The solar water heaters use two common principles for its functioning. a black surface heats up when left in the sun, by absorption of solar radiation; The good absorption property of black surfaces is used to improve solar energy absorption in a solar heater. The inside of car/ bus parked in sun for a long time becomes hot. This is because solar radiation can pass through the glass windows of the bus but cannot come out. It is trapped inside and thus heats up the bus. Similarly water passing through insulated pipes kept in the sun becomes hot.

19. Write the thermal conductivity equation for the good conductor using Forbe's disc method.

The thermal conductivity equation for the good conductor using Forbe's disc method is,

$$\mathbf{K}=\frac{\rho S\int_p^{end}\left(\frac{d\theta}{dt}\right)}{\left(\frac{d\theta}{dt}\right)p}dt$$

20. Suppose that your body is covered with a "perfect" blanket where the heat you release is 100 percent reflected back to you, it does not leak any temperature outside the environment and the blanket is indestructible and can not be deformed in any way. If your body continues to generate heat at 37 degree, and just like the blanket you are indestructible and no matter what will continue to radiate 37 degrees of heat, with enough time, can the temperature inside the blanket match that of the sun why or why not?

Temperature would rise to 41 or 42 degrees at most. Then it leads to death.

21. Does the material molded in a mold with low and high thermal diffusivity freeze faster?

We have a mold material with low thermal diffusivity (plastic) vs a mold material with high thermal diffusivity (silver). So easily freeze the same material in the same freezer.

22. Explain how the water and air can be at the same temperature.

A closed bottle contains water at the bottom and air above it in equilibrium at the same temperature.

23. A 1-kg block of ice at 0ºC is placed into a perfectly insulated, sealed container that has 2 kg of water also at 0ºC. The water and ice completely fill the container, but the container is flexible. After some time one can except that,

The ice will melt so that the mass of the ice will decrease. In accordance to second law of thermodynamics, if an irreversible process occurs in a closed system, the entropy of that system always increases; it never decreases. Entropy measures the state of disorder. As the ice is more order than water, therefore the ice will melt so that the mass of the ice will decrease.

24. Consider an ideal heat pump and a perfect electric heater. The electric heater converts 100% of the electrical energy into heat energy; the heat pump converts 100% of the electrical energy into work, which is the more "efficient" way to heat a home?

The heat pump is more efficient if the outside temperature is not too cold. The value of coefficient of performance **K** becomes larger as the temperature of the two reservoirs becomes more nearly the same. Thus heat pumps are more efficient in temperate climates than in climates where the outside temperature fluctuates between wide limits. From the above observation we conclude that, the heat pump is more efficient if the outside temperature is not too warm.

25. How heat passes through fluids?

Convection is how heat passes through fluids. A fluid is anything that has loosely moving molecules that can move easily from one place to another. Liquids and gases are fluids. One important property of fluids is that they rise when heated. That's because the molecules spread out

and move apart when they get hot. The hot fluid becomes less dense and rises up. Cooler fluid is less dense and so it sinks down. This up-and-down motion creates what are called convection.

26. How the heat transfers through direct contact with objects?

Conduction is how heat transfers through direct contact with objects that are touching. Any time that two objects or substances touch, the hotter object passes heat to the cooler object

27. How the electromagnetic energy converted to heat?

Radiation is how heat moves through places where there are no molecules. Radiation is actually a form of electromagnetic energy.

28. What are green house gases?

Many greenhouse gases occur naturally in the atmosphere, such as carbon dioxide, methane, water vapor, and nitrous oxide, while others are synthetic. Those that are man-made include the chlorofluorocarbons (CFCs), hydrofluorocarbons (HFCs) and Perfluorocarbons (PFCs), as well as sulfur hexafluoride (SF_6).

29. What is the thermal expansion of two different metals combined?

The strip consists of two strips of different metals which expand at different rates as they are heated, usually steel and copper, or in some cases steel and brass known as bimetallic strip. A bimetallic strip is used to convert a temperature change into mechanical displacement.

30. How that heat exchangers works in every day home equipments?

They are widely used in space heating, refrigeration, air conditioning, power stations, chemical plants, petrochemical plants, petroleum refineries, natural-gas processing, and sewage treatment.

31. Define the term temperature?

A measure of the warmth or coldness of an object or substance with reference to some standard value. The temperature of two systems is the same when the systems are in thermal equilibrium. The degree of heat in a living body, normally about 98.6°F (37°C) in humans. What are different temperature scales?

32. What the temperature Scales used?

There are three temperature scales in use today, Fahrenheit, Celsius and Kelvin.

33. State various temperature measurement methods.

Thermocouples. Thermistors. Resistance temperature detector (RTD) Pyrometer. Langmuir probes (for electron temperature of a plasma) Infrared.

34. What is positive temperature coefficient of resistance?

This factor is represented by the Greek lower-case letter "alpha" (α). A positive coefficient for a material means that its resistance increases with an increase in temperature. Pure metals typically have positive temperature coefficients of resistance.

35. What is negative temperature coefficient of resistance?

Negative coefficient for a material means that its resistance decreases with an increase in temperature. Semiconductor materials (carbon, silicon, germanium) typically have negative temperature coefficients of resistance.

36. Temperature measurement Electrical methods What are they?

i) Resistance wire
ii) Thermocouple
iii) Thermistor

37. What is thermistor? What kind of material is used for thermistor?

The thermistor is a kind of resistor whose resistivity depends on surrounding temperature. It is a temperature sensitive device. The word thermistor is derived from the word, thermally sensitive resistor. The thermistor is made of the semiconductor material that means their resistance lies between the conductor and the insulator. The most common materials to be used for these thermistors are Manganese oxide, nickel oxide, cobalt oxide, copper oxide and ferric oxide. Semiconductor thermistors are used for much lower temperatures.

38. State the Seebeck effect.

The Seebeck effect is a phenomenon in which a temperature difference between two dissimilar electrical conductors or semiconductors produces a voltage difference between the two substances.

39. What is meant by renewable energy source?

Resource which are renewed by nature again and again and their supply is not affected by the arte of their consumption are called renewable. E.g., solar, wind, biomass, ocean, geothermal, hydro, etc.

40. Define solar constant.

Solar constant is defined as the amount of energy received in unit time on a unit area perpendicular to the sun's direction at the mean distance of the earth from the sun.

41. Define the term thermal Physics.

Thermal Physics is the science that deals with the energy transfer to practical applications such as energy transfer power generation, refrigeration, gas compression and its effect on the properties of working substance.

42. Define Heat.

Heat is the energy crossing the boundary due to the temperature difference between the system and surroundings.

43. State the law of conservation of energy.

Energy can neither be created nor destroyed, but it can be transferred from one form to another.

44. Define free expansion process.

When a gas expands suddenly into a vacuum through a large orifice is known as free expansion process.

45. What is the difference between a heat pump and a refrigerator?

Heat pump is a device which operating in cyclic process, maintains the temperature of a hot body at a temperature higher than the temperature of surroundings.

A refrigerator is a device which operating in a cyclic process, maintains the temperature of a cold body at a temperature lower than the temperature of the surroundings.

46. Name four important properties of a good refrigerant

- Low boiling point
- High critical temperature & pressure
- Low sp.heat of liquid
- Non – flammable and non explosive.

47. Name any four commonly used refrigerants

- Ammonia (NH_3)
- Carbon di oxide (CO_2)
- Sulphur di oxide (SO_2)
- Freon – 12.

48.Given two examples of heat transfer with internal heat generation.

Chemical Reaction

Nuclear Reaction

Combustion Reaction.

49.Name the law which governs convection heat transfer

Newton's law of cooling

50. Write down the Stefan Boltzmann law with its Unit

The total energy emitted by a black body at a particular temperature is given by

$Eb = \sigma T4$

Where σ – Stefan Boltzmann constant – 5.669 x 10-8 w/m2k4

UNIT IV – QUANTUM PHYSICS

1. **Explain Planck's hypothesis or what are the postulates of Planck's quantum theory?**
   (i) The black body radiation chamber is filled with a large number of oscillating particles. The particles can vibrate in all possible frequencies.
   (ii) The frequency of radiation emitted by an oscillator is the same as that of the frequency of that vibrating particle.
   (iii)The oscillatory particles cannot emit energy continuously. They will radiate energy only in the form of a discrete packet of energy, i.e., a small unit called quantum photon.

2. **What is meant by blackbody radiation?**

A perfect black body is one that absorbs radiation of all wavelengths incident on it. Further, such a body cannot transmit or reflect any radiation and therefore it appears black. A black body can radiate energy in all possible wavelengths when it is heated to a suitable temperature. The radiation emitted from black body is known as black body radiation or total radiation.

3. **Give the importance of Planck`s radiation formula.**
   1. It explains all region of block body spectrum
   2. It is based on quantum theory
   3. It is used derive other laws related to black body radiation

4. **State Compton effect. (Or) Give a brief account on Compton effect.**

When a beam of X – rays is scattered by a substance of low atomic number, the scattered X-ray radiation consists of two components, one component has the same wavelength λ as the incident ray and the other component has a slightly longer wavelength λ'. This phenomenon is known as Compton effect.

5. What is the physical significance of a wave function?

1. It relates the particles and wave nature of matter elastically.
2. The square of the wave function is a measure of the probability of finding the particle at a particular position.It cannot predict the exact location of the particle.
3. The wave function is a complex quantity, where as the probability is a real and positive quantity.
4. The wave function has no physical meaning.
5. $|\psi|2$ = is real and positive, amplitude may be positive or negative but the intensity(square of amplitude) is always real and positive.
6. $|\psi|2$ represents the probability density or probability of finding the particle in the given region.
7. For a given volume dτ, probability $P = \int\int\int |\psi|2\, d\tau$ where $d\tau = dx\, dy\, dz$

6. What is meant by photon?

The discrete energy values in the form of small packets or quantas of definite frequency or wavelength are called photons. Photons propagate like a particle with speed of light as 3×10^8 ms^{-1}.

7. State Wien's displacement law.

According to Wien's displacement law, the product of the wavelength corresponding to aximum energy (λm) and absolute temperature is a constant. i.e λmT = constant.

8.What is a wave function?

A variable quantity which characterizes de-Broglie wave is known as wave function and it is donoted by the symbol Ψ.

9. What is black body and what are its characteristics?

1. A perfect black body is the one which absorbs and also emits the radiations completely.
2. There is no black body in nature. We have to coat the black colour over the inner surface to make a black body.
3. Black body is said to be a perfect absorber, since it absorbs all the wavelengths Of the incident radiation.
4. The black body is the perfect radiator because it radiates the entire wavelength absorbed by it. This phenomenon is called black body radiation.

10. State Planck's quantum theory (or) State Planck's hypothesis (or) What are the postulates of Planck's quantum theory? (or) What are the assumptions of quantum theory of black body radiation? (or) Give the special features of Quantum theory.

1. The electrons in the black body are assumed as simple harmonic oscillators.
2. The oscillators will not emit energy continuously.

3. The emit radiation in terms of quantas of magnitude ‘hυ’, discretely.i.e. E = nhυ, where n = 0, 1, 2, 3, …..

11. What are meant by a degenerate state and Non-degenerate state?

1. For various combinations of quantum numbers, if we get same eigen value but different eigen functions, it is called degenerate state.
2. For various combinations of quantum numbers, if we get same eigen value but same eigen functions, it is called Non-degenerate state.

12. What is the principle electron microscopy?

In an electron microscope, a stream of electrons are passed through the object and the electrons which carry the information about the object are focused by electric and magnetic lenses (or) electromagnetic lenses.

13. For a free particle moving within a one dimensional potential box, the ground state energy cannot be zero. Why?

For a free particle moving within a one dimensional potential box, when n = 0, the wave function is zero for all values of x, i.e., it is zero even within the potential box. This would mean that the particle is not present within the box. Therefore the state with n = 0 is not allowed. As energy is proportional to n2, the ground state energy cannot be zero since n = 0 is not allowed.

14. Mention the applications of scanning tunneling microscope.

1. Scanning tunneling microscope is a microscope commonly used in fundamental and industrial research.
2. STM has a good resolution is 0.1nm lateral resolution and 0.01nm depth resolution
3. The high resolution of STM`s enable researches to examine surface at an atomic level.

15. List out the advantages and disadvantages of STM.

Advantage:

1. It can be used to examine specimens of large thickness.
2. It can be used to get a three dimensional image of the object.
3. Since the image can be directly viewed in the screen, structural details can be resolved in a precise manner.
4. The magnification may be upto 3,00,000 times greater than that of the size Of the object.

Disadvantage:

1. The resolution of the image is limited to about 10-20 nm, hence it is very poor.

16. State wave duality.

Wave–particle duality is the concept that every particle or quantic entity maybe partly described in terms not only of particles, but also of waves. It expresses the inability of the classical concepts "particle" or "wave" to fully describe the behavior of quantum-scale objects.

17. What is mean by electron diffraction?

Electron diffraction refers to the wave nature of electrons. However, from a technical or practical point of view, it may be regarded as a technique used to study matter by firing electrons at a sample and observing the resulting interference pattern.

18. Define wave duality.

The behaviors of the electron does not allow for it to be observable as a particle and as a wave. The two sided nature of the electron is known as the Wave-Particle Duality: The property of particles behaving as waves and the property of waves behaving as particles as well as waves. Although the duality is not very effective in large matter. The wave characteristic of the electron implicates many of the electron's particle behaviors.

19. What is mean by tunneling?

Quantum tunnelling or tunneling (see spelling differences) refers to the quantum mechanical phenomenon where a particle tunnels through a barrier that it classically could not surmount. ... The effect was predicted in the early 20th century and its acceptance as a general physical phenomenon came mid-century.

20. Define STM.

Scanning tunneling microscopy is a microscopical technique that allows the investigation of electrically conducting surfaces down to the atomic scale. In the following I will give a very short overview of the basic principle

21. What are matter waves?

The waves associated with moving particles of matter (e.g., electrons , photons , etc) are known as matter waves or de-Broglie

22. What is meant by photon?

The discrete energy values in the form of small packets or quantas of definite frequency or wavelength are called photons. Photons propagate like a particle with speed of light as 3×10^{8} ms^{-1}

23. Write the principle of TEM.

When the electrons are passed through the specimen, image is produced by using either transmitted (bright field image) or diffracted (dark field image) electron beam through the specimen. This gives the three dimensional image of the specimen.

24. Bring out the differences between Scanning Electron Microscope (SEM) and Transmission Electron Microscope (TEM).

S.NO	SEM	TEM
1	Scattered electrons are used	Transmitted electrons are used
2	SEM provides a 3D image	TEM provides an detailed image of a specimen
3	Magnification upto 2 million times	Magnification upto 50 million times
4	Preparation of sample is easy	Preparation of sample is difficult
5	Thick specimen can also be analyzed	The specimen should be thin
6	SEM is used for surface analysis	TEM is used for analyzing sections

25. Give the special features of Quantum theory.

(i) The electrons in the black body are assumed as simple harmonic oscillators. (ii) The oscillators will not emit energy continuously. (iii)The emit radiation in terms of quantas of magnitude 'hυ', discretely. i.e. $E = nh\upsilon$, where n = 0, 1, 2, 3, …..

26. What are meant by a degenerate state and Non-degenerate state?

(i) For various combinations of quantum numbers, if we get same eigen value but different eigen functions, it is called degenerate state. (ii) For various combinations of quantum numbers, if we get same eigen value but same eigen functions, it is called Non-degenerate state.

27. For a free particle moving within a one dimensional potential box, the ground state energy cannot be zero. Why?

For a free particle moving within a one dimensional potential box, when n = 0, the wave function is zero for all values of x, i.e., it is zero even within the potential box. This would mean that the particle is not present within the box. Therefore the state with n = 0 is not allowed. As energy is proportional to n2 , the . ground state energy cannot be zero since n = 0 is not allowed.

28. Define Quantum Mechanics

Quantum Physics The classical world-view works fine at the everyday (macroscopic) level – much of modern engineering relies on this – but there are things at the macroscopic level that cannot be understood using classical physics, these including the colour of a heated object, the existence of solid objects . . . . So where does classical physics come unstuck? Non-classical behaviour is most readily observed for microscopic systems – atoms and molecules, but is in fact present at all scales. The sort of behaviour exhibited by microscopic systems that are indicators of a failure of classical physics are • Intrinsic Randomness • Interference phenomena (e.g. particles acting like waves) • Entanglement

29. What aspect of all physical theories do the principles of quantum mechanics apply?

The principles must apply to theories as diverse as Newton's Laws describing the mechanical properties of matter, Maxwell's equations describing the electromagnetic field, the laws of thermodynamics .

30. Is particle physics and quantum physics the same?

Quantum physics (also known as quantum mechanics, or QM) is more general. ... Particle physics, on the other hand, involves the application of QM or other fundamental type of theories to specific particles that we see in nature. You probably know about atoms.

31. What is the magnification of the scanning tunneling microscope?

The scanning electron microscope is capable or rendering images at magnifications ranging from 10X to 500,000X, 250 times the limit of the most powerful optical microscopes. A SEM produces a beam of electrons with an electron gun.

32.What is meant by photon?

The discrete energy values in the form of small packets or quantas of definite frequency or wavelength are called photons. Photons propagate like a particle with speed of light as 3 x 108 ms-1.

33.Write the principle of TEM.

When the electrons are passed through the specimen, image is produced by using either transmitted (bright field image) or diffracted (dark field image) electron beam through the specimen. This gives the three dimensional image of the specimen.

34. What is black body and what are its characteristics?

A perfect black body is the one which absorbs and also emits the radiations completely. (ii) There is no black body in nature. We have to coat the black colour over the inner surface to make a black body. (iii) Black body is said to be a perfect absorber, since it absorbs all the wavelengths Of the incident radiation. (iv) The black body is the perfect radiator because it radiates the entire wavelength absorbed by it. This phenomenon is called black body radiation.

35. State Planck's quantum theory (or) State Planck's hypothesis (or) What are the postulates of Planck's quantum theory? (or) What are the assumptions of quantum theory of black body radiation? (or) Give the special features of Quantum theory.

The electrons in the black body are assumed as simple harmonic oscillators. (ii) The oscillators will not emit energy continuously. (iii)The emit radiation in terms of quantas of magnitude 'hυ', discretely. i.e. $E = nh\upsilon$, where n = 0, 1, 2, 3, ….. 11.

36.What are meant by a degenerate state and Non-degenerate state?

For various combinations of quantum numbers, if we get same eigen value but different eigen functions, it is called degenerate state. (ii) For various combinations of quantum numbers, if we get same eigen value but same eigen functions, it is called Non-degenerate state.

37.For a free particle moving within a one dimensional potential box, the ground state energy cannot be zero. Why?

For a free particle moving within a one dimensional potential box, when n = 0, the wave function is zero for all values of x, i.e., it is zero even within the potential box. This would mean that the particle is not present within the box. Therefore the state with n = 0 is not allowed. As energy is proportional to n2, the ground state energy cannot be zero since n = 0 is not allowed.

38.Mention the applications of scanning electron microscope

(i) It is used to examine the structure of specimens in a three dimensional view. (ii) This microscope also has wide range of applications in various fields such as biology, industry, engineering, physics, chemistry etc.

39. List out the advantages and disadvantages of SEM.

Advantages: (i) It can be used to examine specimens of large thickness. (ii) It has large depth of focus. (iii) It can be used to get a three dimensional image of the object. (iv) Since the image can be directly viewed in the screen, structural details can be resolved in a precise manner. (v) The magnification may be upto 3,00,000 times greater than that of the size Of the object.

Disadvantages: The resolution of the image is limited to about 10-20 nm, hence it is very poor.

40. What is the concept of wave-particle duality?

On the basis of experimental facts like interference, diffraction and polarization it is clear that EM radiation possess wave nature. On the other hand there are experimental evidences like photoelectric effect, emission and absorption spectra, black body radiation etc., which shows that EM radiation consists of discrete indivisible packets of energy ($h\upsilon$), photons. Hence we can conclude that EM radiation has dual character, i.e., particle as well as wave. However, both the characteristics can never be observed simultaneously.

41. Why the de-Broglie wave associated with a moving car is not observable.

We know that λ in inversely proportional to m. Since m is very large for a car therefore λ is very small. Consequently, the de-Broglie wave associated with moving car is not visible.

42. Are matter waves electromagnetic?

No. This is because electromagnetic waves are produced by accelerated charge. On the other hand, the de-Broglie wave is independent of the charge of a particle.

UNIT V – CRYSTAL PHYSICS

1. What is a Crystal? (or) What are crystalline materials? Give examples.

Crystalline solids (or) Crystals are those in which the constituent atoms (or) molecules are arranged in an orderly fashion throughout in a three dimensional pattern .Example :Copper, Silver.

2. What is an amorphous solid? Give example. (or) Non-Crystalline materials.

It is a type of solid, in which the atoms (or) molecules are not arranged in an orderly fashion. (ie) the same atomic groups are arranged more randomly. Example: Plastic, rubber

3. What is meant by Crystallography?

The study of the geometric form and other physical properties of crystalline solids, using x-rays (or) electron beam (or) neuron beam etc is termed as the science of crystallography.

4. What is a single crystal?

A crystal in which solid contains only one crystal. These single crystals are produced artificially from their vapor (or) Liquid State.

5. What is a poly crystal? Give example.

A crystal structure in which has an aggregate of many small crystals (or) grains separated by well defined grain boundaries. These crystals will have a sharp melting point. Examples: Diamond, Copper, Platinum, Silver, Polonium, Gold, Aluminum, Nickel, Cadmium, Iron etc.

6. What are the differences between crystalline and non-crystalline materials?

Crystalline material .	Non-crystalline material
They have definite and regular geometrical shapes which extend through the crystal.	They don't have definite geometrical shapes
They are anisotropic	They are isotropic
They are most stable	They are less stable
Example: Nacl , Kcl	Example: Plastic, Glass, rubber

7. What is meant by crystallization and X-ray crystallography?

The phase change from Liquid (or) gas to solid is called Crystallization. The crystal structure gives the arrangement of atoms within a crystal. Determination of crystal structure with the help of X-ray is known as X-ray crystallography.

8. Define Lattice.

Lattice is defined as an array which are imaginarily kept points to represent the position of atoms in the crystal such that every lattice point has got the same environment as that of the other and hence one lattice point can not be distinguished from the other lattice point.

9. Define space lattice (or) crystal lattice. (May 2003, June 2005)

A three dimensional collection of points in space are called space lattice (or) crystal lattice. The environment about any particular point is in every way the same. (or) A geometrical representation of the crystal structure in terms of lattice points are called space lattice, provided the environment about every point is identical to that of every other point.

10. Define lattice points.

Lattice points denote the position of atoms (or) molecules in the crystal (or) The points in the space lattice are called lattice points.

11 .Define Basis.

Basis is an unit assembly of atoms (or) molecules which are identified with respect to the position of lattice points, identical in composition arrangement and orientation.

12 What is meant by structure?

A crystal lattice refers to the geometry of a set of point in space, Where the crystal structure refers to the actual ordering (or) alignment of its constituent ions, atoms (or) molecules in the space. The crystal structure is formed by the addition of basis to every lattice point of the lattice. (ie) Space lattice + Basis (or) Mot if = Crystal structure

13. Define lattice planes?

A set of parallel and equally spaced planes in a space lattice, Which are formed with respect to the lattice points are called lattice planes.

14. Define unit cell.

The unit cell is defined as the smallest geometric figure, the translational repetition of which in all over the three dimensions gives the actual crystal structure.(OR) The unit cell may also be defined as the fundamental elementary pattern with minimum number of atoms, molecules (or) groups of molecules which represents the total characteristics of the crystal.

15.What is called as crystallographic axis?

The lines drawn parallel to the lines of intersection of any three faces of the unit cell which do not lie in the same place called crystallographic axis.

16.What is unit cell parameter (or) lattice parameters?

The three intercept quantities a, b and c are called the fundamental translational vectors (or) axial lengths (or) intercepts (a,b and c) and three inter facial angles are called unit cell parameter (or) lattice parameters.

17.What are primitives (or) Characteristic intercepts?

The intercepts a, b and c are nothing but the edge of the unit cell (ie the distance between two lattice points) which define the dimensions of an unit cell. These intercepts are known as its primitives (or) characteristic intercepts on the axis.

18.What is primitive cell? (May 2003, Nov 2003)

A primitive cell is the simplest type of unit cell which contains only one lattice point per unit cell (contains lattice points at its corner only) Example :Simple Cubic (SC)

19.Which are called Non-Primitive cell.

If there are more than one lattice point in an unit cell, it is called a non-primitive cell.

Example: BCC and FCC.

20.Name of the seven crystal system.

Triclinic, Mono clinic, Orthorhombic, Tetragonal, Hexagonal, Trigonal and Cubic.

21.What are Bravais lattices? (May 2004)

The 14 possible ways of arranging points in space lattice such that, all lattice points have exactly the same surroundings. These 14 lattice in seven crystal structure are called Bravais lattice.

22.What is known number of atoms per unit cell (or) effective number?

The total number of atoms present in (or) shared by an unit cell is known as number of atoms per unit cell.

23.Define atomic radius.

Atomic radius is defined as half of the distance between any two nearest neighbor atoms which have direct contact with each other, in a crystal of a pure element. It is interns of cube edge 'a'.

24.Define co-ordination number.

Co-ordination number is the number of nearest neighboring atoms to a particular atom. (or) Co-ordination number is the number of nearest neighbors directly surrounding a given atom.

25.Define Atomic packing factor (or) Packing density (or) density of packing (May 2004).

Atomic packing factor is defined as the ratio between the volume occupied by the total number of atoms per unit cell (v) to the total volume of the unit cell (V). APF=Volume occupied by the total number of atoms per unit cell(v) Total volume of the unit cell(V) (or) APF=Total Number of atom per unit cell * Volume of one atom (v) Total volume of the unit cell(V) APF=v/V

26.Name the crystal structure of the following materials.

(a).Gold – FCC (b) Germanium –Diamond cubic (c).Barium – BBC (d) Zinc – HCP

27.Bismuth has a=b=c=4.74AU and angle α=β= γ=60ºwhat is its crystal structure?(Nov 2003)

a=b=c=4.74AU α=β=r=60º Bismuth is Trigonal (or) Rhombohedral

28. State the condition imposed on the coll parameter for systems having the largest number of bravais lattices the least number of nearest neighbors.

1. Crystal with least number of nearest neighbours is simple (six neighbours)
2. crystal system with largest number of bravais lattice is orthorhombic (4 bravaais lattices).

29.Define inter atomic distance and inter planer distance.

Inter atomic distance: The distance between any two atoms is called inter atomic distance.

Inter planar distance: The distance between any two planer is called inter – planar distance.

30.What are point defects? (or) Zero dimensional defects?

The defects which take place due to imperfect packing of atoms during crystallization are known as point defects.

31.What are franked (or) Scotty defects?

Franked defect is an conic crystal imperfection that occurs when an ion moves into an interstitial site, thereby creating two defects. Simultaneously (ie) One vacancy and the other self interstitial. A pair of ion vacancies in an ionic crystal is termed as Scotty defect.

32.What is a line defect (or) a one dimensional defect?

The defects which take place due to dislocation (or) distortion of atoms along a line in some direction is called line defect.

33.What is edge dislocation?

It is a region of lattice disturbance extending along on edge inside a crystal due to the insertion of an extra place of atoms.

34. What is screw dislocation?

Screw dislocation results from a displacement of the atoms in one part of a crystal relative to the rest of crystal forming a spiral ramp around the dislocation line.

35.What is meant by stacking fault?

The stacking faults are planar surface imperfection and are caused by faults in the stacking sequence of atomic planer in construction FCC and HCP materials

36. What are the differences between edge and grew dislocation?

Edge dislocation	Screw dislocation
These dislocations arise due to introduction (or) elimination of an extra place of atoms.	Screw dislocation results from a displacement of the atoms in one part of a crystal relative to the rest of

	crystal forming a spiral ramp around the dislocation line.
Region of lattice disturbance extends along an edge inside a crystals	Region of lattice disturbance extends in two separate planes each other.
An edge dislocation can glide and climb	A screw dislocation can glide only
Burger vector is always perpendicular to the dislocation line.	Burger vector is parallel to the dislocation line.
These are formed during deformation and crystallization	These are also formed during deformation and crystallization.

37. Classify the crystal growth techniques.

The crystal growth technique is classified into 6 types. They are

1. Melt growth
2. Low temperature solution growth
3. High temperature solution growth (Flux growth)
4. Hydrothermal growth
5. Gel growth
6. Vapour phase growth

38. Name the two important Melt growth techniques.

Melt growth is the process of crystallization by fusion and resolidification of the starting materials from the melt.

Czochralski's and Bridgman technique are the two important techniques of crystal growth from melt.

39. What is difference between crystal structure and crystal systems?

A crystal structure is described by both the geometry and atomic arrangements within, the unit cell, whereas a crystal system is described only in terms of the unit cell geometry. For example, face-centered cubic and body-centered cubic are crystal structures that belong to the cubic crystal system.

40. What is the difference between atomic structure and crystal structure?

Atomic structure relates to the number of protons and neutrons in the nucleus of an atom, as well as the number and probability distributions of the constituent electrons. On the other hand, crystal structure pertains to the arrangement of atoms in the crystalline solid material.

41. What is meant by structure?

A crystal lattice refers to the geometry of a set of point in space, Where the crystal structure refers to the actual ordering (or) alignment of its constituent ions, atoms (or) molecules in the space. The crystal structure is formed by the addition of basis to every lattice point of the lattice. (ie) Space lattice + Basis (or) Mot if = Crystal structure.

4. Define density of a material.

$$\rho = \frac{mass\ of\ the\ unit\ cell}{volume\ of\ the\ unit\ cell}$$

$$\rho = \frac{nM}{Na3}$$

N-avogadro's number ,

42. Define polymorphism.

Polymorphism is defined as the ability of a material to exist in two or more crystal structure.

43. Define the term allotropy.

If the change in structures reversible,then the polymeric change is known as allotropy.

44. How many lattice point does a primitive cell have?

Primitive unit cells contain only one lattice point, which is made up from the lattice points at each of the corners.

45. How many lattice points are there in FCC?

The face centered cubic lattice contains 4 lattice point per unit cell. The maximum packing density occurs when the atoms have a radius which equals one quarter of the diagonal of one face of the unit cell.

46. What is the lattice structure of diamond?

Diamond is a metastable allotrope of carbon where each carbon atom is bonded covalently with other, surrounding four carbon atoms and are arranged in a variation of the face centered cubic crystal structure, called a diamond lattice.

47. What is the structure of diamond?

Diamond has a giant molecular structure. Each carbon atom is covalently bonded to four other carbon atoms. A lot of energy is needed to separate the atoms in diamond. This is because covalent bonds are strong, and diamond contains very many covalent bonds.

48.What is meant by crystallization and X-ray crystallography?

The phase change from Liquid (or) gas to solid is called Crystallization. The crystal structure gives the arrangement of atoms within a crystal. Determination of crystal structure with the help of X-ray is known as X-ray crystallography.

49.What is meant by Crystallography?

The study of the geometric form and other physical properties of crystalline solids, using x-rays (or) electron beam (or) neuron beam etc are termed as the science of crystallography.

50. What is the need for growing single crystals?

Many electronic devices need single crystals for their operation. Single crystal silicon is used in the fabrication of semiconductors. Another application of single crystal solids is in materials science in the production of high strength materials with low thermal creep, such as turbine blades.

51. Give any two advantages of crystal defects.

Substitutional impurities that are a mismatch in size to the host prevent dislocations from migrating smoothly along a plane. Generally, the higher the concentration of impurities, the more effectively they block migration, and the stronger the material.

If an interstitial impurity forms polar covalent bonds to the host atoms, the layers are prevented from sliding past one another, even when only a small amount of the impurity is present.

52.Draw the Bravais lattices belonging to the orthorhombic crystal system.

There are four orthorhombic Bravais lattices: primitive orthorhombic, base-centered orthorhombic, body-centered orthorhombic and face-centered orthorhombic.

53. Distinguish between Frenkel and schottky defects.

Schottky defect occurs when oppositely charged atoms (cation and anion) leave their corresponding lattice sites and create a pair of Vacancy Defects. Since both cation and anion leave the lattice sites at the same time, so overall electrical neutrality of the crystal is maintained; however, density reduces because of the vacancies. Schottky defects occur in ionic crystals where the size of anion is almost same with the size of the cation.

Frenkel Defect is one type of point defect where an atom (better to say ion, especially cation) leaves its original lattice site and occupies an interstitial position on the same crystal. Usually, this type of defect is observed in ionic solids, where size of anion is substantially larger than the size of cation.

54. Define the terms primitive and non-primitive unit cells.

A primitive cell is the simplest type of unit cell which contains only one lattice point per unit cell (contains lattice points at its corner only).

If there are more than one lattice points in a unit cell, it is called a non-primitive cell.

55. Write Packing factor for BCC and FCC structures.

Packing factor of BCC is 0.68 and packing factor of FCC is 0.74.

56.Name the crystal structure of the following (a).Gold (b) Germanium (c).Barium (d) Zinc

(a).Gold – FCC (b) Germanium –Diamond cubic

(c).Barium – BBC (d) Zinc – HCP

57.What is meant by grain boundaries?

A grain boundary is the interface between two grains, or crystallites, in a polycrystalline material.

58.Bismuth has a=b=c=4.74AU and angle α=β= =60º what is its crystal structure?

a=b=c=4.74AU α=β=r=60º

Bismuth is Trigonal (or) Rhombohedral

59. State the condition imposed on the cell parameter for systems having the largest number of bravais lattices the least number of nearest neighbors.

(i) Crystal with least number of nearest neighbours is simple (six neighbours)

(ii) Crystal system with largest number of bravais lattice is orthorhombic (4 bravais lattices).

60. Define inter – atomic distance and inter planer distance.

Inter atomic distance: The distance between any two atoms is called inter atomic distance.

Inter planar distance: The distance between any two planers is called inter – planar distance.

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ENGINEERING CHEMISTRY

UNIT-I
WATER AND ITS TREATMENT
PART-A

1. What is meant by soft water and hard water?

Hard water:

Water, which does not produce lather with soap solution, but produces white precipitate, is called hard water. This is due to the presence of dissolved Ca and Mg salts.

Soft water:

Water which produces lather readily with soap solution is called soft water. This is due to the presence of Ca and Mg salts.

2. Distinguish between carbonate hardness and non-carbonate hardness.

S.No.	Carbonate hardness	Non-carbonate hardness
1	It is due to bicarbonates of Ca and Mg in water	It is due to chlorides and sulphates of Ca and Mg in water.

2	It can be removed by boiling the water.	It cannot be removed by boiling the water.
3	It is also called as alkaline hardness.	It is also called as non-alkaline hardness.

3. How is hardness of water expressed? (or) Bring out the significance of calcium carbonate equivalents (or) why is hardness expressed in terms of calcium carbonate equivalent?

The concentration of hardness producing salts is usually expressed in terms of an equivalent amount of $CaCO_3$.

Significance:

a) Its equivalent weight is 50.
b) Molecular weight is whole number (100)
c) It is the most insoluble salt and
d) Calculation can be simplified.

If the concentration of hardness producing salt is X mgs/lit, then

$$\text{Amount equivalent to } CaCO_3 = \frac{X \times \text{Molecular weight of Ca}}{\text{Molecular weight of hardness producing substance}}$$

4. Why is water softened before using in boiler? (or) write any disadvantages of hard water in boilers. (Apl-May 2015) (Apl-May 2017)

If hard water obtained from natural sources is fed directly into the boilers, the following troubles may arise.

- ❖ Scale and sludge formation
- ❖ Priming and foaming
- ❖ Caustic embrittlement
- ❖ Boiler corrosion.

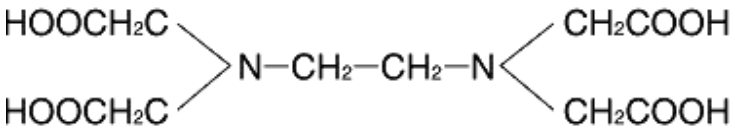
5. What are scales and sludges?

Sludge: If the precipitate is loose and slimy then it is called sludge. Sludges are formed by substances like $MgCl_2$, $MgCO_3$, $MgSO_4$, and $CaCl_2$. They have greater solubilities in hot water than cold water.

Scale: On the other hand, if the precipitate forms hard and adherent coating on the inner walls of the boiler then it is called scale. Scales are formed by substances like $Ca(HCO_3)_2$, $CaSO_4$ and $MgCl_2$.

6. Draw the structure of EDTA. What happens when EDTA is added to hard water?

EDTA – Ethylene Diamine Tetra Acetic Acid



When EDTA is added to hard water, it forms stable complex producing ions like Ca and Mg.

pH=8-10

$$[Ca^{2+}, Mg^{2+}] + \text{EDTA} \longrightarrow [\text{Ca or Mg} - \text{EDTA}] \text{ stable complex}$$

Hard water

7. How does Erichrome Black T indicator function as an indicator in EDTA titration?
(or) Why NH_4Cl-NH_4OH buffer is used in EDTA titration.

This complexation by EDTA and EBT with Ca^{2+} and Mg^{2+} ions are possible only at pH= 8- 10, that can be maintained by this buffer.

$$\underset{\text{Hardness causing ions}}{[Ca^{2+},Mg^{2+}]} + \underset{\text{indicator}}{EBT} \xrightarrow{pH=8\text{-}10} \underset{\text{unstable complex}}{[Ca \text{ or } Mg - EBT]} \text{ wine red colour}$$

$$[Ca \text{ or } Mg - EBT] + EDTA \xrightarrow{pH=8\text{-}10} \underset{\text{stable complex}}{[Ca \text{ or } Mg - EDTA]} + EBT$$

8. What is meant by reverse osmosis? How it is applied in the desalination of water? (Jan 2018)

If pressure in excess of osmotic pressure is applied on the higher concentration side, the solvent flow is reversed. Solvent flows from higher concentration (salt water) to lower concentration (water). This process is called reverse osmosis.

9. Define desalination.

The process of removing common salt from the water is known as desalination. The water containing dissolved salts with a peculiar salty or brackish taste is called brackish water.

10. Mention any two requirements of boiler feed water. (Nov/Dec 2016) (May/Jun 2016) (Apl/May 2018)

1. Boiler feed water should have zero hardness.
2. It must be free from dissolved gases and suspended impurities.
3. It should free from oil, & turbidity, TDS, dissolved & alkalinity salts.

11. Distinguish between hard and soft water.

S.No.	Hard water	Soft water
1	It does not produce lather with soap solution.	It produces very good lather with soap solution.
2	It gives wine red colour with EBT indicator.	It does not give colour with EBT.

12. What are the disadvantages of scale formation?

It acts as thermal insulators. It decreases the efficiency of boiler. Any crack developed on the scale, leads to explosion.

13. Calculate the hardness of water sample containing 2.4 mg of calcium chloride in 500 ml of water.

The amount of $CaCl_2$ =2.4 mgs/500ml (or) The amount of $CaCl_2$ = 4.8 mgs/lit

$$\text{Amount equivalent to } CaCO_3 = \frac{\text{The amount of hardness producing salt}}{\text{Molecular weight of hardness producing salt}} \times 100$$

We know that, the molecular weight of $CaCl_2$ = 111

$$\text{Amount equivalent to } CaCO_3 = \frac{4.8 \times 100}{111} = 4.324 \text{ mgs/lit.}$$

14. Define softening of water. How it is carried out?

The process of removing hardness producing salts from water is known as softening or conditioning of water. It can be done in two methods.

1. External treatment 2. Internal treatment

15. Give some examples for cation and anion exchange resins.

Cation exchange resins: Sulphonated coals, sulphonated polystyrene

Anion exchange resins: Cross linked quaternary ammonium salts, urea – formaldehyde resin.

16. What is calgon conditioning? (May-June 2014) (Nov/Dec 2016) (May/Jun 2016)

Calgon is sodium hexa meta phosphate $Na_2[Na_4(PO_3)_6]$. This substance interacts with calcium ions forming a highly soluble complex and thus prevents the precipitation of scale forming salt.

$$2\,CaSO_4 + Na_2[Na_4(PO_3)_6] \rightarrow Na_2[Ca_2(PO_3)_6] + 2\,Na_2SO_4$$

17. What is blow down operation?

Blow-down operation is a process of removing a portion of concentrated water by fresh water frequently from the boiler during steam production.

18. Why is hard water unsuitable for boilers?

The hard water is not suitable for boilers due to the following reasons. If you use hard water in boilers it may arise

- ✓ Scale and sludge formation,
- ✓ Priming and foaming
- ✓ Caustic embrittlement
- ✓ Boiler corrosion

19. What is phosphate conditioning? (or) What is the role of phosphates in the internal treatment of water?

Scale formation can be avoided by adding sodium phosphate. It is used in high pressure boilers. The phosphate reacts with calcium and magnesium salts to give soft sludges of calcium and magnesium phosphates.

$$3\,CaSO_4 + 2\,Na_3PO_4 \rightarrow Ca_3(PO_4)_2 + 3\,Na_2SO_4$$

20. How is exhausted ion-exchange resins regenerated?

When the cation exchange resin is exhausted, it can be regenerated by passing a solution of dil. HCl or dil. H_2SO_4.

$$RCa + 2HCl \rightarrow RH_2 + CaCl_2$$
$$RNa + HCl \rightarrow RH + NaCl$$

Similarly, when the anion exchange resin is exhausted, it can be regenerated by passing a solution of dil. NaOH.

$$R'Cl_2 + 2NaOH \rightarrow R'(OH)_2 + 2NaCl$$

21. Mention the units to measure hardness.

- Parts per million (ppm)
- Milligrams per litre(mg/lit)
- Clarke's degree(°Cl)
- French degree (°Fr).

22. What is hardness of water?

Hardness is the property or characteristics of water, which does not produce lather with soap solution.

$$2C_{17}H_{35}COONa + CaCl_2 \rightarrow (C_{17}H_{35}COO)_2 Ca + 2NaCl$$

23. Mention any two disadvantages of formation of deposits in steam boilers. (Nov-Dec 2015)

1. Wastage of fuels:

Since scales have low thermal conductivity, the consumption of fuels are more.

2. Decrease in efficiency:

Scales deposits in the valves and condensers of the boiler leads in decreases the efficiency of boiler.

3. Boiler explosion:

Scales develops large amount of steam and high pressure inside the boiler which leads to explosion.

24.100 ml of a sample of water required 25.0 ml of 0.01 M EDTA for the titration using Erichrome Black T indicator. Calculate the total hardness.

We know that,

1 ml of 0.01 M EDTA = 1 mg of $CaCO_3$

25 ml of 0.01 M EDTA = 25mg of $CaCO_3$

100 ml of sample of water required = 25 mg of $CaCO_3$

= 25.0 mgs of $CaCO_3$ equivalent

1000 ml of water = $25 \times \frac{1000}{100}$ mg of $CaCO_3$ equivalent

Total hardness =250 mgs/lit or ppm

25. Distinguish between internal and external conditioning of water. (Nov-Dec 2014)

S.No	Internal conditioning	External conditioning
1	It takes place inside the boiler	It takes place outside the boiler
2	Removal of scale forming substances by adding chemicals directly into the boilers. Ex. Sodium carbonate, Sodium phosphate	Removal of hardness producing salts from the water before feeding into the boiler. eg., Demineralisation process, Zeolite process.

26. What are boiler compounds? (Apl-May 2015)

Removal of scale forming substances by adding chemicals directly into the boilers. eg. Sodium carbonate, Sodium phosphate.

27. Name any two salts that cause temporary hardness. (Jan 2018)

$Ca(HCO_3)$, $Mg(HCO_3)$

28. How you will remove temporary hardness in water? (Apl/May 2018)

- By boiling of water
- By adding lime to the water

PART-B

1. Write short notes on :
   (a) Phosphate conditioning **(Apl-May 2015)** (b) Scales and sludges **(Apl-May 2015) (Apl-May 2017) (Apl/May 2018)** (c) Calgon conditioning **(Nov-Dec 2014)**
2. What is desalination? With a neat diagram, describe the reverse osmosis for the desalination of brackish water. **(Nov-Dec 2014)**
3. What are boiler troubles? How can scale formation be prevented by calgon conditioning? **(Nov-Dec 2014) (Nov-Dec 2015) (Jan 2018)**
4. Distinguish between softening and demineralization of water. Name any two methods of demineralization of water and explain briefly. (or) Draw a suitable diagram and describe the ion exchange process for softening of hard water. **(May-June2014) (Nov-Dec 2015) (Nov/Dec 2016) (May/Jun 2016) (Apl-May 2017)**
5. What is meant by reverse osmosis? Explain the purification of water by reverse Osmosis. **(Apl-May 2015) (Nov/Dec 2016) (May/Jun 2016) (Apl-May 2017)**
6. Explain the softening of water by zeolite process? **(May-June2014) (Nov-Dec 2014) (Apl-May 2015) (Nov/Dec 2016) (May/Jun 2016) (Apl/May 2018)**
7. Describe briefly the various methods of internal conditioning of boiler feed water. **(May-June2014) (Apl-May 2015) (May/Jun 2016) (Apl-May 2017) (Jan 2018)**
8. What are the requirements of boiler feed water? **(Nov-Dec 2015) (Jan 2018)**
9. Distinguish between internal and external conditioning treatment of boiler water**. (Jan 2018).**
10. Explain estimation of hardness of water by EDTA method.

UNIT-II

SURFACE CHEMISTRY AND CATALYSIS

PART-A

1. Define Adsorption and Adsorbate.

(i) Adsorption: The phenomenon of concentration of molecules of a gas or liquid at a solid surface is called adsorption. The adsorption of gas on a solid is sometimes called occlusion.
(ii) Adsorbate: The substance which is held on the surface of the solid is called Adsorbate.

2. What is sorption?

Sorption is the process in which both adsorption and absorption takes place simultaneously.

3. What is chemisorption? Give an example.

Chemisorption (or) chemical adsorption is the one, in which the adsorbed molecule, are held on the surface of the adsorbent by chemical bonds (Covalent bond (or) Ionic bond).

Example: Adsorption of H_2 on Ni.

4. What is physical adsorption (or) physisorption? Give an example.

Physical adsorption is the one, in which the adsorbed molecules are held on the surface of the adsorbate by weak physical (or) vander waal's force of attraction.

Example: Adsorption of H_2 (or) O_2 on charcoal.

5. Mention some important characteristics of adsorption.

1. Adsorption on surface of a solid is always spontaneous.

2. Adsorption is always accompanied by evolution of heat.
3. Adsorption is accompanied both by decrease in enthalpy and entropy of a system.
4. Adsorption is a selective process.

6. How does chemisorption differ from physisorption? (OR)
Write any two differences between Physisorption and Chemisorption.

S.No	Physisorption	Chemisorption
1	It is caused by intermolecular Vander Waal's force (weak).	It is caused by chemical bond formation (strong).
2	Heat of adsorption is low (20-40 k.cal/mole)	Heat of adsorption is high (40-400 k.cal/mole)
3	Adsorption is completely reversible.	Adsorption is irreversible.
4	Adsorption decrease with increase in temperature.	Adsorption increase in temperature.
5	Multilayer adsorption occurs.	Monolayer adsorption occurs.

7. How will you increase the activity of an adsorbent?

Activation leads to increase in the surface area of the adsorbent, which increases adsorption. Activation is achieved by the following ways.

(i) Creation of rough surface

(a) by mechanical rubbing of the solid adsorbents.

(b) by subjecting to some chemical reactions on the solid adsorbent.

(ii) Increasing effecting area of the surface

(a) by sub-dividing the solid adsorbent into finer particles.

(b) by heating of solid adsorbent in superheated steam, now its pores are opened and adsorption increases.

8. Explain the effect of temperature on adsorption.

Physical adsorption: It occurs rapidly at lower temperature and decreases with increase in temperature.

Chemical adsorption: It increases with increase in temperature and then decreases.

9. Explain the function of activated charcoal with example.

(i) It adsorbs coloring matter present in sugar solutions.

(ii) It also adsorbs certain acids like CH_3COOH and $(COOH)_2$ present in water, thereby acid concentration in water decrease.

(iii) It also adsorbs out NH_3 from the solution of NH_4OH and phenolphthalein.

10. Define adsorption. What is adsorption isotherm?

Adsorption: The phenomenon of concentration of molecules of a gas or liquid at a solid surface is called adsorption.

Adsorption isotherm: Adsorption isotherm is a relationship between magnitudes of adsorption with pressure at a constant temperature.

$$\frac{x}{m} = KP^{1/n}$$

11. What is Freundlich's adsorption isotherm?

The relationship between the magnitude of adsorption (x/m) and pressure (P) can be expressed mathematically by an equation known as Freundlich's adsorption isotherm

$$\frac{x}{m}=KP^{1/n}$$

12. Explain the limitations of Freundlich's adsorption isotherm.

1. Freundlich equation is purely empirical and has no theoretical basis.
2. The equation is valid only upto a certain pressure and invalid at higher pressure.
3. The constants K and 'n' are not temperature independents, they vary with temperature.
4. Freundlich's adsorption isotherm fails, when the concentration of adsorbate is very high.

13. What is Langmuir Adsorption isotherm? How it is mathematically represented?

The relation between the amounts of gas adsorbed to the pressure of the gas at constant temperature is known as Langmuir Adsorption isotherm.

It is mathematically represented as $x= K^{'}\frac{KP}{1+KP}$

14. What is the demerit of Langmuir's Adsorption isotherm?

Langmuir's Adsorption isotherm holds good at lower pressure but fails at high pressure.

15. What are promoters?

Promoters are defined as, "the substance which increase the activity of a catalyst".

16. What is catalytic poisoning?

Catalytic poison is defined as, "a substance which destroys the activity of the catalyst to accelerate a reaction". This process is called catalytic poisoning.

17. What is the effect of temperature and pressure on the adsorption of hydrogen gas on charcoal?

Adsorption of H_2 on charcoal is rapid at lower temperature and decrease in temperature, but the rate of adsorption increases with the increase of pressure.

18. How is arsenic poisoning removed from the body?

Colloidal ferric hydroxide is administrated, which adsorbs arsenic poison and is removed from the body by vomiting.

19. Define ion-exchange adsorption.

Ion-exchange adsorption is defined as, "the process of releasing the ion and adsorbing another like ion".

20. How is evaporation of water in lake minimized?

Due to scarcity of water during summer a layer of stearic acid is spread over water lakes and reservoirs. The adsorbed stearic acid on the surface of water minimizes evaporation of water.

21. What are the important applications of catalysis?

1. Catalysis impacts the environment by increasing the efficiency of industrial processes.
2. Many fine chemicals are prepared by using catalysis.
3. Zeolites are used as catalysts in petrochemical industries for cracking of hydrocarbons.

22. What are the differences between homogenous and heterogeneous catalysis?

S.No	Homogenous catalysis	Heterogenous catalysis
1	Reactant, Products and catalyst are present in only one phase ie., either in liquid (or) gaseous phase.	Reactant, Products and catalyst are present in different phase ie., either in liquid & gaseous phase
2	Catalyst cannot be regenerated.	Catalyst cannot be regenerated and activated.
3	Thermal stability of catalyst is low.	Thermal stability of catalyst is high.
4	Temperature increases the rate of the reaction.	Temperature increases the rate of the reaction.

23. Define auto catalysis.

When one of the products of a reaction itself acts as a catalyst for that the reaction catalyst is called auto-catalyst. This process (or) phenomenon is called auto-catalysis.

Examples: 1. Hydrolysis of an ester.
2. Oxidation of oxalic acid.
3. Decomposition of Arsine.

24. Mention the functions of catalytic converter.

The following are some important functions of the catalytic converter.

1. Reduction of nitrogen oxides into elemental nitrogen and oxygen.
   $NO_X \rightarrow N_X + O_X$
2. Oxidation of CO into CO_2.
3. Oxidation of hydrocarbons into CO_2 and H_2O.

25. List any four characteristics of enzyme catalysis. (Jan 2018)

1. More specific
2. More efficiency
3. More reactivity at optimum temperature
4. More reactivity at optimum pH

26. What are auto catalysts? Give example? (Jan 2018)

If one of the product itself act as catalyst then that is known as auto catalyst.

Ex: Hydrolysis of ester.

PART - B

1. Distinguish between physical adsorption and chemical adsorption.
2. Discuss the factors influencing adsorption of gases on solids.
3. Define adsorption isotherm. Explain the various types of adsorption isotherm. **(Jan 2018)**
4. Derive Langmuir adsorption isotherm and interpret the results at i) low pressure & ii) high pressure. **(Jan 2018)**

5. Define the term adsorption and explain its applications.
6. Explain the role of adsorption in catalytic reaction or adsorption theory or contact theory.
7. Discuss the role of activated carbon in pollution abatement of air and waste water.
8. Discuss the kinetics of enzyme catalysis.
9. What are acid-base catalysis? Explain the mechanism of acid-base catalysis.
10. Explain the functions of catalytic convertors with the neat diagram.
11. Derive Michael's – Menton equation. **(Jan 2018)**
12. Give four applications of adsorption. **(Jan 2018)**

UNIT-III

ALLOYS AND PHASE RULE

PART - A

1. What are the uses of a phase diagram? (April/May 2017) (Jan 2018)

i) It is possible to predict from the phase diagrams whether a eutectic alloy or a solid solution is formed on cooling a homogeneous liquid containing mixture of two metals.

ii) The phase diagrams are useful in understanding the properties of materials in the heterogeneous equilibrium system.

iii) The study of low melting eutectic alloys, used in soldering, can be carried out using phase diagrams.

2. Give the condensed / reduced phase rule. (Jan 2014) (April/May 2015)

The system studied under constant pressure in which only the solid and liquid phases are considered and the gas phase is ignored is called a condensed system.

$$F' = C - P + 1$$

3. Define the terms Annealing and Hardening.

Annealing: Annealing mean softening. This is done by heating the metal to high temperature followed by slow cooling in a furnace.

Hardening / Quenching: It is the process of heating steel beyond the critical temperature and them suddenly cooling it either in oil or brine water or some other fluid. Hardening increases the harness of steel. The faster the rate of cooling harder will be the steel produced. Medium and high carbon steels can be hardened, but low carbon steels cannot be hardened.

4. Mention any four properties of Nichrome.

i) It shows good resistance to oxidation and heat.

ii) Steels containing 16 to 20 % chromium with low carbon content (0.06 to 0.15%) possess oxidation resistance up to 900°C.

iii) Steel containing 18% nickel, with small amounts of chromium can withstand Temperature above 900°C.

iv) It possesses high melting point.

5. Calculate the number of components, phases and degrees of freedom for the following

Equilibrium. (NOV / DEC 2016)

$$NH_4Cl_{(s)} \leftrightarrow NH_{3(g)} + HCl_{(g)}$$

The system consists of two phases namely solid NH_4Cl and the gaseous mixture containing NH_3 + HCl. When NH_3 and HCl are present in equivalent quantities the composition of both the phases can be represented by the same chemical compound NH_4Cl and hence the system will be a one component system.

Hence, P = 2; C = 1; Then, Degrees of Freedom, F = C – P + 2;

= 1 – 2 + 2; = 1

So, degree of freedom is one.

6. Mention any five advantages/purposes of alloy making.

i) To increase the hardness of the metal.
ii) To resist the corrosion of the metal.
iii) To lower the melting points of the metal.
iv) To modify chemical activity of the metal.
v) To get good casting of metal.

7. State the number of degrees of freedom for the following systems. (Nov / Dec 2016)

i) PCl_5 (s) ↔ PCl_3 (g) + Cl_2 (g) at 50°C

i) F = C – P + 1;
= 2 – 3 + 2;
= 1.

ii) $CaCO_3$ (s) ↔ CaO (s) + CO_2 (g)

ii) F = C – P + 2;
= 2 – 3 + 2;
= 1.

iii) $MgCO_3$ (s) ↔ MgO (s) + CO_2 (g)

iii) F = C – P + 1;
= 2 – 3 + 2;
= 1.

8. What are the uses of eutectic mixture?

i) Suitable alloy composition can be predicted with the help of eutectic systems.
ii) Eutectic systems are used in preparing solders, used for joining two metal pieces together.

9. What is meant by quenching in heat treatment of metals?

It is the process of heating steel/metal beyond the critical temperature and then suddenly cooling it either in oil or brine water or some other fluid.

10. State phase rule and explain the terms involved?

If the equilibrium between any number of phases is not influenced by gravity, or electrical, or magnetic forces but are influenced only by pressure, temperature, and concentration then the number of degree of freedom (F) of the system is related to number of components (C) and number of phases (P) by the following phase rule equation.

F = C – P + 2

11. Explain the terms component and degree of freedom? (Nov/Dec 2015) (Jan 2018)

Component is defined as "the smallest number of independently variable constituents, by means of which the composition of each phase can be expressed in the form of a chemical equation".

Degree of freedom is defined as, "the minimum number of independent variable factors such as temperature, pressure and concentration, which must be fixed in order to define the system completely".

12. What is eutectic or eutectic point?

It is the point at which two solid and one liquid phase are in equilibrium.

Eg. Solid Pb + Solid Ag ⇆ Melt

13. How is thermal analysis conducted? What are the applications of thermal analysis curves?

Thermal analysis is a method involving a study of the cooling curves of various compositions of a system during solidification. The shapes of the freezing point curves for any system can e determined by thermal analysis. The form of the cooling curve indicates the composition of the solid.

Applications:

i) Melting point and eutectic temperature can be noted from the cooling curve.

ii) Percentage purity of the compounds can be noted from the cooing curve.

14. Define Phase (P).

Phase is defined as, "any homogeneous physically distinct and mechanically separable portion of a system which is separated from other parts of the system by definite boundaries.

15. What is triple point? (May-June 2014)

It is the point at which three phases namely solid, liquid and vapour are simultaneously at equilibrium.

16. How many phases and components are existing at triple point in ice - water – water vapour system?

Phases – 3; Components – 1.

17. Define an alloy. (Jan 2014)

An alloy is defined as "homogeneous solid solution of two or more different elements, one of which at least is essentially a metal".

18. Give the composition of Rose metal.

Bi=50% Pb=28% Sn=22%

19. Name some important heat treatment process.

The main heat treatment processes are,

i) Annealing
ii) Hardening
iii) Tempering
iv) Normalizing
v) Carburizing
vi) Nitriding

20. What is tempering?

It is the process of heating the already hardened steel to a temperature lower than its own hardening temperature and then cooling it slowly.

21. What is heat treatment of alloys?

Heat treatment is defined as, "the process of heating and cooling of solid steel article under carefully controlled conditions".

22. Give the composition of Gun metal

Cu=85% Zn=4% Sn=8% Pb=3%

23. Calculate the number of phases in the following systems

1. Emulsion of oil in water
   Phases=2
2. $MgCO_3(s) \leftrightarrow MgO(s) + CO_2(s)$
   Phases=3
3. Ice(s) $\leftrightarrow$ Water (l) $\leftrightarrow$ Water vapour (g)
   Phases=3

24. Give the composition of Nichrome and its uses. (May-June 2014)

Composition: Ni=60% Cr=12% Fe=26% Mn=2%

Uses: Used in making parts of boilers, Making resistant coils in stoves

25. Define stainless steels. Give its any two applications.

These are alloy steels containing chromium (16% or more) together with other elements such as Nickel, Molybdenum etc. Carbon content ranges from 0.3 % to 1.5%.

Applications:

1. They are very good resistant towards weather and water.
2. They are used in making surgical instruments.

26. How many phase, components and degrees of freedom are available in this equilibrium?

$NH_4Cl_{(s)} \rightarrow NH_{3\,(g)} + HCl_{(g)}$ (April-May 2018)

The system consists two phases and one components P=2, C=1

F=C-P+2 =1-2+2=1

27. Write the chemical compositions of Dutch metal and Gun metal

Dutch metal: 80% Copper and 20% Zinc

Gun metal; 85% Copper, 8% of tin, 4% Zinc and 3% lead

28. What is alnico?

It is an alloy of 12% of Al, 20% of Ni,6% of Co and 62% of Fe.

29. What is 18/8 stainless steel? (NOV / DEC 2016)

If the stainless steel contains 18% of Cr and 8% of Ni then it is called as 18/8 stainless steel**.**

It is widely used than all other stainless steel.

30. A system consists of benzene and water. What is the number of phases?

Two liquid phase and one vapour phase.

31. What is the degree of freedom at eutectic point in lead–silver system? (NOV/DEC 2014)

Solid lead + solid silver ↔ melt

F' = C-P+1 (F' = 2-3+1) F'=0 (Non- variant)

32. Mention any two significance of alloy making. (NOV / DEC 2014)

- To increase the hardness of the metal
- To lower the melting point of the metals.

33. What is hardening/ quenching of steel ? Mention it's is purpose. (April/May 2015)

Process of heating of steel beyond the critical temperature and then sudden cooling it either in oil or brine water.

Purpose: It increases it is resistance to wear and abrasion, ability to cut other metals.

34. Define degree of freedom. (May/June 2016) (April/May 2017)

Degree of freedom is defined as the minimum number of independent variable factors such as temperature, pressure and concentration is must be fixed in order to define the system completely**.**

PART – B

1. State phase rule and explain the terms involved in it. **(May / June 2014)**
2. Draw and explain the phase diagram of one component water system. **(Jan 2014) (Nov / Dec 2014) (April / May 2015) (Nov/Dec 2015) (May/June 2016) (Nov / Dec 2016) (April/May 2017) (April/May 2018)**
3. Explain the two component system with an example. (or) Explain the lead – silver system**.(May / June 2014) (April / May 2015) (Nov/Dec 2015) (May/June 2016) (Nov / Dec 2016) (Jan 2018)**
4. What are the significance of purpose of alloying? **(April / May 2015) (Nov/Dec 2015)**
5. What are the Function and effect of alloying elements? **(May/June 2016)**
6. Explain the heat treatment of steel/alloy. **(Jan 2014) (May / June 2014) (Nov / Dec 2014) (Nov/Dec 2015) (May/June 2016) (Nov / Dec 2016) (April/May 2017) (April/May 2018)**
7. Give the composition, properties and uses of the ferrous alloys. **(May / June 2014) (April / May 2015)**
8. With two cooling curves for pure substance and mixture , discuss briefly about thermal analysis. **(Jan 2018)**
9. What is thermal analysis? Draw the cooling curve of a pure substance and a mixture and discuss.

UNIT- IV

FUELS AND COMBUSTION

PART-A

1.What are the different varieties of coal? How do you rank them in the order of increasing calorific value? (Jan 2018)

Coal is classified on the basis of its rank. The rank of coal indicates its degree of maturity. Various types of coal are Wood-peat-lignite-bituminous coal-anthracite. Calorific value increases from left to right.

2.What is meant by the term "Fixed Carbon"?

It is the pure non-volatile, carbon content present in the coal. Higher the percentage of fixed carbon greater is its calorific value.

3.What is carbonization of coal?

When bituminous coal is heated strongly in the absence of air, the volatile matter escapes out and the mass becomes hard, strong, porous and coherent which is called Metallurgical coke. This process is called carbonization.

4.Give the characteristics of metallurgical coke.

i) Purity: The moisture, ash, sulphur contents in metallurgical coke should be low.
ii) Porosity: Coke should be highly porous.
iii) Strength: It should have high mechanical strength.
iv) Calorific value: The calorific value of coke should be high.

5. How is coke superior to coal? Or Distinguish between coal and coke.

i) Percentage of fixed carbon and hence the calorific value of coke is high.
ii) Percentage of moisture, volatile and ash contents are higher in coal, where as they are low in coke.

6. Why is coke used in metallurgical process than coal?

i) Percentage of fixed carbon in coke is more.
ii) The moisture, ash, sulphur contents are very low.
iii) The mechanical strength, calorific value is high.

7. Explain the term "knocking" of a petrol engine.

Knocking is a kind of explosion due to rapid pressure rise occurring in petrol engine. This can be reduced by adding TEL. ie: Tetraethyl lead.

8. Define octane number of petrol? How it can be improved? (Nov-Dec 2014)

Octane number is defined as, the percentage of iso-octane present in a mixture of Iso-octane and n-heptane. It can be improved by adding anti-knock agents like TEL.

9. Define cetane number (or) cetane rating of a diesel oil? How it can be improved? (Nov/Dec 2016)

It is defined as the percentage of cetane present in a mixture of cetane and alpha-methyl cetane. It can be improved by adding dopes like ethyl nitrate.

10. What is leaded petrol? Give its significance in automobiles.

When the petrol is mixed with TEL it is called leaded petrol. Knocking in petrol engine is minimized.

11. How will you improve the anti-knocking characteristics of diesel?

Anti-knocking characteristics of diesel can be improved by increasing n-cetane value of the fuel. Cetane value can be increased by adding dopes like ethyl nitrate, iso-amyl nitrate.

12. What is ignition temperature?

It is the lowest temperature to which the fuel must be heated, so that it starts burning smoothly.

13. Define spontaneous ignition temperature.

It is defined as "the minimum temperature at which the fuel catches fire (ignites) spontaneously without external heating".

14. Name the important units of calorific values. (Jan 2018)

i) Calorie
ii) Kilocalorie
iii) British thermal unit (B.T.U.)
iv) Centigrade Heat Unit (C.H.U.)

15. What are the reagents used in flue gas analysis? Indicate their functions.

Reagents	Absorbent
KOH solution	CO_2
Alkaline Pyrogallic acid	O_2
Ammoniacal. Cuprous. chloride	CO

16. Mention combustible and non-combustible constituents present in the fuel.

Combustible constituents: C, H, S and O
Non-combustible constituents: N, CO_2

17. Calculate the Dulongs formula for the calculation of GCV and NCV.

GCV= 1/100 (8080C+34 500(H-O/8) + 2240S)K.Cal/Kg

NCV= (HCV- 9/100 H*587) Kcal/Kg

18. Mention the uses of flue gas analysis.

Flue gas analysis gives an idea about the complete or incomplete combustion process. If the flue gases contain considerable amount of CO, it indicates that incomplete combustion. If the glue gases contain considerable amount of O_2, it indicates that complete combustion.

19. What is CNG? Mention its primary component and advantages. (Apl-May 2015) (Apl-May 2017)

When the natural gas is compressed, it is called Compressed Natural Gas (CNG). The primary component present in CNG is methane. It is mainly derived from natural gas.

Advantages:

1. It is cheapest, cleanest and least environmentally impacting alternating fuel
2. Less expensive than petrol and diesel.
3. Its calorific value is 8,000 kcal/m^3.

20. Write the characteristics of good fuel.

The characteristics of good fuel are as follows.

i. It should have a high calorific value.
ii. It is cheap and readily available.
iii. It should undergo spontaneous combustion.
iv. It should have very low non-combustible matter.

21. What are the requisites of good coke for metallurgy?

Purity: The moisture, ash, sulphur contents in metallurgical coke should be low.

Porosity: Coke should be highly porous.

Strength: It should have high mechanical strength.

Calorific value: The calorific value of coke should be high.

22. What is LPG? (Apl/May 2018)

LPG is dehydrated, desulphurised and traces of organic sulphides (mercaptans) are added to give warning of gas leak.

The average composition of LPG is,

Propane	-	24.7 %
Butane	-	38.5 %
Isobutane	-	36.7 %
Others	-	0.1 %

23. What is the significance of analyzing sulphur in coal? (or) what are the drawbacks of sulphur in coal? (Apl-May 2015)

Significance:

Though sulphur increases the calorific value, its presence in coal is undesirable because,

i. The combustion products of sulphur, i.e., SO_2 and SO_3 are harmful and have corrosion effects on equipments.
ii. The coal containing sulphur is not suitable for the preparation of metallurgical coke as it affects the properties of the metal.

24. What do you meant by synthetic petrol? Give the suitable reaction.

The preparation of liquid fuels from solid coal is called hydrogenation of coal or synthetic petrol.

$$nCO + 2n\ H_2 \rightarrow C_nH_{2n} + n\ H_2O$$
$$(2n+1)\ H_2 + n\ CO \rightarrow C_n\ H_{2n+2} + n\ H_2O$$

25. The ultimate analysis of a coal sample indicates Carbon=84%, Sulphur=1.5%, Nitrogen=0.6%, Hydrogen5.5% and Oxygen=8.4%. Calculate the GCV.

GCV=1/100[8080C+34500(H-O/8) +2240S] kcal/kg

GCV= 1/100[8080X84+34500(5.5-8.4/8) +2240X1.5] kcal/kg

= 1/100[678720+34500(4.45) +3360] kcal/kg

= 8356.05 kcal/kg

26. Write the expression for the amount of air required for combustion of 1kg fuel Amount of oxygen equivalent for combustion of 1 kg of fuel

= [32/12*C + 8{H-O/8) + S]kg

Hence, the amount of air required for the combustion of 1 kg of fuel

= 100/23[32/12*C + 8{H-O/8) + S]kg

27. What is the difference between caking coal and coking coal?

S.No	Caking coal	Coking coal
1	When coals are heated strongly, the mass becomes soft, plastic and fuses to give a coherent mass. Such types of coals are called caking coals	If the mass so produced is hard, porous and strong then the coals are called coking coals. It has lower volatile matter and used for manufacture of metallurgical coke.

28. What is the calorific value of coal? (Nov-Dec 2015) (May/Jun 2016) (Jan 2018)

It is defined as "the total amount of heat liberated when a unit mass of coal is burnt completely".

i) Calorie, ii) Kilocalorie

iii) British thermal unit (B.T.U.), iv) Centigrade Heat Unit (C.H.U.)

29. What is power alcohol? Mention the advantage. (May-June2014)

When ethyl alcohol is blended with petrol at concentration of 5-10%, it is called power alcohol.

Advantage :

- It is cheaper than petrol.

- If any moisture is present power alcohol absorbs it.

30. Define explosive range of fuel. (May-June2014) (Apl/May 2018)

Explosive range or explosive limit is the limiting composition of a gas air mixture beyond which the mixture will not ignite and continue to burn is called explosive range or explosive limit.

31. What is bio diesel? Mention its advantage. (Nov/Dec 2016)

Triglycerides of long chain fatty acids derived from natural sources.

Advantage:

Bio degradable
Less pollutant
Renewable energy.

32. Distinguish between Proximate and Ultimate analysis

S. No	**Proximate analysis**	**Ultimate analysis**
1	It involves determinations of physical constituents like moister, volatile, ash and fixed carbon contents in coal	It involves determinations of chemical constituents like carbon, hydrogen, nitrogen, oxygen and sulphur contents in coal
2	It gives approximate composition of main constituent of coal	It gives the exact composition of the elemental constituents of coal

33. What is a natural gas? Mention its uses?

Natural gas is always found above the oil in the oil wells. It is also called Marsh gas. It consist of methane and other saturated hydrocarbons.

It is used as domestic and industrial fuels, raw material for the manufacture of carbon black & hydrogen and for generation of electricity by using fuel cell.

PART – B

1. Discuss briefly proximate analysis of coal and their significance? (or)What is the importance of proximate analysis **(Nov-Dec 2015) (Apl-May 2015) (May/Jun 2016) (Apl/May 2018)**
2. Discuss the ultimate analysis of coal? **(May-June2014) (Apl-May 2017) (Jan 2018)**
3. Describe the Otto-Hoffman method of coke manufacture and the recovery of various byproducts? / Describe any one method of manufacturing metallurgical coke. **(Nov-Dec 2014) (Apl-May 2015) (Nov-Dec 2015) (Nov/Dec 2016) (May/Jun 2016)**
4. What is meant by cracking of petroleum?
5. How synthetic petrol is obtained by Bergius process? **(May-June2014) (Apl/May 2018)**
6. What is CNG? Justify why is it preferred? Write short notes on the following
   i) Compressed Natural Gas ii) Power alcohol. **(Nov-Dec 2015)**
7. Describe the manufacture of producer gas. Give its composition and uses.
8. Describe the manufacture of water gas. Give its composition. Why is it superior to Producer gas? **(Nov-Dec 2014) (Nov/Dec 2016) (May/Jun 2016) (Apl-May 2017)**

9. Explain the flue gas analysis by orsat method. **(Nov/Dec 2016)(May-June2014) (Nov-Dec 2014) (May/Jun 2016) (Apl-May 2017) (Jan 2018)**
10. Write short notes on Octane and Cetane number.
11. Calculate the volume of air required for the complete combustion of $1m^3$ of gaseous fuel having the composition. CO=46% CH_4= 10% H_2=4% C_2H_2=2% N_2=1% and remaining being CO_2 **(May-June2014)**
12. Write a note on (i) power alcohol and (ii) biodiesel **(Nov-Dec 2014)**
13. What is meant by knocking in petrol engine? How is knocking prevented? **(Apl-May 2015)**
14. Write short notes on the following: **(Nov/Dec 2016)** 1. Ignition Temperature 2. Explosive range
15. What is Bio-diesel? Explain transesterification and advantages of Bio-diesel. **(Apl-May 2017) (Apl/May 2018)**

UNIT-V
ENERGY SOURCES AND STORAGE DEVICES

PART-A

1. Define nuclear fission. Write all the nuclear fission reactions of $_{92}U^{236}$. (Nov-Dec 2015) (Jan 2018)

Nuclear fission is defined as the process of splitting of heavier into two (or) more similar nuclei with simultaneous liberation of large amount of energy.

$$_{92}U^{235} + _0n^1 \rightarrow [\ _{92}U^{236}] \rightarrow {}_{56}Ba^{139} + {}_{36}Kr^{94} + 3_0n^1 + \text{Energy}$$

$$\searrow \quad {}_{54}Xe^{144} + {}_{38}Sr^{90} + 2_0n^1 + \text{Energy}$$

$$\searrow \quad {}_{55}Cs^{144} + {}_{37}Rb^{90} + 2_0n^1 + \text{Energy}$$

2. What is nuclear fusion? Give an example. (Jan 2018)

The process of combination of lighter nuclei into heavier nuclei with simultaneous liberation of large amount of energy. Nuclear fusion occurs in Sun.

$$_1H^1 + {}_1H^1 + {}_1H^1 + {}_1H^1 \rightarrow {}_2He^4 + 2\ {}_{+1}e^0$$

3. What is nuclear chain reaction? (Apl-May 2017)

A fission reaction, where the neutrons from the previous step continue to propagate and repeat the reaction is called nuclear chain reaction.

4. Give any two differences between nuclear fission and fusion. (Nov-Dec 2014) (May/Jun 2016)

S.No	Nuclear fission	Nuclear fusion
1	It is the process of breaking a heavier nucleus	It is the process of combination of lighter nuclei.
2	It emits radioactive rays	It does not emits any kind of radioactive rays
3	It occurs at ordinary temperature.	It occurs at high temperature.($>10^6$K)

5. What is Nuclear energy? Explain using a suitable example

The energy released by the nuclear fission is called nuclear fission energy (or) Nuclear energy.

Example: When U^{235} nucleus is hit by a thermal neutron, the following reaction occurs with the release of energy.

$$_{92}U^{235} + {}_0n^1 \rightarrow {}_{56}Ba^{139} + {}_{36}Kr^{94} + 3_0n^1 + \text{Energy}$$

6. Give any two nuclear fission reactions, mention the factors that impede the nuclear chain reaction.

$$_{92}U^{235} + {}_0n^1 \rightarrow [{}_{92}U^{236}] \rightarrow {}_{56}Ba^{140} + {}_{36}Kr^{93} + 3_0n^1$$

The factors that impede the nuclear chain reaction are

1. Some of the neutrons may escape from the surface to the surroundings
2. Some of the neutrons may be absorbed by U^{238} present as impurity

7. What is a nuclear reactor?

The arrangement or equipment use to carry out fission reaction under controlled conditions is called nuclear reactor.

8. What are the types of nuclear fission reaction?

The nuclear fission reactions are of two types.

1. Uncontrolled fission reactions - Atom bomb
2. Controlled fission reactions - Nuclear bomb

9. What are the general components of a nuclear reactor?

1. Fuel rods
2. Control rods
3. Coolants.
4. Moderators
5. Pressure vessel.

10. What is breeder reactor? (Nov/Dec 2016) (Apl/May 2018)

Breeder reactor is the one which converts non-fissionable material($U^{238,}$ Th^{232}) into fissionable material (U^{235}, Pu^{239})

11. What is photo galvanic cell (or) Solar cell?

Photo galvanic cell is the one, which converts the solar energy (energy obtained from the sun) directly into electrical energy. It consists of a p-type semiconductor (such as Si doped with P). They are in close contact with each other.

12. What are the electrodes used in the fuel cells porous?

Compressed carbon containing a small amount of catalyst like Pt, Pd, Ag are used in the fuel cells porous.

13. Write any four methods adopted for harnessing wind energy?

Sky sail

1. Ladder mill
2. Kite ship (large free flying sails)
3. Sky wind power (Flying electric generator)
4. Briza technologies (Hovering wind turbine)
5. Sequoia automation (The kite wind generator)

14. How is the reactor wastes disposed?

Nuclear fission products like $Ba^{139,}$ $Kr^{92,}$, Mo^{98}, Ba^{141} are themselves radioactive. i.e., they emit dangerous radiation for several hundred years. Hence, the reactor wastes are placed in concrete barrels which are buried deep in the sea.

15. What is meant by nuclear binding energy?

Loss of mass defect during the formation of the nucleus from nucleons is converted into energy. The release of energy imparts stability to the nucleus. The energy released when constituent nucleons combine to form a nucleus, is called binding energy of the nucleus.

16. Define light water nuclear power plant.

Light water nuclear power plant is the one, in which U^{235} fuel rods are submerged in water. Here, water acts as both moderator and coolant.

17. What are nuclear moderators? Give two examples

The substances used to slow down speed of the neutrons are called moderator.

Ex - Graphite, Cadmium rod.

18. Explain super critical mass and sub- critical mass.

If the mass of fissionable material is more than the critical mass, it is called super critical mass.

If the mass of fissionable material is smaller than the critical mass, it is called sub-critical mass.

19. What is a primary battery? Give an example. (or) what are primary cells?

Primary cells are cells in which the electrode and the electrode reactions cannot be reversed by passing an external electrical energy. The reactions occur only once and after use they become dead. Therefore, they are not chargeable. Example: **Leclanche's cell**

20. Write the overall equation for the reaction taking place in an alkaline battery?

At anode,

$$Zn_{(s)} + 2OH_{(aq)} \rightarrow Zn(OH)_2 + 2e^-$$

At cathode,

$$2MnO_{2(s)} + H_2O_{(l)} + 2e^- \rightarrow Mn_2O_{3(s)} + 2OH^-$$

Overall cell reaction:

$$Zn_{(s)} + 2MnO_{2(s)} + H_2O_{(l)} \rightarrow Zn(OH)_{2(s)} + Mn_2O_{3(s)}$$

21. State the reaction when a lead storage battery is recharged.

The cell can be recharged by passing electric current in the opposite direction. The electrode reaction gets reversed. As a result, is deposited on anode and PbO2 on the cathode. The density of H_2SO_4 also increases.

The net reaction during charge is,

$$PbSO_4 + H_2O + Energy \leftrightarrow Pb_{(s)} + PbO_{2\,(s)} + 2H_2SO_{4(aq)}$$

22. Write the charging and discharging reaction of lead accumulator.

$$Pb_{(s)} + PbO_{2\,(s)} + 2H_2SO_{4(aq)} \leftrightarrow PbSO_{4\,(s)}\,(s) + 2H_2O + Energy$$

23. Lithium battery is the cell of future. Why? Or what are the advantages of using lithium as anode in batteries? (Jan 2018)

- Its cell voltage is high, 3.0 V.
- Since Li has the most negative E_0 value, it generates a higher voltage than the other types of cells
- Since Li is a light-weight metal, only 7g (1 mole)material is required to produce 1 mole of electrons

24. Mention few characteristics of nuclear fission.

- A heavy nucleus (U^{235} or Pu^{239}) when bombarded by slow moving neutrons, split into two or more nuclei
- Two or more neutrons are produced by fission of each nucleus
- Large quantities of energy is produced s a result of conversion of small mass of nucleus into energy

25. What is wind energy? How is it harnessed?(or) Write how wind energy is generated. (Nov-Dec 2015)

Moving air is called wind. Energy recovered from the fore of the wind is called wind energy. The energy possessed by wind is because of its high speed. The wind energy is harnessed by making use of wind mills.

26. What are non-conventional energy sources?

- ✓ Wind energy
- ✓ Solar energy
- ✓ Hydropower
- ✓ Tidal energy

27. What are fissile and fertile nucleides?

- The fissional nucleides like U^{235} and Pu^{239} are called fissile nucleides.
- The non fissionable nucleides such as U^{238} and Th^{232} are called fertile nucleides.

28. What are the applications of solar cells?

- ❖ Solar cells can be used to drive vehicles.
- ❖ They can be used for lighting purposes.
- ❖ Solar cells are used in boilers to produce hot water for domestic and industrial uses.
- ❖ They can be used to produce hydrogen by electrolysis of water.

29. What are the applications of lithium batteries?

Button sized Li batteries are used in calculators, watches, cameras, mobile phones, laptop, computers.

30. What are the drawbacks of nuclear energy? (May-June2014)

Nuclear fission products like $Ba^{139,}$ $Kr^{92,}$, Mo^{98}, Ba^{141} are themselves radioactive. i.e., they emit dangerous radiation for several hundred years.

31. Will the emf of a battery vary with size? Give reason. (May-June2014)

No, the emf of a battery will not vary with size.

Reason: EMF of a battery depends only on concentration and nature of anode and cathode.

32. What are batteries? (Nov-Dec 2014) (May/Jun 2016)

Batteries are an arrangement of several electrochemical cells, connected in series that can be used as a source of direct electric current.

33. What are the limitations of hydrogen – oxygen fuel cell? (Apl-May 2015)

1. Hydrogen gas is explosive.
2. It is very expensive to be carried out.

34. Point out the advantages of wind energy. (Nov/Dec 2016)

1. No pollution
2. Renewable energy
3. Available at all time

35. Alkaline battery is superior to dry cell. How?

Zn does not dissolve readily in a basic medium

The life of alkaline battery is longer than dry cell

Alkaline battery maintains its voltage, as the current is drawn from it

36. What is fuel cell?

Fuel cell is a voltaic cell, which converts the chemical energy of the fuels directly into electricity without combustion. In these cells, the reactants, products and electrolytes pass through the cell.

37. Write a short note on super capacitors.

Super capacitor is a high capacity capacitor with capacitance value much higher than other capacitor. They store 10 to 100 times more energy per unit volume and deliver charge much faster than batteries. Super capacitors, do not use the conventional solid dielectric, but rather they use electrostatic double layer capacitance.

38. Write two applications of H2-O2 fuel cell. (Apl/May 2018)

- Auxiliary energy sources in space and military vehicles
- Fresh water is produced

PART-B

1. What are the main parts of light water nuclear reactor? What is a breeder reactor? **(May-June2014) (Nov-Dec 2014) (Nov/Dec 2016) (Apl-May 2017) (Apl/May 2018)**
2. State the principle and application of solar batteries.
3. What are fuel cells? Describe the construction and working of H_2-O_2 fuel cell. **(May-June2014) (Nov-Dec 2014) (Nov/Dec 2016) (May/Jun 2016) (Apl-May 2017)**
4. Explain the power generation from light water nuclear reactor. (or) Explain the method of conversion of nuclear energy into electrical energy in a nuclear reactor. **(May-June2014) (Nov-Dec 2015). (Apl-May 2015) (Jan 2018)**
5. Describe the conversion methods of solar energy. **(Nov/Dec 2016) (Apl/May 2018)**
6. What are the components of a nuclear reactor? Write briefly about each component.
7. Write briefly about the advantages and limitations of the wind energy. **(Apl-May 2015) (May/Jun 2016) (Apl/May 2018)**
8. Discuss about the principle and functions of an alkaline battery.
9. Write a note on photovoltaic cell/solar cell. **(Nov-Dec 2014) (Apl-May 2015)**
10. Give an account on the different methods by which solar energy can be harnessed?
11. What is reversible battery? Describe the Construction and working of Lead acid storage battery with reactions occurring during charging and discharging cycles. **(Jan 2013) (May-June2014) (Nov-Dec 2014) (Nov-Dec 2015) (May/Jun 2016) (Jan 2018) (Apl/May 2018)**
12. Define nuclear fission .Explain its characteristics?

13. Explain the nuclear chain reaction with a suitable example.
14. Write a short note on lithium batteries. (or) How is solid state lithium battery constructed .Describe its functioning. **(Nov-Dec 2015) (Apl-May 2017)**
15. Describe the nuclear fusion reaction. **(Nov-Dec 2015)**
16. Explain about breeder reactor**. (May/Jun 2016)**
17. Describe the construction and working of hydrogen-oxygen fuel cell. **(Jan 2013) (Dec 2014)(May 2017)**
20. Explain the construction, working of Lechlanche's cell
21. Brief about the super capacitor.

UNIT I

2 MARKS

1. **Define computers?**

A computer is a programmable machine or device that performs pre-defined or programmed computations or controls operations that are expressible in numerical or logical terms at high speed and with great accuracy. Computer is a fast operating electronic device, which automatically accepts and store input data, processes them and produces results under the direction of step by step program.

2. **What are the basic operations of Computer?**

- It accepts data or instructions by way of input.
- It stores data.
- It controls all operations inside a computer.
- It can process data as required by the user.
- It gives results in the form of output.

3. **What are the components of the computer systems?**

Basic components of the computer system are Input Unit, Central Processing Unit, Secondary Storage Unit and Output Unit.

4. **What is algorithm? (JAN2009)**

Algorithm means the logic of a program. It is a step-by-step description of how to arrive at a solution of a given problem.

5. **What are the steps to solve the problem in a computer system?**

- Problem must be analyzed thoroughly.
- Solution method is broken down into a sequence of small tasks.
- Based on this analysis, an algorithm must be prepared to solve the problem.
- The Computer program is fed to the computer.
- The instruction in the program executes one after another and outputs the expected result.

6. What are the characteristics of an algorithm?

- In algorithms each and every instruction should be precise.
- In algorithms each and every instruction should be unambiguous.
- The instructions in an algorithm should not be repeated infinitely.
- Ensure that the algorithm will ultimately terminate.
- The algorithm should be written in sequence.
- It looks like normal English.

7. What is Flowchart?

A Flowchart is a pictorial representation of an algorithm. It is often used by programmer as a program planning tool for organizing a sequence of step necessary to solve a problem by a computer.

8. What is the need of Flowchart symbols?

Each symbol of different shapes denotes different shapes denote different types of

instructions. The program logic through flowcharts is made easier through the use of symbol that has standardized planning.

9. What is pseudo code?

"Pseudo" means imitation of false and "code" refers to the instruction written in the programming language. Pseudo code is programming analysis tool that is used for planning program logic.

10. What are the languages used in computer generations?

Generations	Languages used
I Generation	Machine Language.
II Generation	Assemble Language
III Generation	High Level Language, BASIC, PASCAL, COBOL, FORTRON

11. How can you measure the quality of algorithm?

The primary factors that are often used to judge the quality of an algorithm are time requirement, memory requirement, and accuracy of solution.

12. How many types the Algorithm can be represented?

- Normal English
- Program
- Flowchart
- Pseudo code

13. What is the need of Flowchart symbols?

Each symbol of different shapes denotes different shapes denote different types of instructions. The program logic through flowcharts is made easier through the use of symbol that has standardized planning

14. What are the rules for drawing a flowchart?

- The standard symbols should only be used.

- The arrowheads in the flowchart represent the direction of flow of control in the problem.
- The usual direction of the flow of procedure is from top to bottom or left to right.
- The flow lines should not cross each other.
- Be consistent in using names and variables in the flowchart.
- Keep the flowchart as simple as possible.
- Words in the flowchart symbols should be common statements and easy to understand.
- Chart main line of logic, and then incorporate all the details of logic.
- If a new page is needed for flowcharting, then use connectors for better representation.

15. **List the different building blocks of algorithms.**

The algorithm can be constructed from basic building blocks. These building blocks are,

- Instructions/statements
- Control flow
- State
- Functions

16. **What is coding?**

Translating an algorithm into a programming language is called coding the algorithm.

17. **Define iteration.**

The iteration is a technique that repeats the execution of a block of instructions. Computers are often used to automate repetitive tasks. Repeating identical or similar tasks without making errors.

18. **Define recursion.**

Recursion is a way of programming or coding a problem, in which a function calls itself one or more times in its body. Usually, it is returning the return value of this function call. If a function definition fulfils the condition of recursion, we call this function a recursive function.

19. **What is the difference between program and algorithm?**

Program: A program is a set of instruction given to a computer to solve a problem**.**

Algorithm: An algorithm is a procedure or finite sequence of steps that provides the solution to a problem.

20. **What are the benefits of pseudocode?**

- Pseudocode is language independent code. It can be used by most programmers.
- It is easy to develop a program from pseudocode. Programmers can focus on the steps to solve a problem rather than on how to use the computer language.
- It is easy to translate pseudocode into a program in high level programming language.

21. **List the factors to judge the the quality of the algorithms?**

- Time
- Memory
- Accuracy

- Sequence
- Generability

22. Types of ways to represent an algorithm.

- Normal English
- Flowchart
- Pseudocode
- Decision table
- Program

23. **what are the basic parts of an algorithm?**

- Description of problem
- Set up
- Parameters
- Execution
- Conclusion

24. **What are rules for writing pseudocode.**

- Write one statement per line
- Capitalize initial keywords
- Indent to show hieraechy
- End multiline structure
- Keep statements language independent

25. **write a pseudocode to calculate student total and average?.**

READ name, class, m1,m2,m3 Total=m1+m2+m3 Average=TOTAL/3

WRITE name, class, total,average

UNIT I 16 MARKS

1. Explain about algorithm, pseudo code and flowchart with an example of finding the sum of 'n' numbers.
2. Draw a flow chart to check whether the given number is zero, positive or negative.
3. What is pseudo code? Write a pseudo code for swapping two numbers without using temporary storage.
4. Discuss the need for logical analysis with an example in brief.
5. Explain the guidelines for preparing flowchart, benefits and limitation of flowcharts.
6. Draw a flowchart to find the highest mark in a set of n marks.
7. Draw the flowchart and write an algorithm to sort a list of numbers in ascending order.
8. Draw and explain the various symbols of flowchart and also draw the flowchart to add an array of N elements.

9. Write a pseudo code to multiply two matrices.

10. What is an algorithm? Write an algorithm to print numbers from 2 to 100.

UNIT- II

2 marks

1. What is interpreter mode in python?

The interpreter mode is the mode where the scripted and finished .py files are in the python interpreter.

2. What is interactive mode in python?

The interactive mode is a command shell which gives immediate feedback for each statement.fed statements in active memory. As new lines are fed into the interpreter. The fed program is evaluated both in part and in whole.

3. Define Keyword.

Keywords are reserved words that have predefined meanings in python. They cannot be used as ordinary identifiers and must be spelled exactly as they are written python3 has 33 keywords.

4. What are escape sequences?

Escape sequences are in non printable characters. It consists of backslash followed by a character both are enclosed within single quotes.

5. Write any four escape sequence in python.

- \\ Backslash(\)
- \' single quote(')
- \" Double quote(")
- \a ASCII Bell(BEL)

6. Define Value.

A Value is the basic units of data, like a number or string that a program manipulates.

7. What are the types used in python?

Types are category of values. Integers(type in), Floating point numbers(type float),strings(type str),Booleans(type bool) and lists are a set of predefined data types in python. They are called as built-in-types.

8. What is a variable?

A Boolean type represents special values True and False. The most common way to produce a Boolean value is with a relational operator.

9. What is expression?

An expression is on combination of values, variables and operators. A value by itself is considered as an expression.Expressions, most commanly,consists of a combination of operators and operands.

Examples:5-(8*k).

10. Give example for simultaneous assignment statements.

Simultaneous Assignment Statements:

Python permits any number of variables to appear as on the left side separated by commas. The same number of expressions must then appear on the right side, again separated by commas.

Examples:

```
>>>x=10

>>>y=5

>>>sum,diff,prod=x+y,x-y,x

>>>sum 15

>>>diff 5

>>>pr 50
```

11. What is a tuples? Give example.

A tuples is a immutable sequence of values. They are comma separated list of values. The values can be any type, and they are indexed by integer.

Example:

```
>>>t='a','b','c','d','e'

>>>t=('a','b','c','d','e')
```

12. How to create a tuple.

Creating a tuple:

i) To create a tuple with a single element, final comma is to be included.

```
>>>t1='a'
```

ii) Another way to create a tuple is the built-in function tuple. With no argument, it creates empty tuple:

```
>>>t=tuple()

>>>t 0
```

iii) If the argument is a sequence(string, list or tuple),the result is a tuple with the elements of the sequence:

```
>>>t=tuple ('jovita')

>>>t ('j','o','v','i','t','a')
```

13. List out the rules to be followed in naming variable.

1. Variable names can be at any length.

2. They can contain both letters and numbers, but they can't begin with a number.
3. Both uppercase letters and lower case letters can be used.
4. The underscore (_) character can appear in a name. It is often used in names with multiple words.
5. A variable name cannot be any one of the keywords.

14. Give the precedence of operators is python.

Parentheses have the highest precedence and can be used to force an expression to evaluate in the order.

Example: 5*(9-3)=30 since expression in parentheses are evaluated first. Exponentiation has the next highest precedence.

Example: 1+2**3=9,not27and 2*3**2=18,not36.

Multiplication and division has higher than Addition and Subtraction. Example:2*3-1=5,not4and

6+4/2=8

15. What is module?

A Python module is a file containing python definitions and statements. The python Standard Library is a collection of built-in modules, each providing specific functionality beyond what is included in the core part of python.

16. Define function.

A function is a named sequence of statements that performs a computation function may or may not take arguments and may or may not produce a result.

17. What are the advantages of using function?

Advantages of using functions in python:

1. Reduction in code redundancy
2. Enabling code reuse
3. Better readability
4. Easy to debug and testing
5. Improved maintainability.

18. Give the syntax of function definition.

Syntax of function defenition

def function_name(parameters): //function header statements(s) //function body

19. Define parameter and argument.

Parameter: A name used inside a function to refer to the value passed as an argument. Argument: A value provided to a function when the function is called. This value is assigned to the corresponding parameter in the function.

20. List out the rules to be followed in naming an identifier.

1. An identifier can be a combination of uppercase letters, lowercase letters, underscores, and digits(0-9)
2. Special characters such as %,@, and $ are not allowed within identifiers.

3. Python keyword cannot be used as identifiers.

21. List python standard/built-in data type.

1. None type
2. Numeric types
3. Sequences
4. Sets
5. Mapping

22. Define python numbers.

Python can handle normal long integers, maximum length determined based on the operating system,python long integers maximum length dependent on available

memory,floating point numbers, octal and hexa decimal numbers and complex numbers,i.e., numbers with an imaginary component.

23. List the python numeric types.

1. Integers
2. Floating point numbers
3. Complex numbers

24. What are the methods for providing data to the program?

There two methods for providing data to the program.

1. Assigning the data to the variables in a program.
2. By usinh the input/output statements.

25. What are the elements of user defined function?

1. Function definition
2. Function invocation

UNIT-II 16 MARKS

1. Explain the following with suitable example.

   1) Tuple assignment

   2) Precedence of operators

2. Define function. Explain function definitions and its uses with examples.

3. Write a python program to exchange the values of two variables.

4. Write a python program to test for leap year. year=int(input("enter a year"); if(year%4)==0:

if(year%100)==0:

if(year%400)==0:

print("{0} is a leap year".format(year)) else:

print("{0} is not a leap year".format(year)) else:

print("{0} is a leap year".format(year)) else:

print("{0} is not a leap year".format(year))

5. Explain the function parameters and arguments in detail.
6. Briefly explain the conditional operators with suitable examples.
7. What are all the decision making statements. Explain in detail.
8. Write a python program circulate the values of n variables.
9. Explain the interactive and interpreter mode in detail.
10. What is iteration and give an example for while condition.

UNIT-III 2 MARKS

1. **What is a bool**?

A bool is a Boolean value that is either true or false.

2. **What are the various type of python operator**?

    1. Arithmetic Operators.
    2. Relational Operators.
    3. Logical Operators.
    4. Bitwise Operators.
    5. Assignment Operators.
    6. Special Operators.
        a) Identity Operators
        b) Membership Operators.

3. **What is floor division or truncating division? Given example**. Floor division or truncating division:

It is denoted by a double slash,//,providing a truncated result based on the type of operands applied to.

When both operands are integer values, the result is a truncated integer referred to as integer division.

Example:7//5=1.

When atleast one of the operands is float type, the result is a truncated floating point.Example:**7//5.0+1.0 and 7.0//5.0 becomes1.0.**

4. **What is the identity operators used in python**?

- Is and is not are the identity operators in python.
- They are used to check if two values (or variables) are location on the same part of the memory.
- Two variables that are equal don't imply that they are identical.
- The identity operators in python are,

Operator Meaning Example

Is True if the operands are identical (refer to x is True The same object)

Is not True if the operands are not identical (do

Not refer to the same object) x is not True

5. **What is the membership operators used in python**?

In and not in are the membership operators in python. They are used to test whether a value or variable is found in a sequence (string, list, tuple, set and dictionary).

In a dictionary they test for presence of key, not the value

Operator Meaning Example

In True if value/variable is found in the 5 in x Sequence

not in True if value/variable is not found in

the sequence. 5 not in x

6. **Differentiate break and continue**.

BREAK: The break statement terminates the loop containing it and the control of the program goes to the statement immediately after the body of the loop.

CONTINUE: The continue statement is used to skip the reset of the code inside a loop for the current iteration only. Loop does not terminate but continues on with the next iteration.

7. What is local and global scope?

A local variable that is only accessible within a given function, such variable is a variable that is defined outside of any function definition. Such variable are said to have global scope.

8. Define composition.

Composition is the ability to make small building blocks (variable, expression, and statement) and compose them.

9. Define recursion.

A function that calls itself is recursive; the process of executing is called recursion. Recursion can be used to solve the problem that can be expressed in terms of similar problems of small size.

10. What is a string? How is it accessed?

A string is a sequence of character. A string can be accessed by the characters one at a time with the bracket operator.

Example:

>>>fruit=’banner’

>>>letter=fruit[i]

>>>letter ‘a’

11. What is a slice? Give example.

A segment of a string is called a slice. The operator [n:m] returns the part of the string from the “m-eth” character, including the first but excluding the first but excluding the last. Selecting a slice is similar to selecting a character.

Example:

>>>a=’python programming’

>>>a[0:05]

‘pytho’

12. What is split function? Give example.

The split function strip leading or trailing white space from a string.

Example:

>>>s=’python programming’

>>>s.split()

[‘python’, ‘programming’]

13. Distinguish between Call by value Call by reference.

Call by value	**Call by reference**
a) In call by value, the value of actual agreements is passed to the formal arguments and the operation is done on formal arguments.	a) In call by reference, the address of actual argument values is passed to formal argument values.
b) Formal arguments values are photocopies of actual arguments values.	b) Formal arguments values are pointers to the actual argument values.
c) Changes made in formal arguments valued do not affect the actual arguments values.	c) Since Address is passed, the changes made in the both arguments values are permanent

14. Give some important points while using return statement.

The return statement is followed by an expression which is evaluated. Its result is returned to the caller as the “fruit” of calling this function. Because the return statement can contain any Python expression we could have avoided creating the temporary variable y and simply used return x*x.

15. Explain default argument in python.

Python's default arguments are evaluated once when the function is defined, not each time the function is called. This means that if you use a mutable default argument and mutate it, you will and have mutated that object for all future calls to the function as well.

16. Explain the anonymous function (or) lamba function in python.

In Python, anonymous function is a function that is defined without a name. While normal functions are defined using the def keyword, in Python anonymous functions are defined using the lambda keyword. Hence, anonymous functions are also called lambda functions.

17. Define the map() function in python.

The map function is the simplest one among Python built-ins used for functional programming. These tools apply functions to sequences and other iterables. The filter filters out items based on a test function which is a filter and apply functions to pairs of item and running result which is reduce.

18. How to access characters in a string?

We can access individual characters using indexing and a range of characters using slicing. Index starts from 0. Trying to access a character out of index range will raise an IndexError. The index must be an integer. We can't use float or other types, this will result into TypeError.

19. How to concatenate two strings?

Joining of two or more strings into a single one is called concatenation. The + operator does this in Python. Simply writing two string literals together also concatenates them. The * operator can be used to repeat the string for a given number of times.

20. How to create a list in python?

In Python programming, a list is created by placing all the items (elements) inside a square bracket [ ], separated by commas. It can have any number of items and they may be of different types (integer, float, string etc.).

21. Define python decorators.

A decorator is function that accepts a function as parameter and returns a function. A decorators are used to add functionality to an existing code. Decorators are powerful feature of thepython programming language.

22. List python’s built-in functions.

1.abs()

2.all()

3.any()

4.ascii()

5.bin()

6.Bool()

23. How to create an array in python.

The arrays can hold similar data type elements. So the type should be specified by using a type code at a time of crating the array object.

Syntax: arrayname= array(type code, [elements])

24. define indexing on arrays.

An index represents the position number of an element in an array. For example when we create the following integer type array:

A=array('I',[1,2,3,4,5])

UNIT III

16 MARKS

1. What are the different operators available in python? Explain with examples.
2. Explain different data type in python with examples.
3. Describe the statements for decision making and looping?
4. Explain the following conditional statements?
    i) if(conditional)
    ii) f-else(alternative)
    iii) if-elif-else(chained conditional).
5. Briefly explain fruitful function in python.
6. Briefly explain local and global scope.
7. Write a program to find the square root of a number using Newton's method.
8. Write python program to find GCD of two numbers.
9. Write a python program to find sum of array of numbers.
10. Explain the string functions and methods with examples.

Unit IV

TWO MARKS

1. Define the python list.

Lists in python are one of the most versatile collection object types available, other two types are dictionaries and tuples, but they are really more like variations of lists. A list is similar to array but a list can store different types of elements, whereas an array can store only one type of elements.

2. Define the uses of lists in python.

List are most often used to store homogeneous values, i.e. a list usually holds names

,numbers, or other sequences that are all one data type. The most common use of a list is to iterate over the list and perform the same action to each object within in the list, hence the use of similar data types.

3. Explain the python updating lists.

It can be used to update single or multiple elements of lists by giving the slice on the left hand side of the assignment operator and it can used to add elements in a list with the append().

4. Explain python delete list elements.

Remove () or del statement is used to remove a list elements. Del statement is used to remove elements which are exactly known. Remove method is used to remove elements which does not known.

5. Define the python list index.

Index operator [] is used to access an item in a list. Index starts from 0, so a list having 5 elements will have index from 0 to 4. Trying to access an element other that this will raise an IndexError.

6. Explain the slice list in python.

Slice list can access a range of items in a list by using the slicing operator(colon). Slicing can be visulalized by considering the index to be between the elements. So if we want to access a range we need two index that will slice that portion from the list.

7. Define Tuples.

A tuple is a sequence of immutable python objects. Tuples are sequences, like lists. The difference between tuples and list are the tuples cannot be changed unlike list and tuples are used paranthesis whereas list use [].

8. Describe buit-in list functions.

Python includes the following list functions −

SN	Function with Description
1	**cmp(list1,list2)** Compares elements of both lists.
2	**len(list)** Gives the total length of the list.

3	**max(list)** Returns item from the list with max value.
4	**min(list)** Returns item from the list with min value.

	list(seq) Converts a tuple into list.
5	

9. Describe buit-in list methods.

Python includes following list methods

SN	Methods with Description
1	**list.append(obj)** Appends object obj to list
2	**list.count(obj)** Returns count of how many times obj occurs in list
3	**list.extend(seq)** Appends the contents of seq to list
4	**list.index(obj)** Returns the lowest index in list that obj appears
5	**list.insert(index,obj)** Inserts object obj into list at offset index
6	**list.pop(obj=list[-1])** Removes and returns last object or obj from list

10. List the various Accessing Elements in a Tuple.?

1. Indexing
2. Negative indexing
3. Changing a Tuple.

11. List the basic tuple operations.

1. Deleting a tuple.

2. python tuple methods.

3. Tuple memebereship operators. 4.Iterating through a tuple.

12. Define methods in a Dictionaries?

Dictionaries have a method called items that returns list of tuples, where each tuple is a key-value pair. The Dictionary method update also takes a list of tuples and adds them as key- value pairs to an existing dictionary.

13. How we create a Dictionary?

Creating a dictionary is as simple as placing items inside curly braces {} separated by comma. An item has a key and the corresponding value expressed as a pair, key:value. While values can be of any data type and can repeat, keys must be of immutable type and must be unique.

14. List the properties of Dictionary keys?

1. More than one entry per key is not allowed.

2. Keys must be immutable.

15. How to change or add elements in a dictionary?

Dictionaries are immutable. We can add new items or change the value of existing items using assignment operator. If the key is already present, a value gets updated, else a new key: value pair is added to the dictionary.

16. How to delete or remove elements from a dictionary?

We can remove a particular item in a dictionary by using the method pop().This method removes an item with the provided key and returns the value. The method removes an item with the provided key and returns the value. The method, popitem() can be used to remove and return an arbitrary item(key, value) from the dictionary. All the items can be removed at once using the clear() method

17. How to process a list in a python using function?.

To process lists in large programs, we have to pass them to the function, and pass individual elements of our list to function like other variables or an entire list can be passed. An entire list can be transferred to a function as a parameter. To transfer list to a function, the list name is enough without subscripts as a parameter within the function call.

18. Define concatenation of two lists?.

The lists can be concatenated by using the '+' operator, for example 'a' and 'b' are two lists then we can concatenate a+b, where the list 'b' is joined at the end of the list.

19. What you meant by repetition of lists?

The list elements can be repeated by using the '*' operator in python. For example, if we write a*n, where the list 'a' will be repeated for 'n' times.

20. Describe list membership?

Python's membership operators test for membership in a sequence, such as strings,lists or

tuples.

There are two membership operators 1.in

2.notin

21. Define list comprehension?

List comprehension is a elegant and concise way to create new list from an existing lists in python. List comprehensions offer a succinct way to create lists based on existing lists. When using list comprehensions, lists can be built by leveraging any iterable, including strings and tuples.

22. Define sorting?

Sorting and searching are fundamental operations in computer science. Sorting refers to the operation of arranging data in some given order.

23. Define Bubble sort?

Bubble sort is one of the easiest sorting methods. In this method, each data item is compared with its neighbor and if it is a decend sorting, then the bigger number is moved to the top of all, the smaller numbers are slowly moved to the bottom position, hence it is called as the exchange sort.

24. List tuple built-in function?

1.len()

2.max()

3.min()

4.sum()

5.any()

What is aliasing?

Aliasing is a circumstances where two or more variables refer to the object and assign b=a, then both variables refer to the same

object. Example:

>>>a=[1,2,3]

>>>b=a

>>>b is a True

25. What is meant by cloning?

Cloning list is to make a copy of the list itself, not just the reference. The easiest way to clone a list is to use the slice operator.

UNIT IV 16 MARKS

1. Explain in detail about python lists.
2. Briefly explain basic tuple operations in python
3. Write a python program to sort using bubble sort.
4. Write a python program to search a element using binary search.
5. Briefly explain python dictionary.
6. Explain python tuples.
7. Explain python list methods.
8. Explain python basic list operations.
9. Write a program to subtract two matrices using nested loop.
10. Write a program to transpose a matrix.

UNIT-5 2 MARKS

What is a file?

A text file is a sequence of characters stored on a permanent medium like hard drive, flash memory or CD-ROM.

What is format operator?

Format operator is an operator (%),that takes a format string and a tuple as input and generates a string that includes the elements of tuple, formatted as specified by the format string. When applied to integers,% is the modulus operator. But when the first operand is a string, % is the format operator.

What is command Line argument?

Python provides a getopt module that helps to parse command-line options and arguments. The command line arguments in python can be processed by using either 'sys' module or 'argparse' module.

$python test.py arg1 arg2 arg3

What is module?

Modules are file containing python statements and definitions. Definitions from a module can be imported into other modules or into the main module, the collection of variables that you have access to in a script executed at the top level and in calculator mode.

What are packages?

A package is just a way of collecting related modules together within a single tree like hierarchy. A well organized hierarchy of directories for easier access. Python has packages for directories and modules for files.

What are exceptions?

Errors can also occur at runtime and these are called exceptions.

Example: A file trying to open does not exists (FileNotFoundError), dividing a number by zero (ZeroDivisionError), module trying to import a module and it is not found(ImportError) etc, are exceptions.

Explain the advantages of files.

When the data is stored in a file, it is stored permanently. This means that even through the computer is switched off, the data cannot removed from the memory since the file is stored on permanent memory. This file data can be utilized as and when required. It is used to store huge amount of data.

Difference between Local and Global variable.

Global: These variables can be accessed (i.e known) by any function comprising the program. They are implemented by associating memory locations with variable names. They do not get recreated if the function is recalled.

Local: These variables only exist inside the specific function that creates them. They are unknown to other functions and to the main program.

Briefly explain python reading and writing files.

A text file is a sequence of characters stored on a permanent medium like a hard drive, flash memory, or CD-ROM. To write a file, you have to open it with mode 'w' as a second parameter:

>>> fout = open('output.txt', 'w')

If the file already exists, opening it in write mode clears out the old data and starts fresh, so be careful! If the file doesn't exist, a new one is created.

The write method puts data into the file.

>>> line1 = "This here's the wattle,\n"

>>> fout.write(line1)

Explain the python file in close() method.

The close() method of a file object flushes any unwritten information and closes the file object, after which no more writing can be done. Python automatically closes a file when the reference object of a file is reassigned to another file.

Write the syntax in python open() function.

Before we read or write a file, we have to open it using python's built-in open() function. This function creates a file object, which would be utilized to call other support methods associated with it.

Syntax: file object= open(file_name [,access_mode][buffering])

Explain the File Object Attributes.

Once a file is opened and one file object, we can get various information related to that file. List of all attributes related to file object:

file.closed

file.mode

file.name

file.softspcae

Explain the python file in read() method.

The read() method reads a string from an open file. It is important to note that python strings can have binary data, apart from text data.

Syntax: fileObject.read([count])

Explain the python file in write() method.

The write() method writes any string to an open file. The write() method does not add a newline character (‘\n’) to end of the string.

Syntax: fileObject.write(string)

Define the package semantics.

A package is a directory that contains modules. Having a directory of modules allows us to have modules contained within other modules. This allows us to use qualified module names, clarifying the organization of our software.

What is assert statement?

Python evaluates the accompanying expression, which is hopefully true. If the expression is false, python raises an AssertionError exception.

Syntax: assert Expression[, Arguments]

State the import statement.

Python source file as a module by executing an import statement in some other python source file. Syntax: import module1[, module2[,….. module]. When the interpreter encounters an import statement, it imports the module if the module is present in the search path.

Explain the raising exception.

Even if a statement or expression is syntactically correct, it may cause an error when an attempt is made to execute it. Errors detected during execution are called *exceptions* and are not unconditionally fatal.

State the variable scope in python.

Scope is the portion of the program from where a namespace can be accessed directly without any prefix. At any given moment, there are at least three nested scopes.

- Scope of the current function which has local names
    - Scope of the module which has global names
    - Outermost scope which has built-in names

Explain the catching exception python.

In Python, exceptions can be handled using a try statement. A critical operation which can raise exception is placed inside the try clause and the code that handles exception is written in except clause.

List the file types.

In Python, a file is categorized as either text or binary, and the difference between the two file types is importany. Text files are structured as a sequence of lines, where each line includes a sequence of characters.

How we perform searching through a file.

When you are searching through data in a file, it is a very common pattern to read through a file, ignoring most of the lines and only processing lines which meet a particular condition. We can combine the pattern for reading a file with string methods to build simple search mechanisms.

Define extracting files in python.

The extractall() method is used to extract all the files and folders from a zip file into the current working directory. It passes a folder name to extractcall() to extract all files and folders in a specific directory.

Define python pickle.

Pickle,which is part of the python library by default, is an important module whenever you need persistence between user sessions. As a module pickle provides for saving of python objects between processes.

How we make a directory in python.

The mkdir() method is used to make a new directory. This method takes in the path of the new directory. If the full path is not specified,the new directory is created in the current working directory.

>>>os.mkdir('NewDir')

>>>os.listdir()

UNIT-5 16 MARKS

1. What is a file? Explain the operations performed on a file.
2. What is exception? How they are handled in python.
3. Explain the following.
4. Modules
5. Packages
6. Explain the reading and writing operations in files.
7. Briefly explain the command line arguments in python.
8. Illustrate the concept of error handling and raising exception in python.
9. Write a program in python to count the words.
10. Write a program in python where a user tries to open a file when they do not have write permission, so it raises an exception.

Engineering Graphics Question Bank

UNIT – I

PLANE CURVES AND FREE HAND SKETCHING

PLANE CURVES (Eccentricity method only)

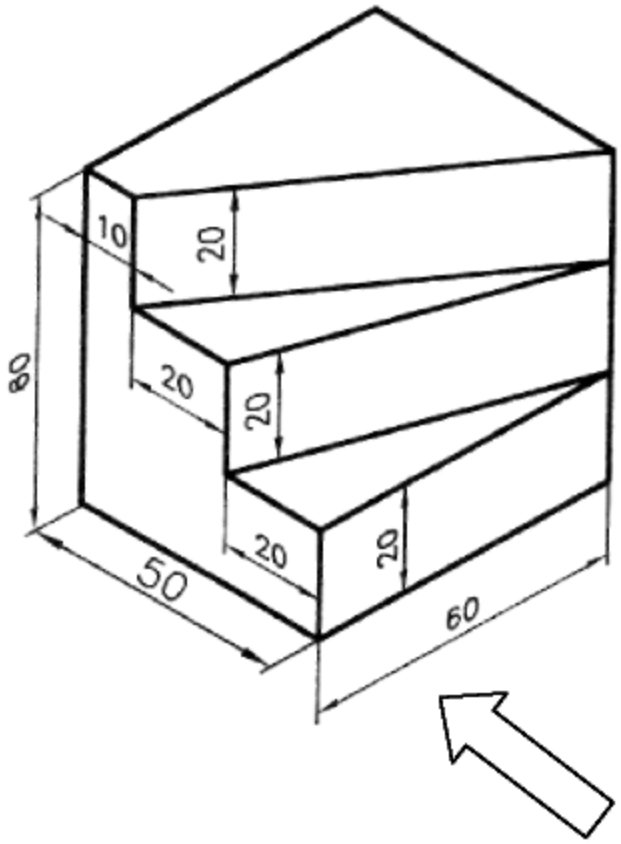
1. Draw an ellipse when the distance of focus from the directrix is 40 mm &eccentricity is ¾. Draw a tangent at a point of the ellipse.
2. Draw the coin cure, if the distance of focus from the directory is 70mm and the eccentricity is ¾. Also draw a tangent and a normal at any point on the curve.
3. The focus of a conic is 50mm front the directory. Draw the locus of a point P moving in such a way that its distance from the directrix is equal to its distance from the focus. Name the cure. Draw a tangent to the curve at a point 60mm from the directrix.
4. Draw a hyperbola when the distance between its focus and directrix is 50mm and eccentricity is 3/2. Also draw the tangent and normal at a point 23mm from the directrix.
5. Draw a parabola when the distance of focus from the directrix is 40 mm. Draw a tangent at a point on the parabola.
6. Construct a curve when the distance of focus from the directrix is 35 mm and eccentricity is 4/3. Draw a tangent to a point on the curve. Name the curve.
7. A fixed point F is 75 mm from a fixed straight line. Draw the locus of a point P moving in. such a way that its distance from the fixed straight line is 3/2 times its distance from F. draw normal and tangent at any point G on the curve.
8. Draw the locus *o*f a point P which moves in a plane in such a way that the ratio of its distances from a fixed point F and a fixed straight line AB i*s* always 2/3. The distance between the fixed point F and fixed straight line is 50 mm. Also draw a tangent and normal on a point on the locus at a horizontal distance of 55 mm from the fixed straight line.

CYCLOIDS AND INVOLUTES

9. Construct a cycloid given the diameter of the generating circle radius is 30 mm. Draw a tangent at point on the cycloid.
10. Construct the path traced by appoint on a circular disc radius of 30 mm rolls on a circular path of radius 100 mm.
11. Draw the involutes of the following. (i) A square of side 30 mm (ii) Circle of radius 25 mm.
12. A circular disc of radius 24 mm rolls on a plane surface. Draw the locus of a point which is at a distance of 300 mm from the centre of the disc, which rolls for one revolution.
13. Construct a cycloid having a rolling circle of 60 mm diameter. Also draw a tangent and normal at any point P on the curve.
14. A circle of 50mm diameter rolls as a horizontal line for ¾ of a revolution clockwise. Draw the path traced by point P on the circumference of the circle. Also draw a tangent and normal at any point on the curve.
15. Draw the involutes of a circle of diameter 40mm and draw the tangent and the normal to the involutes at any points on the curve.
16. Draw the involutes of a Square of side 50mm and draw the tangent and the normal to the involutes at any points on the curve.

17. Draw free hand sketches of the front, top and right side views of the object shown

b)



UNIT-II

PROJECTION OF POINTS, LINES AND PLANE SURFACES

PROJECTION OF POINTS

1. A point M is 35mm above HP and 40mm in front of VP. Draw its projections

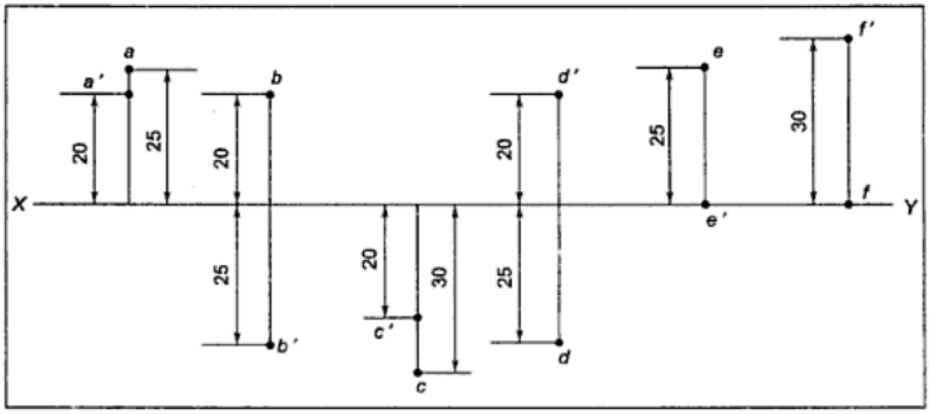
2. Draw the projections of a point B lying on H.P and 55mm in front on V.P

3. Draw the projections of a point A lying on V.P and 55mm in front on H.P

4. Draw the projections of a point Q lying on V.P and 60mm in front on V.P

5. Draw the projections of a point F which lies in both the H.P and the V.P

6. Locate the points in the given figure below



PROJECTION OF STRAIGHT LINES

7. A line AB 60mm long has its end A10mm above HP and 20mm in front of VP. The line is kept perpendicular to HP and parallel to VP. Draw its projections.

8. A line BC 40mm has its end B 25mm above HP and 30mm in front of VP. The line is kept perpendicular to VP and parallel to HP. Draw its projections.

STRAIGHT LINE INCLINED TO ONE PLANE ONLY

9. A line DE 35mm long has its end D 5mm above HP and 10mm in front of VP. The line is kept inclined at 30° to HP and parallel to VP. Draw its projections.

10. A line FG 65mm long has its end F 25mm above HP and on VP. The line is inclined at 45° to HP. Draw its projections.

11. A line GH 35mm long has its end G 20mm in front of VP and on HP. The line is inclined at 35° to VP. Draw its projections.

12. A line QR 65mm long has its end Q 20mm above HP and 15mm in front of VP. Its front view has a length of 35mm. Draw the projections and find the inclination of the line with VP.

STRAIGHT LINE INCLINED TO BOTH HP AND VP

13. A straight line AB of length 50 mm is inclined at 30º to HP and 45º to VP. Point A is 15mm above HP and 20mm in front of VP. Draw the projections of the straight line.

14. A line BC 80mm long has its end B, 15mm from both HP and VP. Other end C is 40mm above HP and 50mm in front of VP. Draw the projections of the line and determine the inclination of the line with HP and VP.

15. A line DE 75mm long has its end D on both HP and VP. The line is kept inclined at 45° to HP and 30° to VP. Draw its projections.

16. One end E of a line EF, 75mm long is 20mm above HP and 25mm in front of VP. The line is inclined at 30° to HP and the top view makes 45° with VP. Draw the projections of the line and find its true inclination with VP.

17 .A line QR has its end Q 20mm above HP and 25mm in front of VP. The other end R is 45mm above HP and 55mm in front of VP. Distance between the end projectors is 60mm. Draw its projections and also find the true length and true inclinations of the line with HP and VP.

18. The midpoint of the line RS 90mm long is 60mm above HP and 50mm in front of VP. It is inclined at 30° to HP and 45° to VP. Draw its projections.

19. A Square plane of side 40mm has its surface parallel to VP and perpendicular to HP. Draw its projections when one of the sides is inclined at 30° to HP.

20. A Circular plane of diameter 50mm has its surface parallel to HP and perpendicular to VP. Its Centre is 20mm above HP and 30mm in front of VP. Draw its projections.

21. A Pentagonal lamina of side 30mm is placed with one side on HP and the surface inclined at 50° to HP and perpendicular to VP. Draw its projections.

22. A Hexagonal plane of side 30mm is placed with a side on VP and the surface inclined at 45° to VP and perpendicular to HP. Draw its projections.

23. A pentagon of side 30mm rests o the ground on one of its corners with the sides containing the corner being equally inclined to the ground. The side opposite to the corner on which it rests is inclined at 30° to the VP and is parallel to HP. The surface of the pentagon makes 50° with the ground. Draw the top and front views of the pentagon.

24. Pentagonal lamina of side 30mm is resting on one of its edges on HP which is inclined at 45° to VP. Its surface is inclined to 30°. Draw its projections.

UNIT – III

PROJECTION OF SOLIDS

SOLIDS IN SIMPLE POSITION

1. A Cube of 40mm side rests on HP on one of its ends with a face parallel to VP and 25mm in front of VP. Draw the top and front views of the cube.
2. A Cube of side 40mm rests on HP on one of its ends with a vertical face inclined at 40° to VP. Draw its projections.
3. A Cube of side 40mm rests on HP on one of its base with its vertical faces equally inclined to VP. Draw its projections.
4. Draw the top and front view of a square pyramid of base side 30mm and altitude 40mm when it is resting on HP on its base with one of the edges of the base inclined at 60° to VP.
5. Draw the projection of a pentagonal prism of base side 20mm and axis length 35mm when it is resting on HP on its base with one of the edges of the base inclined at 30° to VP.
6. Draw the projections of a hexagonal pyramid of side of base 30mm and altitude 60mm when it is resting on its base on a HP with an edge of the base inclined at 30° VP.
7. A square prism of base edge 25mm and axis length 60mm is lying on HP on one of its longer edges with its rectangular faces equally inclined to HP. Draw its projections when the axis is perpendicular to VP and parallel to HP.
8. A Pentagonal pyramid of base side 30mm and axis length 60mm rests on HP on one corner of the base. The base edge containing that corner makes 40° with HP. Draw its projections when the axis is perpendicular to VP and parallel to HP.
9. A hexagonal prism of base side 20mm and axis length 50mm lies on HP on one of its rectangular faces with the axis parallel to both HP and VP. Draw its projections.
10. A pentagonal prism of base side 40mm and axis 60mm long lies on HP on one of its longer edges with its axis parallel to both HP and VP. One of the rectangular faces containing the resting edge is inclined at 30° to HP. Draw its plan and elevation.

AXIS INCLINED TO ONE PLANE AND PARALLEL TO THE OTHER

11. A hexagonal prism of base side 30mm and axis length 60mm lies on HP on one of its base edges with its axis inclined at 60° to HP and parallel to VP. Draw its projections.
12. Draw the projections of a cube of side 40mm when it rests on HP on one of its corners and the face containing that corner is inclined at 30° to HP and parallel to HP.
13. A square prism of base side 35mm and axis length 60mm rests on one of its base edges on HP with its axis inclined at 30° to HP and parallel to VP. Draw its projections.
14. A cylinder of base diameter 30mm and axis length 50mm is resting on HP on a point of base so that its axis is inclined at 45° to HP and parallel to VP. Draw its front view and top views.
15. A hexagonal pyramid of base edge 40mm and height 80mm lies on one of its base edges on HP with its axis inclined at 30° to HP and parallel to VP. Draw its projections.
16. A square pyramid of base side 30mm and height 60mm lies on HP on one of its triangular faces with its axis parallel to VP. Draw its projections.
17. A pentagonal pyramid of base edge 25mm and axis 60mm long rests on a base side on HP such that the highest base corner is 20mm above HP and its axis is parallel to VP. Draw its projections.
18. A cone of base diameter 40mm and altitude 60mm rests on HP with its axis inclined at 30° to HP and parallel to VP. Draw its projections.
19. Draw the projections of a cone of base diameter 50mm and axis length 70mm when it lies on the ground on one of its generators with its axis parallel to VP.
20. A hexagonal pyramid of base side 30mm and axis length 60mm is resting on HP on one of its base corners with its axis inclined at 35° to VP and parallel to HP. Draw its projections when the base sides containing the resting corner are equally inclined to HP.

UNIT-IV

SECTION OF SOLIDS AND DEVELOPMENT OF SURFACES

1. A square prism of base side 30mm and axis length 60mm is resting on HP on one of its faces with a base side inclined at 25° to VP. It is cut by plane inclined at 40° to HP and perpendicular to VP and is bisecting the axis of the prism. Draw its front view, sectional top view and true shape of the section.
2. A pentagonal pyramid of base side 40mm and axis length 75mm is resting on HP on its base with one of its base sides parallel to VP. It is cut by a plane inclined at 35° to HP and perpendicular to VP and is bisecting the axis. Draw its front view, sectional top view and true shape of the section.
3. A hexagonal prism of base side 30mm and axis length 60mm is resting on HP on its base with two of the vertical faces perpendicular to VP. It is cut by a plane inclined at 50° to HP and perpendicular to VP and passing through a point at a distance of 12mm from the top face. Draw its front view, sectional top view and true shape of the section.
4. A cylinder of base diameter 45mm and height 65mm lies on its base on HP. It is cut by a plane perpendicular to VP and inclined at 30° to HP. And meeting the axis at a distance of 30mm from the base. Draw its front view, sectional top view and true shape of the section.

5. A cone of base diameter 50mm and axis length 75mm is resting on HP on its base .It is cut by a plane inclined at 45° to HP and perpendicular to VP and is bisecting the axis. Draw its front view, sectional top view and true shape of the section.
6. A hexagonal pyramid of base side 30mm and height 60mm is resting on HP on its base with two of the base sides perpendicular to VP. It is cut by a section plane perpendicular to VP and parallel to and 25mm above HP. Draw its front view and sectional top view.
7. A square pyramid of base side 30mm and axis length 60mm is resting on HP on its base with one side of the base inclined at 30° to VP. It is cut by a plane perpendicular to VP and parallel to and 15mm above HP. Draw its front view and sectional top view.
8. A pentagonal prism of base side 30mm and axis length 60mm is resting on HP on one of its rectangular faces with its axis perpendicular to VP. It is cut by a section plane inclined at 40° to VP and perpendicular to HP and passing through a point 25mm from the rear face of the prism. Draw its front view, sectional top view and true shape of the section.
9. A cone of base diameter 60mm and axis length 70mm is resting on HP on its base. It is cut by a plane perpendicular to VP and parallel to an end generator (contour generator) and is 10mm away from it. Draw its front view, sectional top view and true shape of the section.
10. A cube of 70mm long edges has its vertical faces equality inclined to the VP. It is cut by an auxiliary inclined plane in such a way that the true shape of the cut part is a regular hexagon. Determine the inclination of the cutting plane with the HP. Draw front view, sectional top view and true shape of the section.
11. A cylinder of diameter 40mm and height50mm is resting vertically on one of its end on the hp. It is cut by a plane perpendicular to the VP and inclined at 30®to the hp. The plane meets the axis at a point 30mm from the base. Draw the development of the lateral surface of the lower portions of the truncated cylinder.
12. A right circular cone, 40mm base and 50mm height, rests on its base on HP. A section plane perpendicular to VP and inclined to HP AT 45^0 cuts the cone bisecting axis. Draw projections of the truncated cone and develop its lateral surface.
13. A pentagonal pyramid of 40mm edge of base and height 70mm rests with its base on HP. One of the bases edges is perpendicular to VP and line on the left of axis of the pyramid. A section plane perpendicular to VP and inclined at 30^0 to VP cut the axis of the pyramid at a point 30mm above the base of the pyramid. Draw the isometric projection of the truncated pyramid.
14. A pentagonal pyramid of base edge 25mm and height 60mm rests vertically on its base on the HP such that one of its base edge parallel to VP. It is cut by a plane, inclined at 60 to HP and passes through a point 35mm from the apex. Draw the development of the lateral surface of the pyramid.
15. A cone of base diameter 50mm and height 75mm is resting on its base on the ground. It is cut by a plane perpendicular to the V.P. and parallel to H.P and meeting the axis at a distance of 20mm from the vertex. It is also cut by a plane inclined at 350 to the base and meeting the axis at a point 20mm above base. Draw the development of the lateral surface of the cut cone.
16. A hexagonal prism of base side 50mm and axis length 80mm rests on the HP on one of its Rectangular faces with its axis perpendicular to the VP. It is cut by a vertical plane inclined at 30˚ to the VP. The cutting plane meets the axis at a distance of 30mm from one end. Draw the top view, sectional front view and the true shape of the section.
17. A cylinder of diameter 45 mm and height 70 mm is resting vertically on one of its ends on the HP. It is cut by a plane perpendicular to VP and inclined at 45º to HP. The plane meets

the axis at a point 35 mm above the base. Draw the development of the lateral surface of the lower portion of the truncated cylinder.

18. A cone of diameter 60mm and axis 80mm long is cut by a plane perpendicular to VP and inclined at 40º to HP and bisects the axis. Draw the front view, sectional top view and true shape of the section.
19. A pentagonal prism of base side 30mm and height 65mm is cut by a plane perpendicular to VP and inclined at 30º to HP and bisecting the axis. Draw the development of prism.
20. A square prism of base side 40 mm and axis length 50 mm is resting on HP on its base with a side of base inclined at 35 ° to VP. It is cut by a plane inclined at 30 ° to HP and perpendicular to VP and bisecting the axis. Draw the development of remaining portion of the prism.

UNIT – V

ISOMETRIC AND PERSPECTIVE PROJECTION

1. A hexagonal prism of base side 20mm and height 40mm has a square hole of side 16mm at the centre. The axes of the square and hexagon coincide. One of the faces of the square is parallel to the face of the hexagon. Draw the isometric projection of the prism with hole to full scale.
2. A pentagonal pyramid of 40mm edge of base and height 70mm rests with its base on HP. One of the bases edges is perpendicular to VP and line on the left of axis of the pyramid. A section plane perpendicular to VP and inclined at 30®to VP cut the axis of the pyramid at a point 30mm above the base of the pyramid. Draw the isometric projection of the truncated pyramid.
3. An object consists of a hemispherical vessel of 80mm diameter which is placed centrally over a cylinder of 50mm diameter and height of 60mm. the cylinder in turn is placed centrally over a square prism of 60mm base side and 20mmh height. Draw the isometric projection of the object.
4. A regular hexagonal pyramid of base edge 20mm and height 35mm rests on its base on the ground plane with one of its base edges touching the picture plane. The station point is 30mm above the ground plane and 40mm in front of the pp. The center plane is 30mm to the right of the axis. Draw the perspective projection of the pyramid.
5. A square prism of 25mm side of base and height 40mm rests with its base on ground such that one of the rectangular faces is inclined at 30^0 to the picture plane. The nearest vertical edge touches the picture plane. The station point is 50mm in front of the picture plane, 60mm above the ground and lies opposite to the nearest vertical edge the touches the picture plane. Draw the perspective view.
6. Draw the perspective view of a square prism of edge of base 40mm and length 60mm lying on a rectangular face on the ground, with a corner on PP and the bases equally inclined to PP. the station point is 60mm in front of PP and 80mm above GL and lies in a central plane, which is passing through the centre of the prism.
7. Make free hand sketches of front, top and right side views of the 3D object shown below.
8. Draw the perspective projection of a cube of 25mm edge, lying on a face on the ground plane, with an edge touching the picture plane and all vertical faces equally inclined to the picture plane. The station point is 50mm in front of the picture plane, 35mm above the ground plane and lies in a center plan which is 10mm to the left of the cube.
9. Draw the perspective projection of a cub of 25mm edge, lying in a face on the ground plane, with

an edge touching the picture plane and all vertical faces equally inclined to the picture plane. The station point is 50mm in front of the picture plane, 35mm above the ground plane and plane and lies in a central plane which is 10mm to the left of the centre of the cube.

10. A regular hexagonal pyramid of base edge 20mm and height 35mm rests on its base on the ground plane with one of its base edges touching the picture plane. The station point is 30mm above the ground plane and 40mmin front of the PP. the center plane is 30mm to the right of the axis. Draw the perspective projection of the pyramid.
11. A cylinder of height 65 mm and diameter 40 mm is resting on its base on the HP. It is cut by a plane perpendicular to VP and inclined at 30º to the HP. The plane passes through a point on the axis located at 25 mm from the top. Draw the isometric projection of the cut cylinder.
12. A frustum of a square pyramid of bottom edge 50 mm, top edge 25 mm and height 50 mm. Draw the isometric projection of the frustum.

13. A hexagonal pyramid of base 25 mm and height 60 mm stands with its base on the HP with an edge of base parallel to VP. A horizontal plane cuts the pyramid and passes through a point on the axis at a distance of 30 mm from the apex. Draw the isometric projection of the frustum of the pyramid.
14. A pentagonal pyramid of base side 30 mm and height 65 mm stands with its base on HP with a side of base perpendicular to VP. It is cut by a plane inclined at 30º to HP and perpendicular to VP and passes through a point at a distance of 30 mm from the apex. Draw the isometric view of the bottom portion of the pyramid.
15. Draw the isometric projection of a hexagonal prism of base side 25 mm and height 50 mm when it rests on one of its ends on HP with two its base sides parallel to VP.
16. A cone of 50 mm diameter and height 70 mm stands on HP with its base. It is cut by a cutting plane perpendicular to VP and inclined at 30º to HP, cutting the axis of the cone at a height of 40 mm from the base. Draw the isometric view of the remaining part of the cone.
17. A pentagonal pyramid of edge of base 40 mm and axis 70 mm long, is resting on its base on HP. One of the base edges of the pyramid is perpendicular to VP. A section plane, perpendicular to VP and inclined to HP at 30^0, passes through the axis, at a height of 30 mm from the base. Draw the isometric view of the truncated pyramid.
18. A cylinder of diameter of base 60 mm and height 70 mm rests with its base in HP. A section plane perpendicular to VP and inclined at $45^{0\ to}$ HP cuts the cylinder such that it passes through a point on the axis above 50 mm. Draw the isometric projection of the truncated cylinder showing the cut surface.
19. A cone of diameter of base 60 mm and height 65 mm rests with its base on HP. Acutting plane perpendicular to VP and inclined at 30^0 to HP cuts the cone such that it passes through a point on the axis at a distance of 30 mm above the base of the cone. Draw the isometric projection of the truncated cone showing the cut surface.
20. A cube of side 40mm is resting on the ground on one of its faces, with a vertical face in PP and the rest behind it. The central plane is located 50mm to the left of the axis of the cube. This station point is 40mm in front of PP and 60mm above GP. Draw the perspective view of the solid.
21. A square pyramid of side of base 50mm and altitude 70mm stands on the ground vertically with

an edge of base parallel to and 20mm behind PP. The station point is 40mm in front of PP and 70mm above the ground. The central plane is located 45mm to the left of the axis of the solid. Draw the perspective view of the solid.

22. A Pentagonal pyramid of 30mm base side and axis height 40mm is standing on its base on the ground Plane with a base side parallel to and 25mm behind PP. The central plane is 35mm to the left of the apex and the station point is 40mm in front of PP and 20mm above the GP. Draw the perspective view of the solid.
23. A cylinder of diameter 40mm and height 65mm rests with its base on the GP such that the axis is 25mm behind the PP. The station point is 30mm in front of the PP and 110mm above the GP and lies in a central plane which is 65mm to the right of the axes of the solids. Draw the perspective view of the cylinder.
24. Draw the perspective projection of a square prism of base side 40 mm and height 50 mm. One of the vertical lateral faces is parallel to PP and 30 mm behind it. The station point is 80 mm from the PP and 80 mm above the ground and 60 mm to the right of the axis of the prism. (Use visual ray method).
25. Draw the perspective view of a square prism of base side 20mm and height 35mm resting on ground with a rectangular face parallel to the picture plane. The axis of the prism is 25 mm behind the picture plane and 25 mm to the right of the eye. The eye is 50 mm infront of the PP and 50 mm above the ground.

26. A square prism of base 25 X 25 mm and height 40 mm is resting on the GP on its square base with a right side rectangular face making 60^0 with picture plane. The corner nearest to the PP is 40 mm to the left of the station point and 20 mm behind the PP. the station point is 60 mm above the GP and 50 mm infront of the PP. Draw the perspective view of the prism by using visual ray method.